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Polarization control devices in silicon photonics

Dispositifs de contrôle de la polarisation en photonique sur silicium

Thèse de doctorat de l'université Paris-Saclay

École doctorale n°575: electrical, optical, bio : physics and engineering
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université PARIS-SACLAY
ÉCOLE DOCTORALE
Physique et ingénierie : Electrons, Photons, Sciences du vivant (EOBE)

Titre: Dispositifs de contrôle de la polarisation en photonique sur silicium
Mots clés: Polarisation, Photonique silicium, Séparateur de faisceaux polarisant, Séparateur et rotateur de polarisation, Rotateur de polarisation, Photonique sub-longueur d'onde

Résumé: Contrôler la polarisation de la lumière est une exigence fondamentale en photonique intégrée, car de nombreux systèmes optiques reposent sur la capacité à manipuler et à séparer les états de polarisation. La technologie silicon-on-insulator (SOI) constitue l'une des plateformes les plus prometteuses pour l'optique intégrée, en raison de son fort contraste d'indice, qui permet une intégration dense de composants optiques, tout en étant compatible avec les procédés de fabrication CMOS. Cependant, l'absorption du silicium devient importante pour des longueurs d'onde inférieures à 1200 nm, ce qui limite son utilisation, par exemple pour les applications quantiques exploitant souvent des longueurs d'onde plus courtes que celles typiquement utilisées en télécommunications (1300 nm, 1550 nm). Cette limitation a motivé l'adoption du nitrure de silicium (SiN), pouvant fonctionner à des longueurs d'onde plus courtes, offrant par ailleurs des pertes de propagation potentiellement plus faibles que le silicium, tout en restant compatible avec une fabrication à grande échelle. Malgré ces avantages, le contrôle de la polarisation en photonique sur SiN, en particulier pour des longueurs d'onde inférieures à 1200 nm, a jusqu'à présent suscité relativement peu d'attention.

Cette thèse vise à combler cette lacune en développant des dispositifs de contrôle de la polarisation compacts et performants sur la plateforme SiN, fonctionnant à des longueurs d'onde en dessous des longueurs d'onde typiques de télécommunication, mais également d'autres fonctionnant à des longueurs d'onde de télécommunication. Plusieurs composants ont été conçus, fabriqués et validés expérimentalement, notamment des séparateurs de faisceaux polarisants, des séparateurs-rotateurs de polarisation et des rotateurs de polarisation. Différentes stratégies de conception ont été explorées afin d'améliorer la compacité et la robustesse des dispositifs, notamment l'utilisation de guides d'ondes sub-longueur d'onde et de géométries trapézoïdales. Ces approches démontrent que la plateforme SiN peut supporter des dispositifs performants de contrôle de la polarisation, malgré son contraste d'indice plus faible comparé à celui de la plateforme SOI.

Dans l'ensemble, ce travail établit une base complète pour le contrôle de la polarisation sur la plateforme SiN, démontrant son potentiel pour une intégration large bande, à faibles pertes et compacte. Les résultats ouvrent la voie à la réalisation de circuits photoniques intégrés complexes, capables d'assurer une gestion active et passive de la polarisation pour des applications allant des télécommunications au traitement de l'information quantique.

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ÉCOLE DOCTORALE
Physique et ingénierie : Electrons, Photons, Sciences du vivant (EOBE)

Title: Polarization control devices in silicon photonics

Keywords: Polarization, Silicon photonics, Polarization beam splitter, Polarization splitter-rotator, Polarization rotator, Subwavelength photonics

Abstract: Controlling light polarization is a fundamental requirement in integrated photonics, as many optical systems rely on the ability to manipulate and separate polarization states. Silicon-on-insulator (SOI) technology is one of the most promising platforms for integrated optics due to its high refractive index contrast, which enables high-density integration of optical components, and its compatibility with CMOS fabrication processes. However, silicon absorption becomes high at wavelengths below 1200 nm, limiting its use for example in quantum applications that often exploit wavelengths shorter than the typical telecommunication ones (1300 nm, 1550 nm). This limitation has motivated the adoption of silicon nitride (SiN), which allows operation at shorter wavelengths while potentially offering lower propagation losses compared to silicon and being compatible with large-scale fabrication processes. Despite these advantages, polarization control in SiN photonics, particularly below 1200 nm, has received relatively little attention.

This thesis addresses this gap by developing compact and high-performance polarization-control devices based on the SiN platform operating at sub-telecom and telecom wavelengths. Several components were designed, fabricated, and experimentally validated, including polarization beam splitters, polarization splitter–rotators, and polarization rotators. Different design strategies were investigated to enhance device compactness and robustness, including subwavelength waveguides and tapered geometries. These approaches demonstrate that SiN platform can effectively support advanced polarization management devices, despite its lower refractive index contrast compared to SOI platform.

Overall, this work establishes a comprehensive foundation for polarization control on the SiN platform, demonstrating its potential for broadband, low-loss, and compact integration. The results pave the way toward the realization of complex, fully integrated photonic circuits capable of active and passive polarization management for applications ranging from telecommunications to quantum information processing.

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This thesis could not have been completed without the constant support of my supervisors, Daniele and Carlos. Throughout these years, they have guided my progress both as a researcher and as a person. They introduced me to the key concepts of integrated photonics and helped me develop a structured way of thinking when approaching complex problems. They also taught me to stay patient, curious, and focused, even when experiments failed. Their trust gave me the confidence to explore new ideas, take initiative, and learn from every step of the process. I am truly grateful for their mentorship and for the supportive environment they created around this project.

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Résumé en français

Introduction

La polarisation décrit l'orientation des oscillations du champ électrique d'une onde électromagnétique. Selon l'évolution du champ électrique, elle peut être linéaire, circulaire ou elliptique. Longtemps considérée comme un phénomène fondamental, la polarisation est aujourd'hui au cœur d'un large éventail d'applications allant de la détection à l'encodage de l'information [1–4]. À titre d'exemples, elle est exploitée en télédétection et en imagerie, où l'analyse polarimétrique révèle des contrastes invisibles aux mesures d'intensité seule [5–7] ; en télécommunications, où le multiplexage en polarisation permet d'augmenter significativement le débit de transmission [8–10] ; en optique quantique, via l'encodage de qubits dans les protocoles de distribution quantique de clés de chiffrement (QKD) [11–13] ; ainsi qu'en biologie et en médecine, pour l'analyse et la caractérisation des tissus biologiques [14–16].

Au cours des dernières décennies, une tendance forte à la miniaturisation des systèmes optiques s'est développée. Dans ce contexte, la photonique sur silicium s'est imposée comme une plateforme de premier plan grâce à son fort indice de réfraction, ainsi qu'à la disponibilité d'infrastructures CMOS déjà largement établies, permettant une fabrication dense, reproductible et à grande échelle. La plateforme standard SOI, constituée d'un cœur en silicium (Si) entouré de dioxyde de silicium (SiO_2), permet de réaliser des dispositifs compacts mais présente une biréfringence importante lorque la forme du coeur est rectangulaire. Cela est due au contraste d'indice élevé entre Si et SiO_2 ($n_{Si} = 3.48$ contre $n_{SiO_2} = 1.44$). Cette sensibilité à la polarisation peut devenir problématique, en particulier lors du couplage avec des fibres optiques présentant une polarisation aléatoire, d'où l'importance de dispositifs dédiés à son contrôle: séparateurs de polarisation, rotateurs et séparateurs-rotateurs. Par ailleurs, la plateforme SOI est limitée aux longueurs d'onde supérieures à ~1200 nm en raison des pertes d'absorption croissantes à plus courte longueur d'onde.

C'est ainsi que le nitrure de silicium (SiN) s'est imposé comme une alternative particulièrement attractive, grâce à sa large fenêtre de transparence (≈ 370–3700 nm), ses faibles pertes de propagation, sa biréfringence réduite et sa faible non-linéarité. Ces propriétés permettent de viser des applications à courte longueur d'onde telles que l'absorption saturée du rubidium (780 nm) [17], le contrôle de l'état de polarisation de photons à 950 nm émis par des boîtes quantiques InP [18], ou encore la génération de seconde harmonique dans des couches de MoS_2 déposées sur SiN (755 nm) [19]. Le SiN est également pertinent en télécommunications (1300–1550 nm) en raison de ses faibles pertes, notamment pour les (dé)multiplexeurs de modes [20] et les systèmes de tomographie en cohérence optique utilisant des réseaux à guides d'onde (AWG) [21, 22]. Toutefois, son contraste d'indice plus faible rend la séparation ou la conversion TE/TM plus difficile, ce qui complique la réalisation de dispositifs compacts et performants.

Ainsi, l'objectif général de cette thèse est de développer des dispositifs de contrôle de polarisation sur la plateforme SiN, à la fois dans les bandes télécom (1300 nm) et à courte longueur d'onde (950 nm), dans le cadre d'un projet mené en collaboration avec le Conseil National de Recherches du Canada (CNRC) visant à réaliser des circuits photoniques quantiques intégrant des sources de photons

uniques à base de boîtes quantiques InP.

Séparateur de faisceaux polarisant

L'un des éléments essentiels pour la gestion de la polarisation est le séparateur de faisceaux polarisant (PBS), un dispositif capable de séparer les modes TE et TM à l'entrée vers deux ports de sortie distincts, comme illustré sur la Fig. 1.

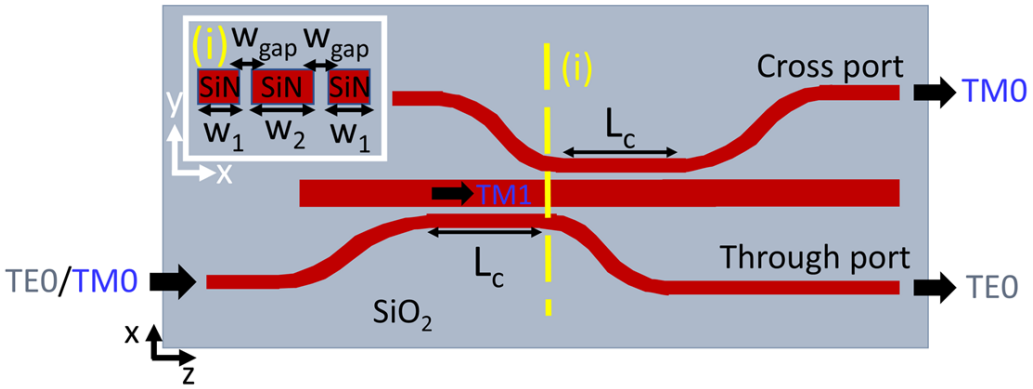

Figure 1: Schéma en vue de dessus de notre dispositif, il s'agit d'un PBS basé sur une cascade de coupleur directionnel asymétrique, avec la vue de coupe indiquée en position (i) par la ligne pointillée jaune.

Pour évaluer les performances d'un PBS, les métriques utilisés sont les pertes d'insertion (IL) et le taux d'extinction de polarisation (PER), définis comme suit:

$$PER_{TE} = 10 \cdot \log_{10}\left(\frac{T_{through}}{T_{cross}}\right) \tag{1}$$

$$PER_{TM} = 10 \cdot \log_{10}\left(\frac{T_{cross}}{T_{through}}\right) \tag{2}$$

$$IL_{TE} = -10 \cdot \log_{10}(T_{through}) \tag{3}$$

$$IL_{TM} = -10 \cdot \log_{10}(T_{cross}) \tag{4}$$

où $T_{through}$ et T_{cross} représentent les transmissions mesurées aux ports through et cross indiqué Fig. 1. Le PER quantifie la qualité de séparation : un PER élevé correspond à une meilleure discrimination entre les deux polarisations. Les IL mesurent, quant à elles, les pertes introduites par le dispositif lors du processus de séparation. Au cours de la thèse, nous avons développé deux PBS fonctionnant à différentes longueurs d'onde : l'un centré à 1300 nm et l'autre à 950 nm. Le schéma du dispositif est présenté Fig. 1. Il s'agit d'un PBS basé sur une cascade de deux coupleurs directionnels asymétriques (ADCs).

Principe de fonctionnement

Le mode TM_0 provenant du guide inférieur, de largeur w_1, est progressivement rapproché du guide central de largeur w_2 au moyen d'une courbe en forme de S. Lorsque la condition d'accord de phase $n_{eff}^{TM_0}(w_1) = n_{eff}^{TM_1}(w_2)$ est satisfaite, le mode TM_0 est converti en mode TM_1 dans le guide central après une certaine longueur de couplage (L_c). Ce mode TM_1 est ensuite reconverti en mode TM_0 dans le guide supérieur (également de largeur w_1) grâce à la même condition d'accord de phase. Après une séparation progressive via une second courbe en S, le mode TM_0 ressort au port cross. En parallèle, le mode TE_0 n'est pas couplé au guide central puisque la condition $n_{eff}^{TE_0}(w_1) \neq n_{eff}^{TE_1}(w_2)$ n'est pas satisfaite. Il continue donc sa propagation dans le guide d'entrée et ressort au port through. Les deux polarisations TE_0 et TM_0 sont ainsi efficacement séparées.

Version effilée (tapered)

Afin d'améliorer la tolérance aux variations de largeur des guides, nous avons également étudié une version effilée du dispositif. Comme montré Fig. 2, la largeur du guide d'entrée varie linéairement de $w_1 + \Delta w$ à $w_1 - \Delta w$, tandis que la largeur du guide central varie de $w_2 - \Delta w$ à $w_2 + \Delta w$, tout en conservant un écart constant entre les guides. Cette configuration vise à réduire l'impact des erreurs de fabrication.

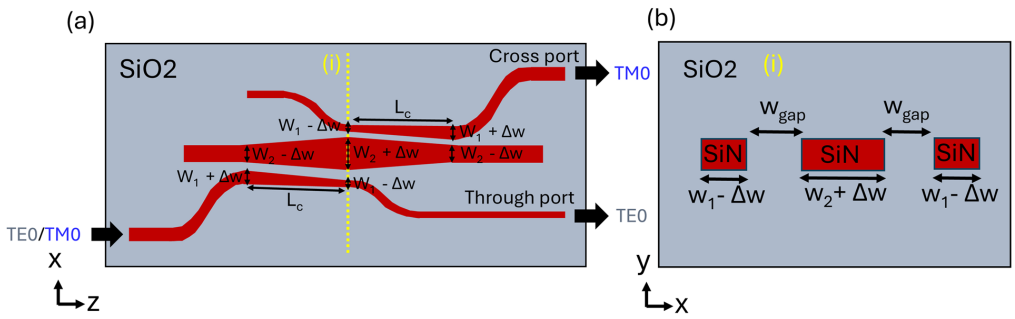

Figure 2: Schéma de notre second dispositif : un PBS basé sur un coupleur directionnel asymétrique effilé. (a) Vue de dessus ; (b) vue de coupe indiquée par la ligne pointillée jaune à la position (i).

Résultats expérimentaux

Les résultats expérimentaux des PBS à 950 nm et 1300 nm sont présentés dans la Fig. 3. Les transmissions, normalisées à l'aide d'un guide de référence, sont représentées en fonction de la longueur d'onde pour différentes variations de largeur de guide (δw). Les résultats montrent que l'ensemble des dispositifs assure correctement la séparation de polarisation, les modes étant mesurés au port approprié avec de faibles pertes d'insertion (< 1 dB). Les deux meilleurs dispositifs sans structure effilée ($\delta w = +10$ nm à $\lambda = 950$ nm et $\delta w = -10$ nm à $\lambda = 1300$ nm) présentent respectivement un PER de 29 dB (TE) et 24 dB (TM) à $\lambda = 930$ nm, ainsi qu'un PER de 32 dB à $\lambda = 1310$ nm. Ces performances sont obtenues pour des longueurs de dispositifs de 110 µm et 124 µm aux longueurs d'onde de 950 nm et 1300 nm, respectivement. Les dispositifs avec structure effilée présentent un $PER_{TE/TM}$ maximal plus faible : 26/16 dB pour les dispositifs opérant autour de 950 nm et 29/22 dB pour ceux opérant autour de 1300 nm. Cependant, cette réduction de performance s'accompagne d'une nette amélioration de

la largeur de bande : le PER reste supérieur à 10 dB sur l'ensemble de la plage spectrale pour la quasi-totalité des dispositifs effillés. Les longueurs de dispositifs correspondantes sont de 242 µm et 290 µm pour les longueur d'onde de 950 nm et 1300 nm, respectivement.

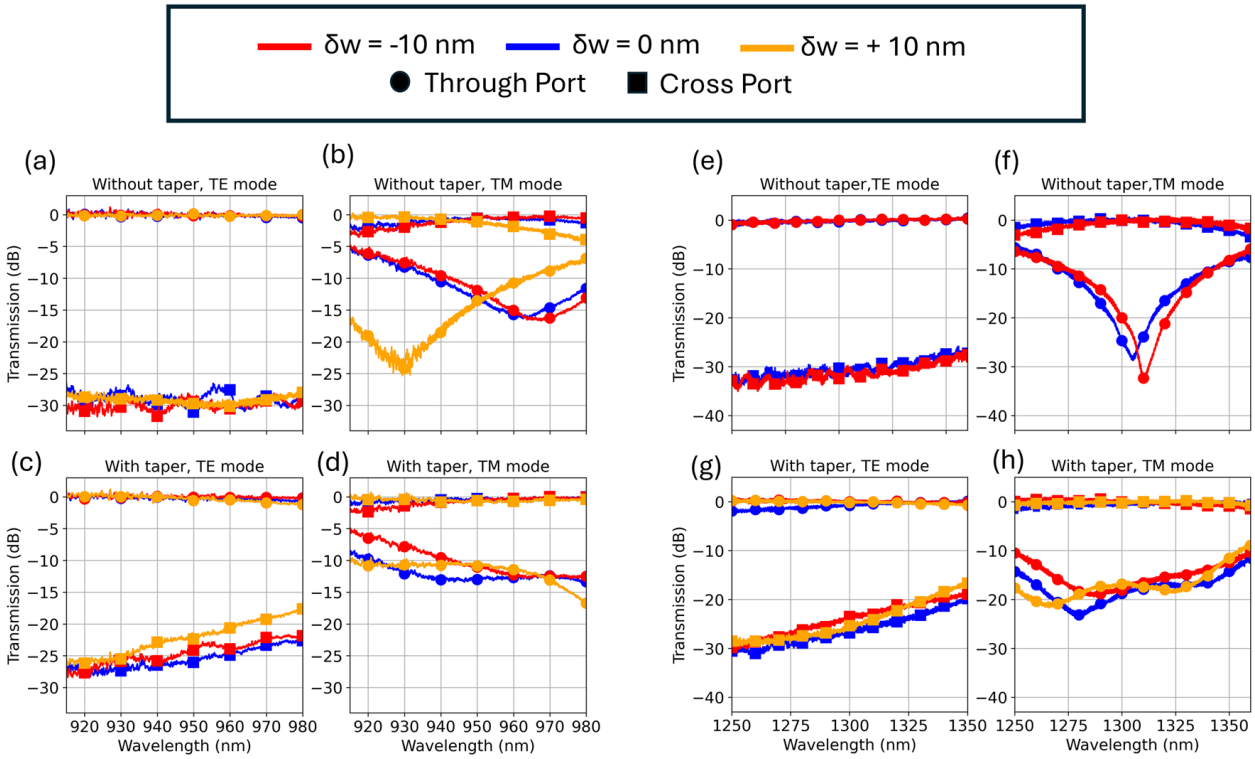

Figure 3: Spectres de transmission normalisée des PBS. Les cercles et carrés représentent respectivement les mesures aux ports through et cross. Les mesures ont été réalisées pour différentes variations de largeur de guide (δw). Les sous-figures (a)–(d) correspondent aux mesures des dispositifs opérant autour de λ = 950 nm, et les sous-figures (e)–(h) à celles des dispositifs opérant autour de λ = 1300 nm. Les sous-figures (a), (b), (e) et (f) montrent les résultats de la version sans effilement du PBS, tandis que (c), (d), (g) et (h) présentent ceux de la version effilée.

Séparateur de faisceaux polarisant sub-longueur d'onde

La structure du PBS développé dans ce chapitre est présentée Fig. 4, avec (a) une vue de dessus dans le plan x–z et (b) une vue en coupe dans le plan x–y. La principale différence par rapport à la structure précédemment étudiée (Fig. 1) est le remplacement du guide central conventionnel par un guide sub-longueur d'onde (SWG). Cette modification apporte deux avantages majeurs.

Le premier avantage est la réduction significative de la longueur du dispositif, pouvant diminuer celle-ci d'un facteur deux. En effet, dans une structure SWG, il est possible de coupler directement le mode TE_0 du guide conventionnel vers le mode TE_0 du guide SWG, tandis que le mode TM_0 reste non couplé. Sur une plateforme SiN, cela constitue un atout majeur, car la séparation TE/TM peut être réalisée sans recourir à un mode d'ordre supérieur.

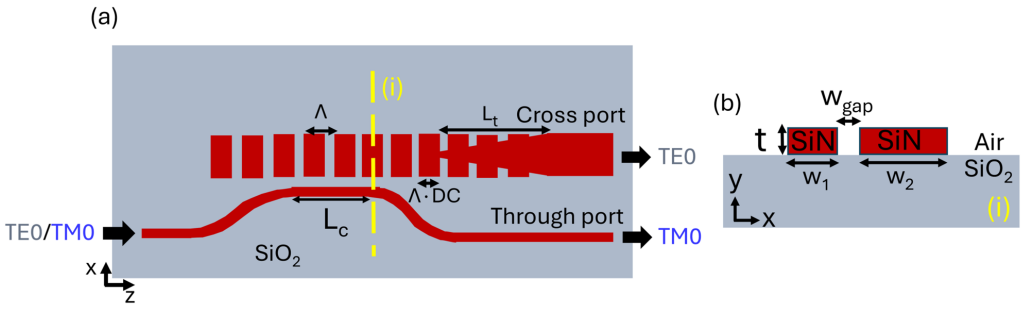

Figure 4: Schéma de notre dispositif, composé d'un PBS basé sur un coupleur directionnel SWG. (a) Vue de dessus dans le plan x–z. (b) Vue en coupe dans le plan x–y à la position (i), indiquée par la ligne en pointillé jaune.

Le second avantage réside dans l'absence volontaire de cladding supérieur. Celle-ci rend les modes TM faiblement guidés dans les guides SWG, ce qui contribue à améliorer le taux d'extinction de polarisation du dispositif. De plus, l'absence de cladding est nécessaire pour certaines applications, notamment la déposition de boîtes quantiques InAsP/InP sur guides SiN [23].

Principe de fonctionnement

Le mode TE_0 du guide d'entrée est couplé vers le mode TE_0 du guide SWG lorsque la condition d'accord de phase est satisfaite $n_{eff}^{TE_0}(w_1) = n_{eff_{SWG}}^{TE_0}(w_2)$. Le guide SWG est ensuite reconverti progressivement en guide conventionnel au moyen d'un guide d'onde inversement effilé. En parallèle, le guide SWG ne supporte pas de mode TM_0 guidé. Le mode TM injecté reste donc entièrement dans le guide d'entrée et ressort au port through. Tout couplage parasite éventuel vers le SWG est rapidement filtré, ce qui améliore fortement le PER du dispositif.

Résultats experimentaux

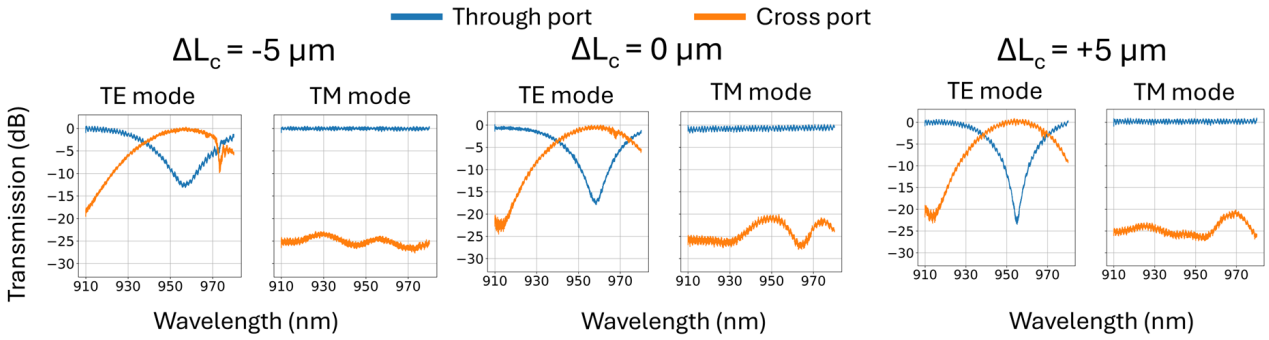

Figure 5: Transmission normalisée du PBS SWG mesurée au ports through et cross en fonction de la longueur d'onde, pour des dispositifs opérant autour de 950 nm, avec une excitation du mode TE ou TM à l'entrée. Les différentes figures correspondent à des déviations de la longueur de couplage comprises entre −5 et +5 μm.

Les performances expérimentales du PBS SWG à 950 nm sont présentées dans la Fig. 5. Les courbes montrent les transmissions normalisées aux ports through et cross lorsque le mode TE ou TM

est injecté, pour différentes déviations de la longueur de couplage du dispositif. En cas de séparation correcte, le mode TE doit être mesuré au port cross et le mode TM au port through, avec de faibles pertes, ce qui est bien observé ici. Les résultats montrent une séparation nette de la polarisation pour les trois déviations de longueur de couplage testées. Le meilleur dispositif correspond à celui dont la longueur de couplage est augmentée de +5 μm. Pour ce dispositif, nous obtenons un PER maximal de 24 dB pour TE et de 26 dB pour TM à $\lambda = 955$ nm, tandis que les pertes d'insertion restent inférieures à 1 dB pour les deux polarisations. Avec une longueur totale de seulement 63 µm, l'utilisation d'un guide SWG permet de réduire la taille du dispositif d'un facteur d'environ deux tout en conservant des performances comparables à celles des PBS conventionnels (sans SWG).

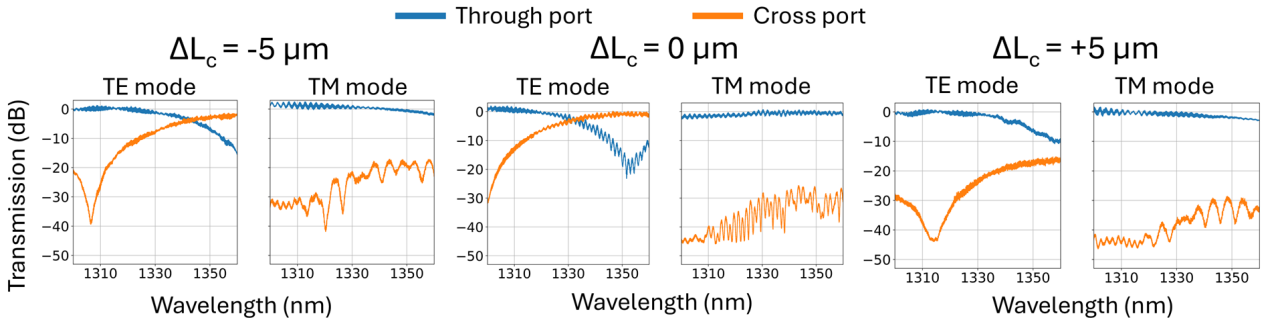

Figure 6: Transmission normalisée des dispositifs PBS SWG mesurée au ports through et cross en fonction de la longueur d'onde, pour des dispositifs opérant autour de 1300 nm, avec excitation en mode TE ou TM à l'entrée. Les différentes figures correspondent à des déviations de la longueur de couplage comprises entre –5 et +5 μm.

Les mesures effectuées pour le dispositif conçu autour de 1300 nm sont présentées dans la Fig. 6. Comme précédemment, la longueur de couplage a été déviée entre –5 μm et +5 μm. Pour deux dispositifs sur trois, une bonne séparation est observée : le mode TE ressort au port cross, tandis que le mode TM reste au port through. Cependant, l'ensemble des courbes présente un décalage spectral d'environ 50 nm vers le rouge, les dispositifs fonctionnant autour de 1355 nm au lieu de 1300 nm. Ce décalage peut s'expliquer par des variations de largeur du guide de l'ordre de 30 nm par rapport à la valeur nominale, observé sur les images SEM du dispositif. Malgré cette erreur de fabrication, le PBS opère correctement autour de 1355 nm pour $\Delta L_c = 0$ μm, atteignant un PER de 20 dB pour TE et 30 dB pour TM.

Séparateur et rotateur de polarisation

Une autre fonction essentielle pour la gestion de la polarisation consiste à séparer deux modes orthogonaux et à convertir l'un en l'autre, par exemple en transformant une polarisation TM en une polarisation TE et en l'acheminant vers un port différent, comme illustré Fig. 7. Pour évaluer les performances d'une telle structure, les métriques couramment utilisées sont la perte de conversion de polarisation (PCL), la perte d'insertion (IL) et le taux d'extinction de polarisation (PER), définis comme

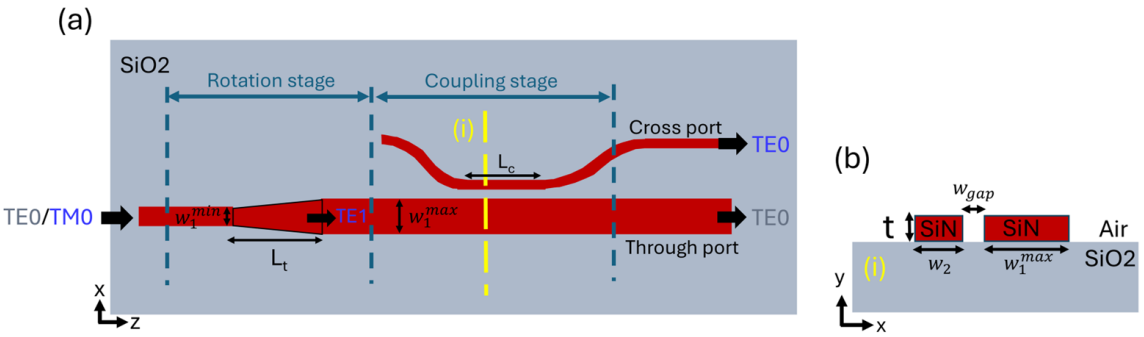

Figure 7: Schéma de notre dispositif, il s'agit d'un PSR basé sur la technique d'évolution de mode combinée à un ADC. (a) Vue de dessus. (b) Vue de coupe à la position (i), indiquée par la ligne pointillée jaune.

suit :

$$PER_{TE} = 10 \cdot \log_{10}\left(\frac{T^{TE-TE}_{through}}{T^{TE-TE}_{cross}}\right), \tag{5}$$

$$PER_{TM} = 10 \cdot \log_{10}\left(\frac{T^{TM-TE}_{cross}}{T^{TM-TM}_{through}}\right), \tag{6}$$

$$IL = -10 \cdot \log_{10}\left(T^{TE-TE}_{through}\right), \tag{7}$$

$$PCL = -10 \cdot \log_{10}\left(T^{TM-TE}_{cross}\right). \tag{8}$$

Ici, $T^{TE-TE}_{through}$ et T^{TE-TE}_{cross} représentent la transmission du mode TE mesurée respectivement aux ports through et cross lorsque TE est injecté à l'entrée. De même, $T^{TM-TM}_{through}$ correspond à la transmission du mode TM mesurée au port through lors d'une excitation TM, tandis que T^{TM-TE}_{cross} désigne la transmission du mode TM converti en mode TE au port cross. Ainsi, comme pour un PBS, le PER évalue la qualité de la séparation des polarisations. La PCL quantifie l'efficacité de conversion du TM en TE (une PCL nulle indiquant ainsi une conversion complète). L'IL mesure quant à elle les pertes introduites par le dispositif.

Dispositif et principe de fonctionnement

Nous avons réalisé un séparateur-rotateur de polarisation (PSR) reposant sur la technique d'évolution de mode et opérant autour de $\lambda = 1300$ nm. Dans notre conception, nous choisissons de travailler sans cladding supérieur en SiO_2, ce qui introduit naturellement une asymétrie nécessaire à la rotation. Le schéma du dispositif est présenté Fig. 7. Son fonctionnement est le suivant : lorsque le mode TM_0 est injecté dans le guide d'entrée, il est progressivement converti en mode TE_1 via un effilement adiabatique du guide d'onde. Ce mode TE_1 est ensuite couplé vers le mode TE_0 du guide supérieur grâce à un coupleur directionnel asymétrique, permettant au TE_0 de sortir par le port cross. À l'inverse, lorsqu'un mode TE_0 est injecté, aucune hybridation n'a lieu : le mode n'est ni converti par l'effilement du guide d'onde ni efficacement couplé par le coupleur directionnel. Il ressort ainsi inchangé au port through.

Résultats expérimentaux

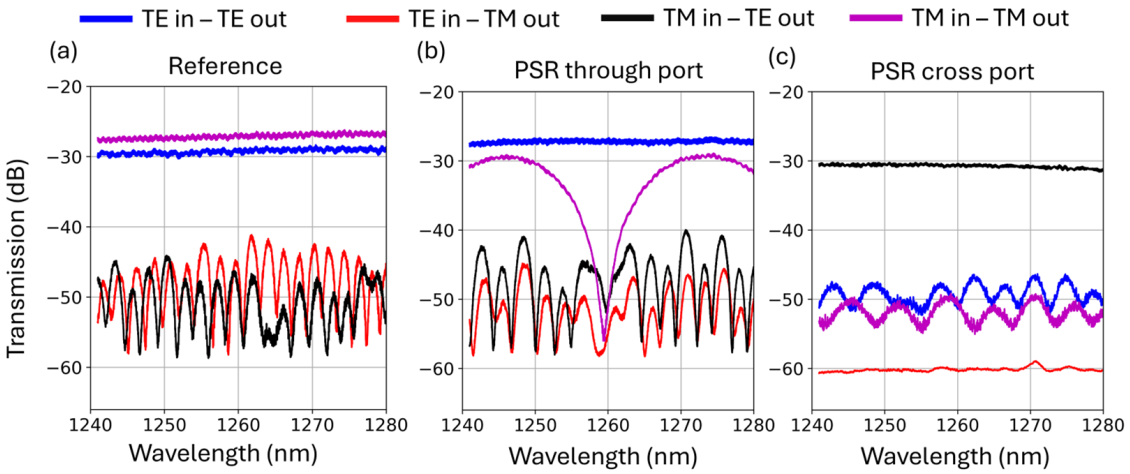

Figure 8: (a) Transmission d'un guide de référence (sans PSR) pour des excitations TE et TM, contrôlées par l'orientation du polariseur d'entrée. Un polariseur de sortie analyse la polarisation TE ou TM. (b) Mesures correspondantes au port through du PSR. (c) Mesures correspondantes au port cross.

La Figure 8 (a) présente les mesures d'un guide de référence (sans PSR), tandis que les Fig. 8 (b) et (c) montrent respectivement les mesures aux ports through et cross. Le dispositif réalise correctement la fonction de séparation et de rotation : au port through, la polarisation dominante mesurée est bien un TE (Fig. 8 (b), courbe bleue), et au port cross, la polarisation dominante mesurée est également un TE issu du TM injecté (Fig. 8 (c), courbe noire). Le dispositif présente un PCL d'environ 2 dB, une IL inférieure à 1 dB, ainsi qu'un $PER_{TE/TM}$ de 28/20 dB pour une longueur totale de 685 µm.

Rotateur de polarisation

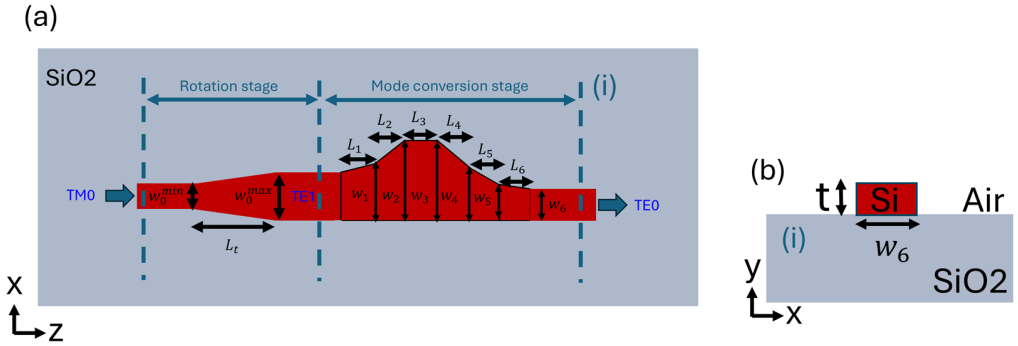

Figure 9: (a) Schéma de notre rotateur de polarisation (PR) basé sur la technique d'évolution de mode et sur un convertisseur de mode TE_1–TE_0 utilisant des guides effilés asymétriques en cascade. (b) Vue de coupe à la position (i), indiquée par la ligne pointillée près de la sortie.

Un rotateur de polarisation (PR) est un dispositif destiné à modifier l'état de polarisation de la lumière en entrée mais, contrairement à un PSR, il ne réalise pas d'étape de séparation. Dans ce contexte, un PR est donc conçu pour agir sur une seule polarisation à la fois. Par exemple, un dispositif peut être spécifiquement optimisé pour convertir TM_0 en TE_0. Utilisé dans le sens opposé, il peut également reconvertir TE_0 en TM_0.

Principe de fonctionnement

La Figure 9 (a) présente le schéma général du dispositif, constitué de deux sections. La première section est un guide d'onde effilé permettant de convertir progressivement le mode TM_0 au mode TE_1 via la technique d'évolution de mode. Cette conversion est rendue possible par l'absence de cladding supérieur, qui introduit l'asymétrie géométrique nécessaire (voir Fig. 9 (b)). Contrairement aux dispositifs précédemment décrits en plateforme SiN, la plateforme SOI est ici utilisée car elle offre un contraste d'indice nettement plus élevé, favorisant une hybridation plus forte entre les modes et permettant ainsi de réduire significativement la longueur de l'effilement du guide d'onde par rapport à celle du PSR présenté plus haut. La seconde section assure ensuite la conversion du mode d'ordre supérieur TE_1 en mode fondamental TE_0, comme illustré dans la Fig. 9 (a). Pour cela, nous combinons le rotateur avec un convertisseur de modes TE_1–TE_0 basé sur une structure effilée asymétrique. L'asymétrie latérale crée une différence de longueur optique entre les deux lobes du mode TE_1 (initialement en opposition de phase). Cette différence permet de les remettre en phase à la sortie, générant ainsi un mode TE_0.

Résultats expérimentaux

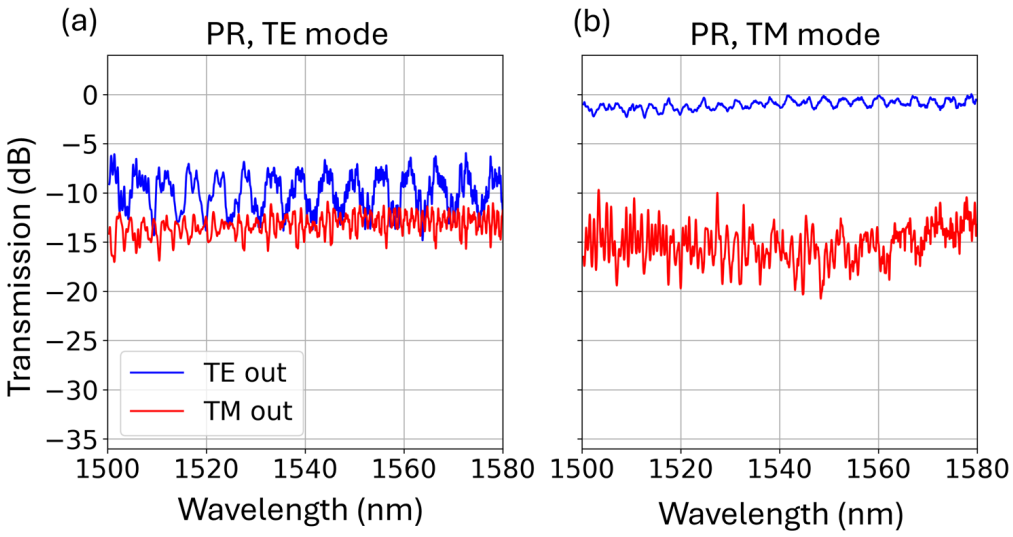

Figure 10: Mesures de transmission normalisées du dispositif PR : (a) pour une excitation TE et (b) pour une excitation TM. La normalisation est effectuée à partir du guide de référence (sans PR). Pour les mesures de même polarisation en entrée et sortie (TE–TE ou TM–TM), la normalisation utilise la transmission de référence correspondante. Pour les mesures en polarisation croisée (TE–TM ou TM–TE), la normalisation repose sur la transmission moyenne des références TE–TE et TM–TM.

La Figure 10 présente les transmissions normalisées du rotateur de polarisation : (a) pour une excitation TE et (b) pour une excitation TM. Le dispositif présente un PCL minimal de 0.5 dB et un PER de 15 dB, pour une longueur totale de 110 µm. Pour ce type de dispositif, le PER est redéfini : il correspond à la différence, en dB, entre le signal TE résultant de la conversion du mode TM injecté en entré et le résidu de TM qui n'a pas été converti en TE. Cette définition correspond à l'écart entre les courbes bleue et rouge de la Fig. 10 (b).

Conclusion

Dans cette thèse, nous avons présenté et réalisé plusieurs dispositifs de gestion de la polarisation sur plateformes SiN et SOI : séparateurs (PBS), séparateurs-rotateurs (PSR) et rotateurs (PR). Ces composants constituent des éléments clés pour les circuits photoniques intégrés, en particulier pour les systèmes quantiques, télécoms et d'imagerie. Deux PBS en SiN ont été démontrés à 950 nm et 1300 nm. La version classique (ADC sans effilement) atteint des PER supérieurs à 29 dB avec des pertes d'insertion < 1 dB pour des longueurs de seulement 110–124 µm. La version effilé offre une meilleure tolérance aux variations de fabrication, au prix d'un PER maximal légèrement réduit mais avec une bande passante plus large. Un PBS ultra-compact basé sur un guide sub-longueur d'onde (SWG) a ensuite été développé. Il permet de réduire par environ deux la longueur du dispositif (63 µm) tout en maintenant un PER de 24–26 dB à 950 nm et jusqu'à 30 dB autour de 1355 nm malgré des écarts de fabrication. Nous avons également démontré un PSR sur SOI, combinant la technique d'évolution de mode et un coupleur directionnel asymétrique. Ce dispositif atteint un PER de 28/20 dB, une IL < 1 dB et une PCL de 2 dB pour une longueur de 685 µm. Enfin, un rotateur de polarisation compact sur SOI a été réalisé, convertissant TM_0 en TE_0 en seulement 110 µm, avec un PCL de 0.5 dB et un PER de 15 dB grâce à l'hybridation modale renforcée offerte par la plateforme SOI. Ces résultats démontrent la faisabilité de dispositifs de polarisation performants, compacts et intégrables, essentiels pour les futurs circuits photoniques complexes, notamment ceux développés dans le cadre du projet CNRC autour des sources de photons uniques InP sur SiN.

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1 - Introduction

1.1 . Polarization in integrated photonics

Polarization is a fundamental property of electromagnetic waves that describe the orientation of the electric fields oscillations (see Fig. 1.1). Depending on how the electric field vector evolves, polarization can be linear, circular, or elliptical. In linear polarization, the electric field oscillates along a fixed direction, while in circular and elliptical polarizations, the field vector rotates around the propagation axis, tracing a circle or an ellipse, respectively.

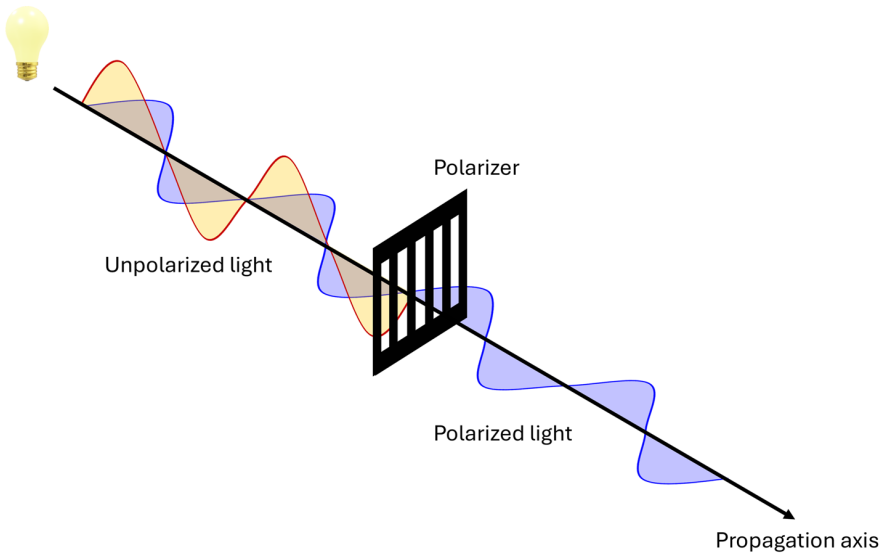

Figure 1.1: Schematic of light polarization

While historically studied as a basic phenomenon in optics, polarization is now widely exploited for applications in control, detection, and information encoding [1–4]. Moreover, polarization control plays a central role in many fields. For instance, in remote sensing and imaging, polarimetric analysis enables the extraction of contrasts invisible to intensity-only measurements, enhancing the detection of surfaces and biomaterials [5–7]. In telecommunications, polarization multiplexing increases data transmission capacity [8–10]. In quantum optics, polarization states can encode qubits, forming the basis of quantum key distribution (QKD) protocols for secure communication [11–13]. In biology and medicine, polarization provides insights into tissue anisotropy, supporting the detection of fibrous structures in cancers [14–16].

The current trend is toward miniaturization and integration of such functionalities. In this context, silicon photonics has emerged as one of the most promising platforms for two main reasons. First, the high refractive index of silicon allows the integration of

dense optical circuits. Second, silicon photonics benefits from established CMOS fabrication infrastructure, enabling large-scale production at reduced cost [24–26]. The standard silicon-on-insulator (SOI) platform consists of a silicon waveguide core surrounded by silicon dioxide (SiO_2) [27, 28]. Figure 1.2 (a) illustrates a typical SOI cross-section. Silicon photonics foundries fabricate such devices on wafers [29, 30], where a wafer is a silicon substrate patterned with multiple photonic chips (see fig 1.2 (b)). It should be noted that the thickness values of the silicon substrate and the SiO_2 cladding shown in Fig. 1.2 (a) are provided as examples, as they depend on the wafer supplier. Although the term silicon photonics originally referred specifically to silicon-core waveguides, it is now broadly used to describe waveguides made from other silicon-based materials, such as SiN [31–33], SiGe [34–36], and SiC [37–39].

In a waveguide, light is confined and propagates as guided modes, which are typically classified as transverse electric (TE) or transverse magnetic (TM). TE modes have their dominant electric field component oriented horizontally (along the x-axis in Fig. 1.3 (a)), while TM modes are primarily vertical (along the y-axis). In Fig. 1.3 (b) and (c), we show the electric field components of the TE_0 and TM_0 modes from the SOI waveguide represented in Fig. 1.2 (a) for a given width w = 0.6 μm. As we can observe from their amplitudes, the major component for the TE_0 mode is indeed E_x, whereas for the TM_0 mode it is indeed E_y. TE and TM modes generally experience birefringence, propagating with different effective indices and therefore at different velocities. The effective index here refers to the modal propagation constant rather than the bulk material index. This birefringence arises from both the waveguide geometry, when the core waveguide is rectangular, and the refractive-index contrast between the core and the cladding. In particular, in silicon waveguides, this difference is high since the refractive index of silicon is $n_{Si} = 3.48$ at $\lambda = 1550$ nm, while that of silica is $n_{SiO_2} = 1.44$. This large index contrast enables compact device designs but simultaneously enhances birefringence, which can be problematic in many applications. For example, in fiber-optic communication, the light from an optical fiber typically has a random polarization state, which can significantly degrade the signal-to-noise ratio when coupling through a polarization sensitive photonic chip [40]. Therefore, the development of polarization management devices is essential.

Typically, in integrated optics, three main types of devices are used to control light polarization. The first is the polarization beam splitter, which separates the TE and TM modes into two distinct output ports. The second is the polarization rotator, designed to convert one polarization state into the other, for example transforming a TM mode into a TE mode (and vice versa). Finally, the polarization splitter and rotator combines the functionalities of both previous devices, simultaneously splitting and rotating the polarization states so that both output ports carry light in the same polarization state.

Although the SOI platform offers many advantages for the development of passive photonic circuits, it also has a major limitation: absorption grows significantly at wavelengths approximately below 1200 nm. Consequently, for certain applications that require operating at wavelengths shorter than those typically used in telecommunications (i.e., λ = 1300 and 1550 nm), this limitation becomes critical.

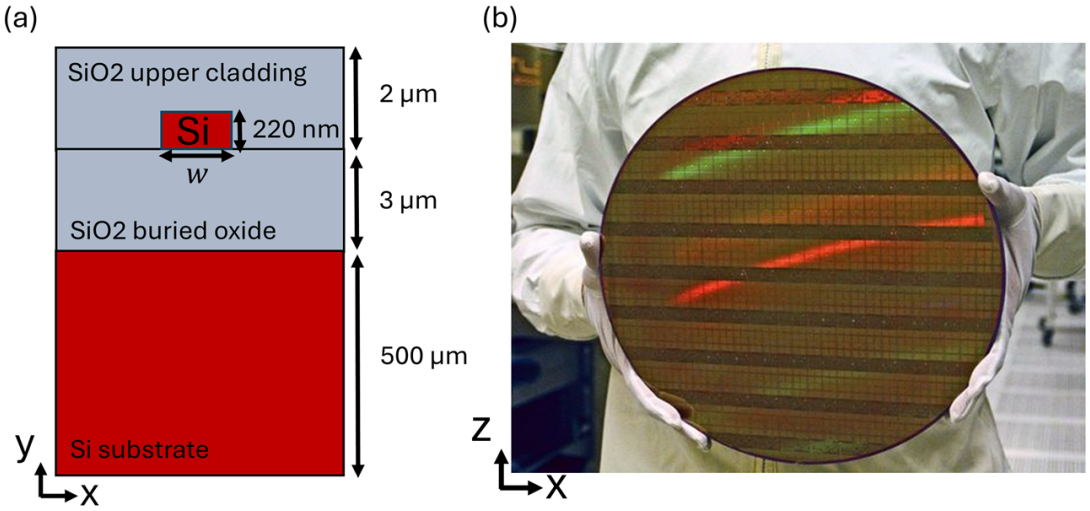

Figure 1.2: (a) Schematic of Silicon-on-Insulator (SOI) platform. The silicon (Si) waveguide is typically surrounded by a silicon dioxide (SiO_2) cladding and rests on a buried oxide (BOX) layer of SiO_2, which sits on a silicon substrate. (b) Silicon photonic wafer.

1.2 . The Silicon Nitride platform

A platform with characteristics complementary to SOI is obtain by substituting the silicon core with a silicon nitride one (SiN). This substitution offers several advantages: a broad transparency window ($\lambda \sim$ 370–3700 nm), reduced birefringence, lower propagation losses and reduced nonlinear effects [31, 32]. These properties significantly expand the range of photonic applications. Figure 1.4 (a), adapted from [41], summarizes application domains across different wavelength ranges, while Fig. 1.4 (b) shows the transparency windows of various waveguide materials.

From Fig. 1.4 (b), it is evident that the short-wavelength applications shown in Fig. 1.4 (a) cannot be implemented in silicon. Silicon nitride therefore offers a promising alternative. Among the various short-wavelength applications, many also rely on polarization control. Examples of such applications include saturated absorption spectroscopy at the rubidium line (780 nm) [17], polarization control of photons at 950 nm generated by InP quantum dots integrated on SiN [18], and TE/TM polarization states at 755 nm from second-harmonic generation in MoS_2 layers deposited on SiN [19]. In addition, the low propagation losses of SiN make it attractive not only at short wavelengths but also in the telecommunications band, for example in waveguide mode (de)multiplexers [20] or in optical coherence tomography (OCT) systems using AWGs around 1300 nm [21, 22], where polarization control is once again essential.

Some characteristics that are usually considered advantageous for the SiN platform can also become drawbacks. For instance, its lower birefringence (compared to the SOI platform) poses challenges when designing polarization control devices, as these often rely on birefringence to separate polarization states. Moreover, the lower refractive in-

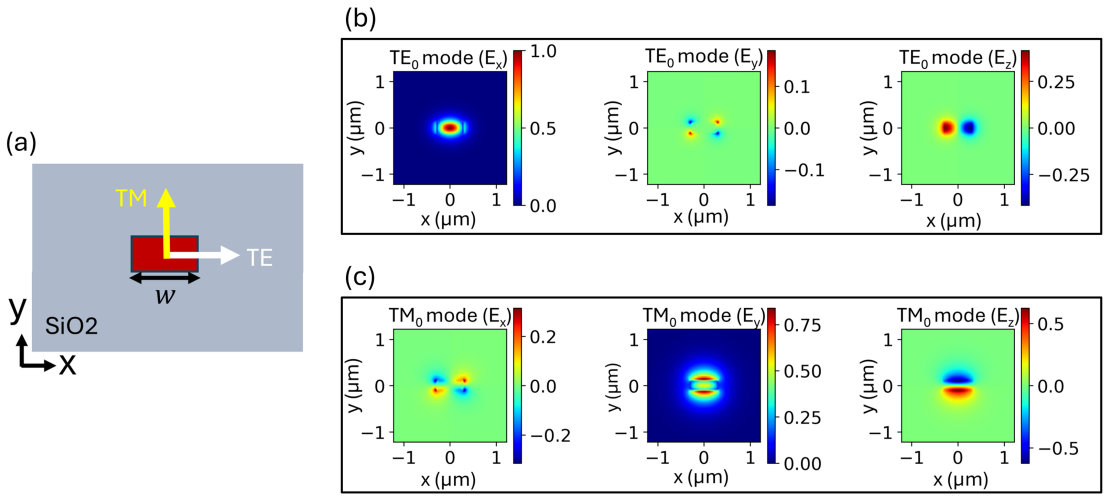

Figure 1.3: (a) Schematic cross-sectional view of a rectangular waveguide, showing the orientation of the TE and TM modes. (b) Electric field components of the TE_0 mode, calculated using a finite-difference eigenmode solver for the SOI platform shown in Fig. 1.2 (a), with a waveguide width of w = 0.6 μm at a wavelength of λ = 1550 nm. (c) Same as (b), but for the TM_0 mode.

dex of silicon nitride (compared to silicon) makes mode rotation more difficult, since efficient mode conversion requires structural asymmetry, a concept that will be further discussed in the following sections. These factors often make it challenging to achieve high-performance polarization devices while maintaining compact size in the SiN platform.

For the devices experimentally demonstrated as part of this thesis, two SiN platforms were exploited: one from Applied Nanotools (a Canadian company offering commercial fabrication services) and the other from C2N. Both platforms feature a 400 nm thick SiN core, but differ in their buried oxide thickness, 4.5 μm for Applied Nanotools and 3 μm for C2N. Applied Nanotools offers the possibility to deposit a silica upper cladding via plasma enhanced chemical vapor deposition (PECVD), a possibility that has not been explored within the C2N cleanroom. Both platforms relies on electron beam lithography.

1.3 . Thesis objectives and manuscript organization

A wide range of polarization control devices has already been demonstrated on the SOI platform. In contrast, significantly fewer results exist for SiN, mainly due to the greater challenges associated with its lower refractive index compared to silicon. As discussed previously, and as will be further shown in the following sections, polarization control often relies on the effective index difference between TE and TM modes. In SiN the reduced index difference typically results in devices that are larger than their SOI counterparts and more prone to TE/TM crosstalk when attempting to separate the modes. Although polarization separation is therefore more challenging in SiN, the lower index contrast also mitigates performance variations due to fabrication errors, representing a genuine advantage

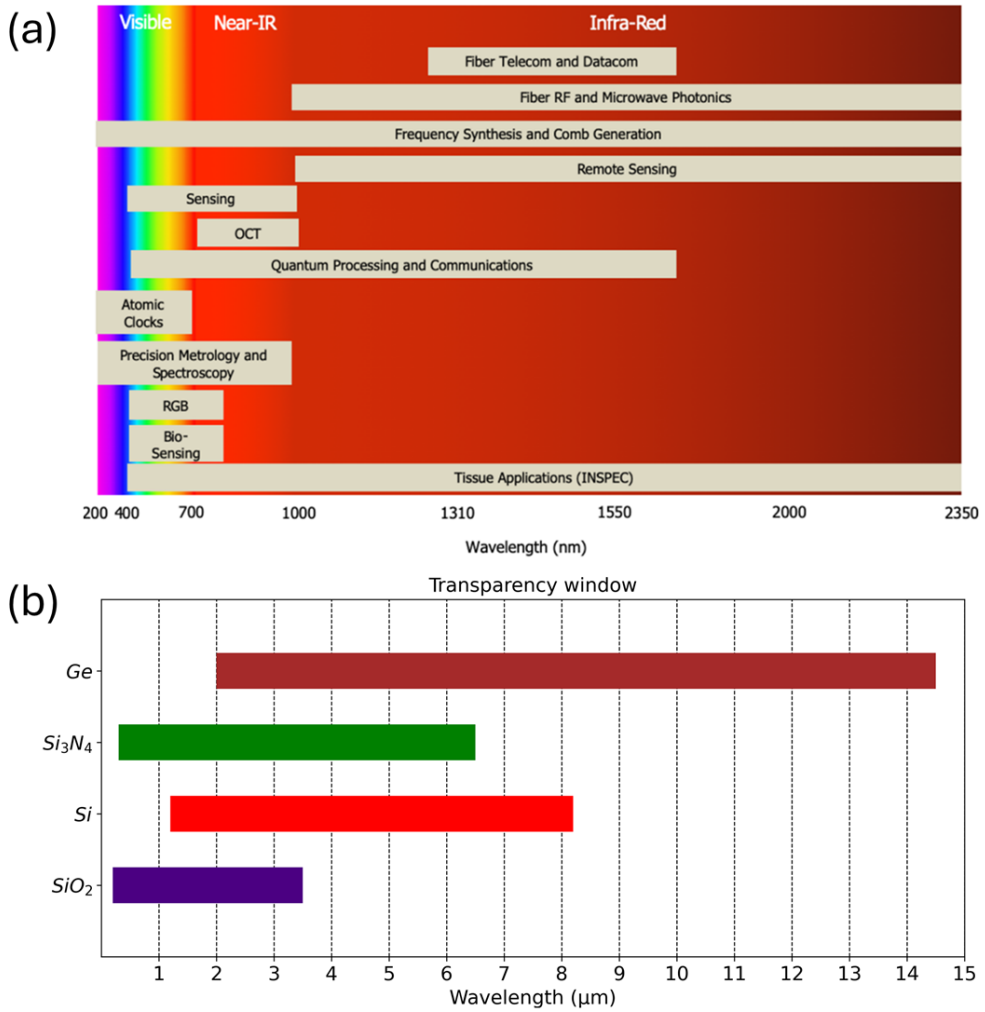

Figure 1.4: (a) Overview of applications across UV, visible, and IR wavelengths, adapted from [41]. (b) Transparency wavelength ranges for different materials.

of the platform.

Existing demonstrations of polarization control devices in the SiN platform overwhelmingly focus on typical telecommunication wavelengths in the near infrared (1300 nm or 1550 nm bands). On the contrary, despite growing interest, very scarce examples exist in the literature of polarization management and polarization control devices on the silicon nitride platform at shorter wavelengths.

My thesis aims at addressing these challenges by demonstrating polarization control devices in the silicon nitride platform operating at both telecommunication wavelengths and shorter ones more common in quantum systems.

The work is part of a larger project conducted in collaboration with the National Research Council Canada (NRC) and focusing on the realization of quantum photonic circuits in which single photons are also generated on-chip using InP quantum dots deposited on

photonic waveguides [23]. This quantum dots emit photons at a wavelength of 950 nm. Moreover, the generated photons possess an unknown polarization, hence the need for polarization control devices. As a part of this project, we designed and experimentally demonstrated several polarization control devices with the goal of extending the operational wavelength range available for polarization control components and improving device performance while maintaining compact footprints. Two main wavelengths were considered during the thesis. The first at 950 nm since this is the wavelength of single photons emitted by the quantum dots at the core of the project. We decided to simultaneously work also at a wavelength of 1300 nm, which is typically used for telecommunication applications. Developing PBS devices at this wavelength is particularly interesting due to the low propagation losses offered by the SiN platform and the limited number of existing devices compared to those operating at 1550 nm. Moreover, it offered the opportunity to benchmark the performance obtained at shorter wavelengths with those achieved in a more largely explored band.

The manuscript is organized as follows. In the first part, chapter 2, we focus on the realization of polarization beam splitters (PBS). A review of the state of the art is provided, followed by the design, fabrication, and characterization of two generation of devices device, along with a comparison to existing works. The developed PBS achieved ultra-low insertion losses of 0.1 dB and polarization extinction ratios up to 32 dB at 1310 nm, with a footprint of 124 μm. The 950 nm PBS device reached PER values up to 24 dB with insertion losses below 1 dB and a compact footprint of 110 μm.

In chapter 3, we introduce a modified version of the PBS employing subwavelength metamaterial waveguides. Its design, fabrication, and characterization are detailed in the corresponding section, together with a comparison to the state of the art. This novel approach enabled the realization of compact SiN PBSs with footprints as small as 63 μm, achieving polarization extinction ratios above 20 dB and insertion losses below 1 dB. To the best of our knowledge, these are among the shortest and best-performing SiN-based PBSs reported to date.

Chapter 4 investigates a polarization splitter-rotator (PSR), beginning with a review of the state of the art, followed by the design, fabrication, and characterization of the device, and concluding with a performance comparison. The fabricated PSR achieved polarization conversion losses around 2 dB, insertion losses below 1 dB, and polarization extinction ratios up to 20 dB, with a simplified single-step fabrication process.

Finally, as part of another project initiated at C2N, a polarization rotator device operating at a wavelength of 1550 nm was developed on the SOI platform, described in Chapter 5. This platform was chosen for an initial proof-of concept demonstration as it offers advantages in implementing devices capable of achieving efficient polarization rotation. The SOI-based PR exhibited polarization conversion losses below 1 dB and extinction ratios of about 15 dB, with a footprint of 110 μm, demonstrating state-of-the-art performance using only a single lithography fabrication step.

2 - Polarization beam splitter

One of the key components for polarization control is the polarization beam splitter (PBS), a device capable of separating the TE and TM modes at the input into two different output ports, as represented in Figure 2.1. Note that this figure does not represent a real device; it is only provided to illustrate the operation. Practical PBS designs can be implemented using various photonic structures, such as multimode interferometers, Mach–Zehnder interferometers, and directional couplers.

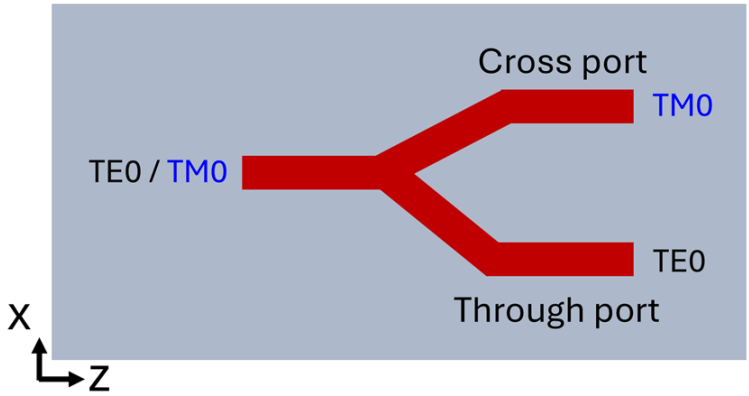

Figure 2.1: Schematic illustrating the operating principle of a polarization beam splitter.

To evaluate the performance of a PBS, common key metrics are the insertion loss (IL) and the polarization extinction ratio (PER), defined as follows:

$$PER_{TE} = 10 \cdot \log_{10} \frac{T_{through}}{T_{cross}} \tag{2.1}$$

$$PER_{TM} = 10 \cdot \log_{10} \frac{T_{cross}}{T_{through}} \tag{2.2}$$

$$IL_{TE} = -10 \cdot \log_{10} T_{through} \tag{2.3}$$

$$IL_{TM} = -10 \cdot \log_{10} T_{cross} \tag{2.4}$$

where $T_{through}$ and T_{cross} are the transmission measured at the through and cross ports (see Figure 2.1) and the subscript indicate the polarization state of the input light. When a mode is excited at the input (either TE or TM), the PER is defined as the ratio between the transmission at the desired output port and that at the undesired port. It therefore quantifies the separation quality: a higher PER indicates improved separation efficiency. The IL measures the losses induced by the device during the separation process. These definitions are used in the rest of the chapter.

A theoretical description of different PBS devices, together with practical examples, is provided in the following section. A state of the art review of PBS devices is also discussed. While some devices designed on SOI platforms are presented to ensure a more comprehensive overview of existing methodologies, a particular attention is dedicated to devices designed for SiN platforms, the main focus of the thesis.

2.1 . State of of the art

2.1.1 . Multi-mode interference couplers

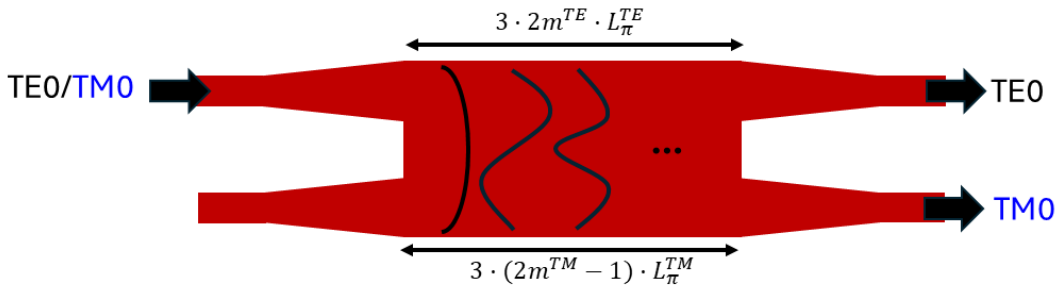

Figure 2.2: Schematic of a polarization beam splitter based on a multimode interferometer

A common approach to implement a PBS is through the use of a multimode interference coupler (MMI). An MMI is a waveguide designed to support a large number of modes (typically > 3), as illustrated in Fig. 2.2. Initially, one of the fundamental modes (TE_0 or TM_0) is launched at the input port of the MMI. Upon entering the wide central section of the waveguide, higher-order modes are excited. These modes propagate with their own phase velocities and constructive/destructive interference produces self-images of the input field after a certain propagation length [42, 43].

One used as a PBS, the MMI is designed such that for the fundamental TE_0 mode, the self-image appears, for example, at the upper output port. This happens when the MMI length satisfies $L_{MMI} = 3 \cdot 2m^{TE} \cdot L_\pi^{TE}$, where m^{TE} is an integer and L_π^{TE} is the beat length between the TE_0 fundamental mode and the first higher-order TE_1 mode, given by:

$$L_\pi^{TE} = \frac{\lambda}{2\left(n_{\text{eff}}^{TE_0} - n_{\text{eff}}^{TE_1}\right)}. \tag{2.5}$$

Here, $n_{\text{eff}}^{TE_0}$ and $n_{\text{eff}}^{TE_1}$ denote the effective indices of the fundamental and first higher-order TE modes, respectively. At the same time, for the fundamental TM_0 mode, the self-image would need to be formed at the lower output port, which requires $L_{MMI} = 3 \cdot \left(2m^{TM} - 1\right) \cdot L_\pi^{TM}$, where m^{TM} is an integer and L_π^{TM} is the beat length between the TM_0 fundamental mode and the first higher-order TM_1 mode:

$$L_\pi^{TM} = \frac{\lambda}{2\left(n_{\text{eff}}^{TM_0} - n_{\text{eff}}^{TM_1}\right)}. \tag{2.6}$$

To achieve the desired polarization separation, the MMI length must therefore satisfy [44]:

$$3 \cdot 2m^{TE} \cdot L_{\pi}^{TE} = 3 \cdot \left(2m^{TM} - 1\right) \cdot L_{\pi}^{TM}. \tag{2.7}$$

The PBS-MMIs have been considered promising devices due to the low losses, good fabrication tolerance, and ease of implementation [44]. However, one major limitation is their length, often reaching to several hundred micrometers. This issue arises because L_{π}^{TE} and L_{π}^{TM} tend to be very close in value. This is true for SOI platforms, but also for SiN platforms [45]. As a result, condition (Eq. 2.7) can only be satisfied with large m^{TE} and m^{TM} values. This leads to impractically long devices, motivating alternative designs on both silicon and silicon nitride platforms to reduce device footprint significantly. The PBS reported in [46] (on an SOI platform) is a direct example. Its operating principle does not rely on satisfying condition (Eq. 2.7), but instead on a quasi-state imaging approach. A quasi-state image is a state formed before and after the perfect images along the propagation direction, with the distance between a perfect image and a quasi-state image being about one-fifth of the distance between perfect images. The imperfect profile of the quasi-state image induces loss, but significantly reduces device length. To provide an order of magnitude, if the direct method were used (i.e., without quasi-state imaging), simulations in [46] demonstrated that the device length would be about 5000 μm. In [46], the quasi-state imaging approach was therefore adopted, reducing the device length to 1034 μm, as shown in the fabricated device schematic in Fig. 2.3 (a). This device was characterized and achieved a PER of 24 dB for TE and 45 dB for TM at $\lambda = 1550$ nm. A theoretical study of tolerance to MMI width variations also showed that, for both TE and TM modes at $\lambda = 1550$ nm, the PER remains above 10 dB even with $\pm$ 10 nm width variations. However, due to the quasi-state imaging, the device exhibits relatively high IL, exceeding 3 dB for both TE and TM over a 35 nm bandwidth (1530–1565 nm). Over this same range, the PER remains above 15 dB for both modes.

Another PBS-MMI design was proposed in [47] based on a slot MMI (again for the SOI platform), as shown in Fig. 2.3 (b). For the input TE mode of the slot waveguide, its dominant component E_x is mostly confined within the slot region. Thus, if this mode is launched into the MMI region, although both fundamental and higher-order modes are excited, only the TE_2 mode of the MMI similar to the input TE_0 mode of the slot waveguide dominates, corresponding to the highest excitation coefficient. As a result, the input TE mode passes almost directly through the MMI into the bar port. For the input TM mode, its dominant component E_y resembles that of a strip silicon waveguide. The self-imaging effect occurs as usual in the multimode region, producing a mirror image at the cross port. Consequently, the input TE and TM modes can be separated. This device has not been fabricated, but only simulated. Simulations show PER values above 10 dB and 15 dB for TE and TM modes, respectively, over the 1500–1600 nm range. IL values are below 4 dB and 2.5 dB over the same range for TE and TM modes. At $\lambda = 1550$ nm, PER values of 17 dB and 21 dB were obtained for TE and TM, with minimum IL of 1.37 dB and 0.81 dB. Although theoretical, these results represent a clear improvement in IL compared to the

device in Fig. 2.3 (a), albeit with reduced PER. In terms of footprint, the device in Fig. 2.3 (b) is drastically shorter, measuring only 9 μm compared to 1034 μm for Fig. 2.3 (a).

In Fig. 2.3 (c) from [45], another approach was proposed to overcome the length limitation, based on cascading two MMIs on the SiN platform. The first MMI directs the self-image of the TM mode to the upper output port, while the second MMI collects the TE-mode light, reflected to the lower port of the first MMI, and redirects its self-image to the lower output port. This reduces the device length to about 390 μm. This device has not been fabricated, but only simulated. It shows PER values above 10 dB and 12 dB for TE and TM modes, respectively, over the 1500–1600 nm wavelength range. Over the same range, IL remains below 1 dB for both modes. Compared to the two previous designs, this device demonstrates the best IL performance, remaining below 1 dB across the considered wavelength range. However, the PER is lower, particularly when compared to the device in Fig. 2.3 (a), which achieved the highest PER among the three. On the other hand, the device length in Fig. 2.3 (c) is 390 μm, representing more than a twofold reduction compared to the 1034 μm length of the device in Fig. 2.3 (a).

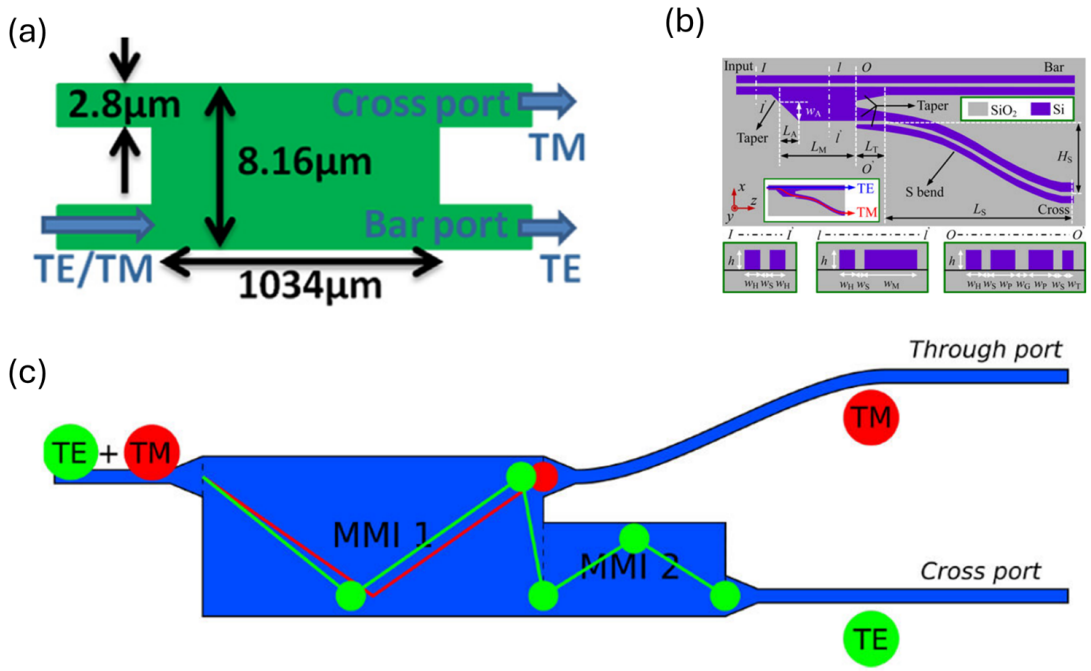

Figure 2.3: Different PBS designs based on MMI structures: (a) Regular MMI from [46], (b) slot MMI from [47], (c) cascaded MMI from [45].

Figure 2.4 from [48] shows an alternative approach that has been proposed and realized on a SiN platform. In this case the MMI does not act directly as a polarization splitter, but rather as a power splitter. The two output waveguides are subsequently directed to Bragg filters that reflect only one of the two modes. In this configuration, the TM_0 mode is transmitted while the TE_0 mode is reflected, after which they are recombined: for the TE

case, into the lower input port of the input MMI, and for the TM case, into the upper port of the output MMI. The fabricated device shows a PER above 20 dB across a wavelength range of 1545 nm – 1573 nm for both TE and TM modes. At λ = 1550 nm, it achieves a PER of about 38 dB and 32 dB for the TE and TM modes, respectively. Within the range of 1548–1570 nm, the insertion losses are approximately 0.6 dB for TE and 1.1 dB for TM. Overall, the device demonstrates impressive performance in terms of PER and insertion loss: the PER is comparable to that of the device in Fig. 2.3 (a), while maintaining much lower IL. However, the device also has a relatively large length of 820 μm.

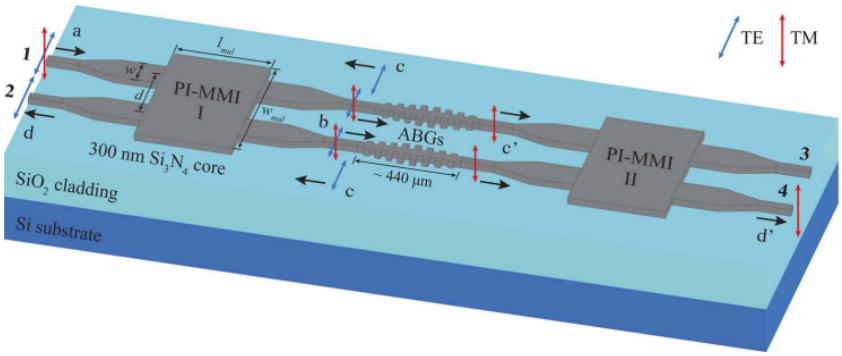

Figure 2.4: PBS based on MMI and Bragg filters in the SiN platform from [48]

2.1.2 . Mach-Zehnder interferometers

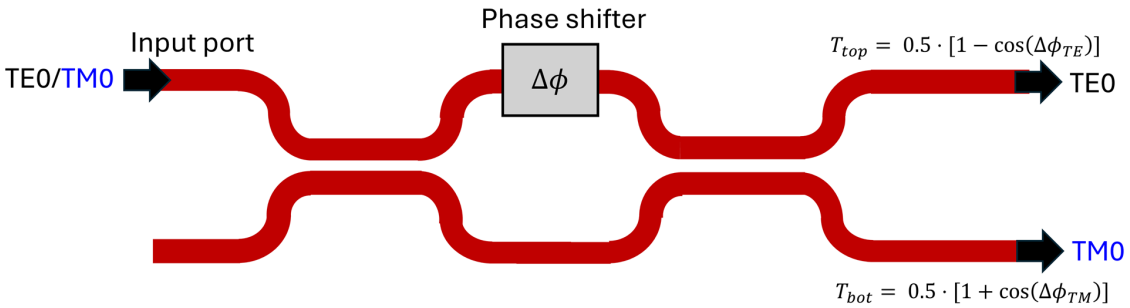

Figure 2.5: Schematic of a PBS based on a Mach-Zehnder interferometer

Another approach that has been investigated to integrate polarization splitting functionalities exploits Mach–Zehnder interferometers (MZIs), as schematically shown in Fig. 2.5. The concept is as follows: the fundamental mode (either TE_0 or TM_0) is injected into the input port and its power is equally split (50%) between in two different waveguides using a directional coupler. A phase shift $\Delta\varphi$ is then introduced between the two arms, for example using the thermo-optic effect by means of a micro-heater. The light is subsequently recombined with a second directional coupler. In the ideal case, where the splitting ratio is exactly 50% and the process is lossless, the transmissions at the upper and lower

output ports are given by:

$$T_{\text{top}} = 0.5 \cdot \left[1 - \cos(\Delta\varphi)\right] \tag{2.8}$$

$$T_{\text{bot}} = 0.5 \cdot \left[1 + \cos(\Delta\varphi)\right] \tag{2.9}$$

To achieve polarization separation, i.e., ensuring that the TE polarized light exits from the upper port and TM lights from the lower port, it is necessary to generate different phase delays $\Delta\varphi_{TE}$ and $\Delta\varphi_{TM}$, typically with $\Delta\varphi_{TE} = \pi$ and $\Delta\varphi_{TM} = 2\pi$.

This approach was implemented in [49] on a SiN platform using a passive device (see Fig. 2.6), where no heater was employed. Instead, a phase control section was used to introduce the phase shift passively. The PBS consists of two identical asymmetric directional couplers at the input (Port 1 and Port 2) and output (Port 3 and Port 4), along with two arms of different widths (the phase control section) to provide distinct phase changes for TE and TM modes. The fabricated device shows a PER higher than 20 dB for both TE and TM modes over the wavelength range of 1530–1560 nm. Within this range, the ILs for both modes are below 1 dB, with minimum values of 0.12 dB and 0.34 dB for TE and TM, respectively (the exact wavelength at which these occur is not specified). The total device length is 113 μm.

In the device shown in Fig. 2.7 from [50], an MZI was employed but with a slightly different method: the TM mode was fully (100%) transferred into the opposite waveguide with respect to the input one immediately after the first directional coupler, while the TE mode was split at 50% between the output waveguides of the first directional coupler. This was achieved by precisely adjusting the gap and the coupling length of the directional coupler. For the TE mode, since the two waveguides after the first coupler are identical (same width) with equal optical path lengths, the phase shift is assumed to be zero. According to Eq. 2.8 and Eq. 2.9, this ensures that the TE mode exits at the lower output port (cross port in Fig. 2.7). However, in practice, phase errors occur due to fabrication imperfections; therefore, heaters are used to correct the phase imbalance. When the TM mode is injected into the upper input port, it is entirely transferred to the lower waveguide after the first directional coupler. The influence of the heaters on this mode is negligible, so the TM mode is fully transferred again after the second directional coupler, exiting at the upper output port (through port in Fig. 2.7).

The device, realized on a SiN platform, exhibits a PER of 20 dB for both TE and TM modes over the wavelength range of 1535–1570 nm. Within this range, the ILs remain below 1 dB. The minimum ILs are 0.1 dB for TE and 1 dB for TM, while the maximum PER reaches 37 dB for TE and 32 dB for TM at λ = 1554 nm. In terms of PER and IL, the device demonstrates excellent performance; however, it is limited by its relatively large size of about 1000 μm.

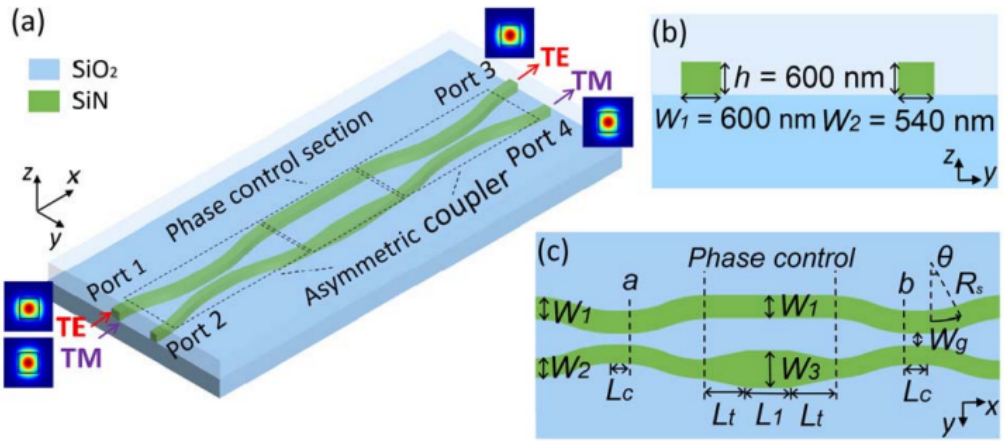

Figure 2.6: The PBS based on a MZI using a phase control section from [49]

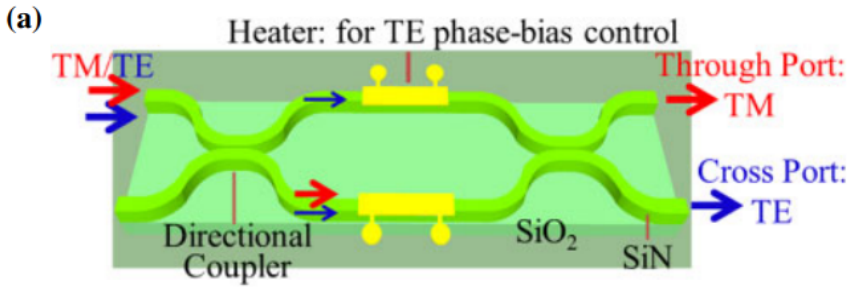

Figure 2.7: PBS using MZI with phase-bias control from [50]

2.1.3 . Directional couplers

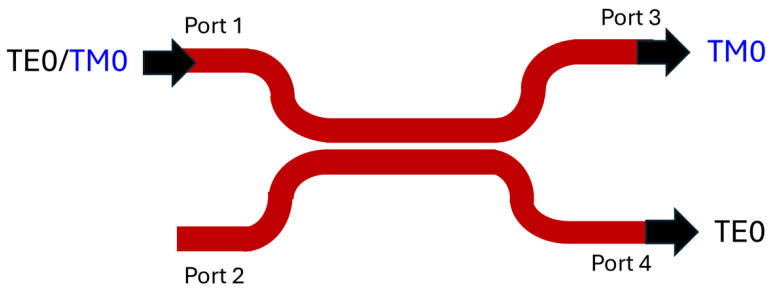

Figure 2.8: Schematic of a directional coupler used as a PBS.

A final method that has been vastly investigated in the literature to achieve polarization splitting is based on directional couplers. Directional couplers consist of two waveguides separated by a small gap, enabling light transfer from one waveguide to the other via evanescent field tails overlap (weak-coupling regime). Fig. 2.8 shows a typical schem-

atic of a directional coupler. As an example of how a directional coupler could be used as a PBS, the fundamental TE_0 mode is injected into the input port. The same principle could also be applied to other modes (TM_0, TE_1, TM_1, ...). In general, the device operates as follows: the mode (here TE_0) is injected into the upper waveguide (port 1). The upper and lower waveguides approach each other via an S-bend structure. By appropriately choosing the waveguide widths, the gap, and the coupling length (L_c), it becomes possible to fully transfer the TE_0 mode from the upper waveguide to the TE_0 mode of the lower waveguide, with the light ultimately exiting through port 4. The key condition enabling this transfer is phase matching: the effective index of the TE_0 mode in the upper waveguide of the coupler must equal that of the TE_0 mode in the lower waveguide. In the case of TM_0 mode excitation at input port 1, the waveguides are intentionally designed so that the phase-matching condition is not satisfied for this mode, resulting in the light exiting through port 3. A more theoretical discussion about the operating principle of directional couplers can be found in Sec. 2.2.1, emphasizing the necessity of phase matching. It should be noted that this conversion principle applies not only to fundamental-to-fundamental mode conversions (i.e., TE_0 to TE_0, TM_0 to TM_0), but also to higher-order mode conversions (e.g., TE_0 to TE_1, or TE_1 to TE_2). Moreover, under certain conditions, it is even possible to achieve conversions between TM and TE modes, which will be detailed later in chapter 4 on polarization rotating devices.

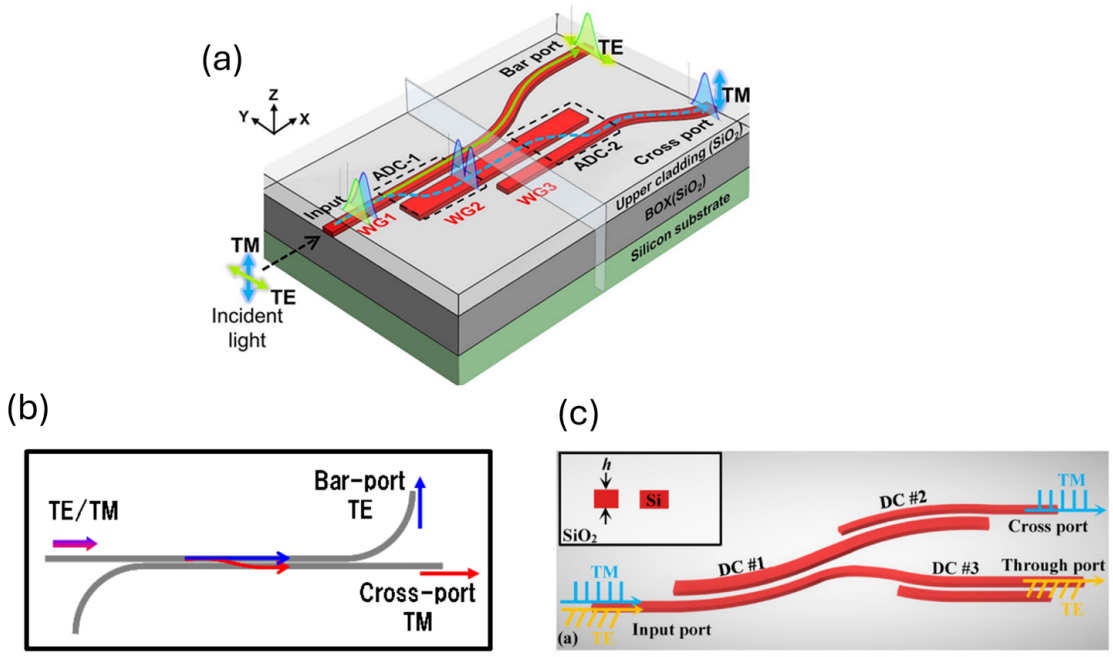

Figure 2.9: Different PBS designs based on directional couplers: (a) asymmetrical cascaded directional coupler from [51], (b) standard directional coupler from [52], (c) cascaded bent directional coupler from [53].

Polarization separation using a directional coupler on a SiN platform is shown in Fig. 2.9

(a) from [51]. More accurately, this approach exploits two cascaded asymmetric directional couplers. The TM_0 mode is injected into the input waveguide (WG1). Phase matching between the TM_0 and TM_1 modes is achieved, enabling complete conversion of the TM_0 mode from the upper waveguide (WG1) to the higher-order TM_1 mode of the central waveguide (WG2). Symmetrically, the TM_1 mode is then reconverted from the central waveguide (WG2) to the TM_0 mode of the lower waveguide (WG3) and exits from the cross port. The TE_0 mode remains instead in the input waveguide and exits from the bar port since phase matching between the TE_0 mode of the upper waveguide (WG1) and guided modes of the central waveguide (WG2) is not satisfied. In this way, the TE_0 and TM_0 modes can be separated. This device demonstrates a PER exceeding 20 dB and 10 dB over the wavelength range of 1520–1600 nm, for TE and TM modes respectively. It achieves a PER of 28 dB and 32 dB at $\lambda = 1550$ nm. The insertion losses are about 1.8 dB and 2.1 dB for TE and TM modes, respectively, across the 1520–1600 nm range. The overall device length is approximately 112 μm.

The device shown in Fig. 2.9 (b), from [52], also employs a directional coupler but this time on an SOI platform. Here, the coupler transfers the TM_0 mode from the upper waveguide to the lower waveguide, reaching the cross-port under phase-matching conditions. This occurs because both waveguides have identical widths, ensuring phase matching not only for the TM_0 modes but also for the TE_0 modes. One might therefore expect significant conversion of the TE_0 mode from the upper waveguide into the TE_0 mode of the lower waveguide. However, this conversion is relatively negligible due to the high index contrast between the Si core and the SiO_2 cladding, which makes the propagation constants of TE and TM modes very different. As a result, the coupling length required to transfer the TM_0 mode to the lower waveguide is much shorter than that for the TE_0 mode, thereby enabling polarization separation. This device was fabricated and characterized, showing a PER greater than 10 dB over 1500–1560 nm for the TE mode and 1500–1600 nm for the TM mode. At $\lambda = 1550$ nm, the PER values are 12 dB and 15 dB for the TE and TM modes, respectively. The insertion losses are below 0.5 dB for both modes across the 1500–1600 nm range. The device length is about 16 μm. Thus, the device in Fig. 2.9 (b) is approximately 10 times smaller than the one in Fig. 2.9 (a), but with lower PER. Furthermore, the larger size of the device in Fig. 2.9 (a) arises from the use of the SiN platform, where the lower index contrast compared to SOI leads to a longer device. In particular, conversion through an intermediate mode was required since the propagation constants of TE and TM modes are much closer in SiN waveguides. As a result, it is not possible to directly couple one polarization without affecting the other one.

A third device employing directional couplers is shown in Fig. 2.9 (c), from [53]. The device is composed of a cascade of three bent directional couplers. Also in this case, an intermediate assisting waveguide with a different width is used. The TM_0 mode in the input waveguide of width w_1 is transferred to the TM_0 mode of a second waveguide of width w_2. To achieve this, the waveguide widths w_1 and w_2 and the bend radii R_1 and R_2 are selected so as to satisfy the phase-matching condition for TM modes. In bent waveguides,

Table 2.1: Performance comparison for several experimentally demonstrated polarization beam splitters in the SOI and SiN platforms.

	Wavelength range (nm)	**Max PER TE-TM (dB)**	**Min IL (dB)**	**Size (μm)**	**$BW_{PER>10dB}$ (nm)**	**platform**	**References**
MMI	1530 - 1565	24-45	> 3	1034	35-35	SOI	[46]
Slot MMI	1500 - 1600	17-24	~ 1.2	9	100-100	SOI	[47]
Asymmetric MZI	1500 - 1650	25-17	TE:1.5 - TM:0.5	200	120-120	SOI	[55]
DC	1500 - 1600	12-15	< 0.5	16	60-100	SOI	[52]
Cascaded bent DC	1400 - 1650	37-43	TE:0.35 - TM:0.1	20	200-200	SOI	[54]
Cascaded MMI	1500 - 1600	14-14	0.6	390	100-100	SiN	[45]
MMIs and Bragg Filters	1530 - 1590	38-32	TE:0.6-TM:1.1	820	32-60	SiN	[48]
Angled MMI	1525 - 1625	30-30	~ 0.7	80	100-100	SiN	[56]
Passive MZI	1530 - 1610	27-23	TE:0.13 - TM:0.34	113	50-50	SiN	[49]
Active MZI	1520 - 1620	37-32	TE:0.1 - TM:1	1000	100-100	SiN	[50]
Phase control DC	1260 - 1350	20-35	< 0.6	136	90-90	SiN	[57]
ADC (no taper)	1520 - 1600	28-32	TE ~ 1.8 - TM ~ 2.1	112	80-80	SiN	[51]
ADC (taper)	740 - 860	30-26	< 1	315	120-40	SiN	[17]
ADC (no taper)	1250 - 1350	32-32	TE:0.1 - TM: 0.1	124	100-60	SiN	Our work [58]
ADC (no taper)	910 - 980	29-24	< 1	110	70-45	SiN	Our work [58]
ADC (taper)	1250 - 1350	29-22	< 1	290	100-100	SiN	Our work [58]
ADC (taper)	910 - 980	26-16	< 1	242	70-70	SiN	Our work [58]

the phase-matching condition for transferring the TM_0 mode becomes [53, 54]:

$$R_1 \cdot n_{eff}^{TM_0}(w_1) = R_2 \cdot n_{eff}^{TM_0}(w_2) \tag{2.10}$$

where $n_{eff}^{TM_0}(w_1)$ and $n_{eff}^{TM_0}(w_2)$ are the effective indices of the TM_0 mode in the bent waveguide. For TE modes, a large phase mismatch is automatically introduced due to the strong waveguide birefringence, which makes the cross-coupling weak. As illustrated in Fig. 2.9 (c), the authors used multiple cascading directional couplers to enhance the PER of both TE and TM modes by filtering out the residual undesired polarization. This device was fabricated on an SOI platform and exhibits a PER higher than 20 dB for both TE and TM modes over a wavelength range from 1475 to 1610 nm. The insertion losses for the TE and TM modes are below 0.4 dB around the central operating wavelength of the device ($\lambda \sim 1550$ nm). The PER exceeds 35 dB for both TE and TM modes around this central wavelength. The device achieves these performances with a length of only 20 μm, making it significantly more efficient (in terms of PER and IL) and compact than the two previous designs. However, in the case of the device shown in Fig. 2.9 (a), the use of the SiN platform must be taken into account, since the device in Fig. 2.9 (c) relies on the very strong birefringence of SOI to provide a large phase mismatch for TE modes. Such a device could also be implemented on SiN, provided that properly designed bend radii are employed to maximize the phase mismatch for TE modes and to reduce bending losses.

The table 2.1 provides a summary of the performance and size of the PBS devices discussed above with some additional examples, implemented on both SOI and SiN platforms. It also reports as a comparison the devices demonstrated as part of the thesis and discussed in the next sections 2.2, 2.3 and 2.4.

2.2 . Straight asymmetric coupler

2.2.1 . Device principle and directional coupler theory

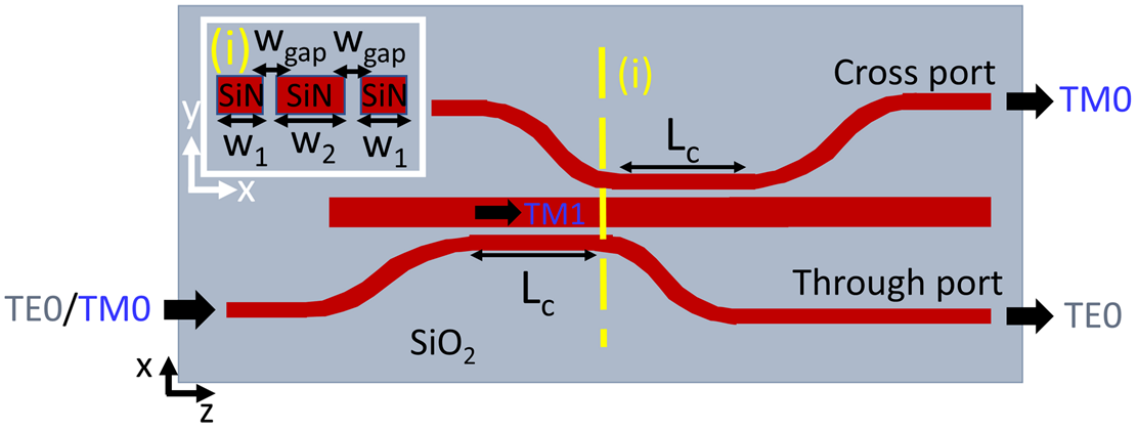

Figure 2.10: Top-view schematic of our device, consisting of a PBS based on a cascaded asymmetrical directional coupler, with the cross-section indicated at position (i) by the yellow dashed line.

In this section, as previously discussed in the introduction (see Chapter 1), we developed two PBS designs, each operating at different wavelengths: one centered at 1300 nm and another at 950 nm. The schematic of the device is shown in Fig. 2.10; it is a PBS based on a cascade of two asymmetrical directional couplers (ADCs).

The operating principle is as follows: the TM_0 mode from the lower input waveguide of width w_1 is gradually brought closer to the central waveguide of width w_2 using an S-bend. This TM_0 mode is then coupled to the TM_1 mode of the central waveguide by satisfying the phase-matching condition, i.e., $n_{eff}^{TM_0}(w_1) = n_{eff}^{TM_1}(w_2)$. The TM_1 mode of the central waveguide is subsequently reconverted into the TM_0 mode of the upper waveguide (also of width w_1) by again satisfying the same phase-matching condition. An S-bend is then employed to gradually separate the central and upper waveguides, allowing the TM_0 mode to exit through the cross port. At the same time, the TE_0 mode in the input waveguide is not coupled into the central waveguide, since the phase-matching condition is not satisfied, i.e., $n_{eff}^{TE_0}(w_1) \neq n_{eff}^{TE_1}(w_2)$. In the absence of conversion, the TE_0 mode therefore continues to propagate and exits through the through port. The TE_0 and TM_0 modes are thus separated.

In order to design the devices, it is useful to detail precisely the theoretical operating principle of directional couplers, which form the basis of the PBS device shown in Fig. 2.10. The theory underlying the operation of directional couplers is the coupled-mode theory [59]. To illustrate this, we consider a simplified version of a directional coupler without S-bends, as represented in Fig. 2.11.

We consider two waveguides: the upper waveguide of width w_2 and the lower waveguide of width w_1, separated by a gap w_{gap}. We denote by $A(z)$ the field propagating in the upper waveguide along the z direction with propagation constant β_a, and by $B(z)$ the

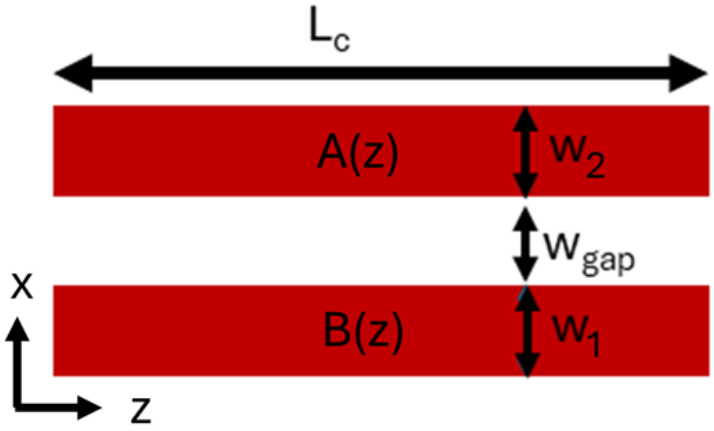

Figure 2.11: Schematic of a simplified directional coupler without the presence of S-bends.

field in the lower waveguide with propagation constant β_b. The coupled-mode equations can then be written as follows:

$$\frac{dA}{dz} = -i\beta_a A - i\kappa_{ab} B \tag{2.11}$$

$$\frac{dB}{dz} = -i\beta_b B - i\kappa_{ba} A \tag{2.12}$$

Here, κ_{ab} is the coupling coefficient between the fields A and B, defined as follows [59]:

$$\kappa_{ab} = \frac{1}{4}\sqrt{\frac{\varepsilon_0}{\mu_0}} \int \left(n^2(x,y) - n_b^2(x,y)\right) A^* B \, dxdy \tag{2.13}$$

where ε_0 and μ_0 are the vacuum permittivity and permeability, while $n(x,y)$ and $n_b(x,y)$ denote, respectively, the refractive index profiles of the two-waveguide system and the isolated lower waveguide. Symmetrically, the coupling coefficient between the fields B and A is defined as:

$$\kappa_{ba} = \frac{1}{4}\sqrt{\frac{\varepsilon_0}{\mu_0}} \int \left(n^2(x,y) - n_a^2(x,y)\right) B^* A \, dxdy \tag{2.14}$$

with $n_a(x,y)$ denoting the refractive index profile of the isolated upper waveguide.

We consider the case where the propagation constants β_a and β_b are different. If the coupled-waveguide system is lossless, then self-consistency requires that Eqs. (2.11) and (2.12) satisfy the law of power conservation. The total guided power can be written as:

$$P(z) = |A|^2 + |B|^2 \tag{2.15}$$

Thus, the following condition on the total guided power must be satisfied:

$$\frac{dP}{dz} = \frac{d}{dz}\left(|A|^2 + |B|^2\right) = 0 \tag{2.16}$$

$$\frac{dP}{dz} = \frac{d}{dz}\left(A^*A + B^*B\right) = A\frac{dA^*}{dz} + A^*\frac{dA}{dz} + B\frac{dB^*}{dz} + B^*\frac{dB}{dz} = 0 \tag{2.17}$$

Multiplying Eqs. (2.11) and (2.12) by A^* and B^* , respectively, yields:

$$A^*\frac{dA}{dz} = -i\beta_a|A|^2 - i\kappa_{ab}A^*B \tag{2.18}$$

$$B^*\frac{dB}{dz} = -i\beta_b|B|^2 - i\kappa_{ba}B^*A \tag{2.19}$$

Substituting Eqs. (2.18) and (2.19) and their complex conjugates into Eq. (2.17) gives:

$$\frac{dP}{dz} = i\left[(\kappa^*_{ab} - \kappa_{ba})B^*A + (\kappa^*_{ba} - \kappa_{ab})A^*B\right] = 0 \tag{2.20}$$

Since Eq. (2.16) is independent of the initial conditions, it follows that $\kappa_{ab} = \kappa^*_{ba} = \kappa$ [60].

In order to simplify Eqs. (2.11) and (2.12) and solve the system, we introduce the substitutions $\tilde{A} = Ae^{i\beta_0 z}$ and $\tilde{B} = Be^{i\beta_0 z}$, with $\beta_0 = (\beta_a + \beta_b)/2$. Defining $\delta = (\beta_a - \beta_b)/2$, we obtain:

$$\frac{d\tilde{A}}{dz} = -i\delta\tilde{A} - i\kappa\tilde{B} \tag{2.21}$$

$$\frac{d\tilde{B}}{dz} = i\delta\tilde{B} - i\kappa^*\tilde{A} \tag{2.22}$$

To solve the differential equations, we use the matrix form $\psi = \begin{pmatrix}\tilde{A}\\ \tilde{B}\end{pmatrix}$ and $H = \begin{pmatrix}\delta & \kappa\\ \kappa^* & -\delta\end{pmatrix}$, so that the system becomes:

$$\frac{d\psi}{dz} = -iH\psi \tag{2.23}$$

This reduces the problem to an eigenvalue problem, and since the matrix is Hermitian, it has the following real eigenvalues and orthogonal eigenvectors:

$$\lambda_\pm = \pm\sqrt{\delta^2 + |\kappa|^2} = \pm\Omega \tag{2.24}$$

$$v_\pm = \begin{pmatrix}1\\ -\frac{\delta - \lambda_\pm}{\kappa}\end{pmatrix} \tag{2.25}$$

The general solution to Eq. 2.11 and Eq. 2.12 can then be expressed as a linear combination of the two eigenvectors:

$$\psi(z) = C_+v_+e^{-i\lambda_+ z} + C_-v_-e^{-i\lambda_- z}, \tag{2.26}$$

With C_+ and C_- being constants determined by the initial conditions of the problem. The general solutions for $\tilde{A}$ and $\tilde{B}$ are:

$$\tilde{A}(z) = C_+e^{-i\lambda_+ z} + C_-e^{-i\lambda_- z} \tag{2.27}$$

$$\tilde{B}(z) = C_+\frac{-\delta + \lambda_+}{\kappa}e^{-i\lambda_+ z} + C_-\frac{-\delta + \lambda_-}{\kappa}e^{-i\lambda_- z} \tag{2.28}$$

With the initial conditions $\tilde{A}(0) = 1$ and $\tilde{B}(0) = 0$, which physically correspond to all the power being initially in the upper waveguide, the constants C_+ and C_- can be determined, yielding the final expressions:

$$A = e^{-i\beta_0 z}\left[\cos(\Omega z) - i\frac{\delta}{\Omega}\sin(\Omega z)\right] \tag{2.29}$$

$$B = -i\frac{\kappa^*}{\Omega}e^{-i\beta_0 z}\sin(\Omega z) \tag{2.30}$$

where $\Omega = \sqrt{\delta^2 + |\kappa|^2}$. As a consequence, the power in the two waveguides is:

$$P_A = \cos^2(\Omega z) + \frac{\delta^2}{\delta^2 + |\kappa|^2}\sin^2(\Omega z) \tag{2.31}$$

$$P_B = \frac{|\kappa|^2}{\delta^2 + |\kappa|^2}\sin^2(\Omega z) \tag{2.32}$$

This result allows us to determine the transmitted power into the lower waveguide. To ensure an efficient power transfer from the upper to the lower waveguide, it is necessary that $\delta \ll |\kappa|$. From Eq. (2.32), the coupling length required to reach the maximum power transfer is:

$$L_c = \frac{\pi}{2\Omega} = \frac{\pi}{2\sqrt{\delta^2 + |\kappa|^2}} \tag{2.33}$$

In the case of perfect phase matching ($\delta = 0$ meaning that $\beta = \beta_a = \beta_b$, synchronous directional coupler), $L_c = \pi/2|\kappa|$ and complete power transfer is achieved.

The system of equations Eq. 2.11 and Eq. 2.12 can also be rewritten in the following matrix form:

$$\frac{d}{dz}\begin{bmatrix}A\\B\end{bmatrix} = -i\begin{bmatrix}\beta & \kappa\\ \kappa & \beta\end{bmatrix}\begin{bmatrix}A\\B\end{bmatrix} \tag{2.34}$$

The eigenvalues of this matrix can be calculated as

$$\beta_{sym} = \beta + k \tag{2.35}$$

$$\beta_{asym} = \beta - k \tag{2.36}$$

and the corresponding eigenvectors:

$$v_{sym} = \begin{pmatrix}1\\1\end{pmatrix} \tag{2.37}$$

$$v_{asym} = \begin{pmatrix}1\\-1\end{pmatrix} \tag{2.38}$$

which yields two supermodes for the coupled waveguides, one symmetric and one antisymmetric. From Eqs. 2.35 and 2.36, κ can be isolated and the coupling length also expressed as:

$$L_c = \frac{\pi}{\beta_{\text{sym}} - \beta_{\text{asym}}} \tag{2.39}$$

2.2.2 . PBS design and simulation

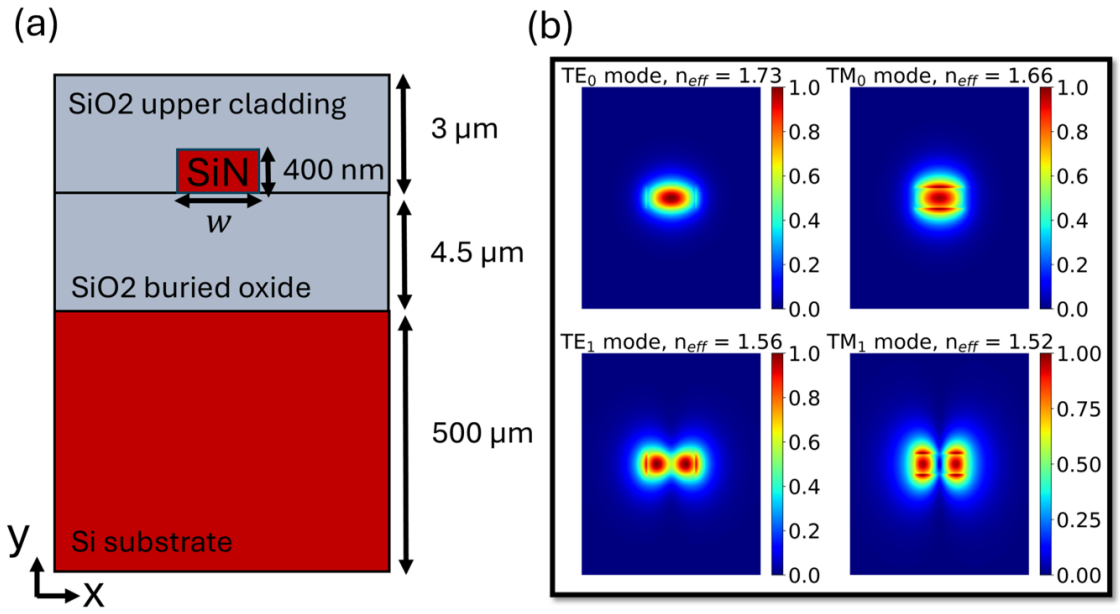

Figure 2.12: (a) Cross-sectional view of the SiN waveguide used for the design of the PBS. (b) Electric-field amplitude of the fundamental and first higher-order TE and TM modes, calculated with a finite-difference eigenmode solver using the SiN platform shown in (a) for a waveguide width $w = 1.2$ μm and a wavelength $\lambda = 1300$ nm.

As detailed in Sec. 1.2, to realize our devices we exploit a SiN platform with a core thickness of 400 nm, a 3 µm SiO_2 upper cladding, and a 4.5 µm bottom cladding (ANT platform), as shown in Fig. 2.12 (a). The refractive index of the SiN material is 2.01 and 2.00 for wavelengths of 950 nm and 1300 nm, respectively. Taking into account the parameters of the used platform (SiN and SiO_2 thickness and index), we can determine the effective indices of the modes as a function of the waveguide width. For this purpose, we use a finite difference eigenmode solver. As an example, Fig. 2.12 (b) shows the electric field amplitudes of the two fundamental modes TE_0, TM_0, and the first higher-order modes TE_1 and TM_1 of the isolated waveguide shown in Fig. 2.12 (a) for a width $w = 1.2$ μm at a wavelength $\lambda = 1300$ nm. Their respective effective indices calculated using the finite difference eigenmode solver are reported as well. By sweeping the waveguide width, we calculate the effective indices of the TE_0 and TM_0 modes, as well as the TE_1 and TM_1 modes, as a function of the width. Simulation results are reported for a wavelength of 950 nm, Fig. 2.13 (a), and 1300 nm, Fig. 2.13 (b).

The widths for the bottom waveguide (w_1) and the central waveguide (w_2) shown in Fig. 2.10 must be chosen in order to ensure that the phase matching condition between the TM_0 and TM_1 modes is satisfied. For this purpose, it is preferable to select values such that $n_{\text{eff}}^{TE_0} \neq n_{\text{eff}}^{TM_0}$ for the narrowest waveguide since this condition could favor spurious conversions between TE_0 and TM_0 within the waveguide. We also ensure that the

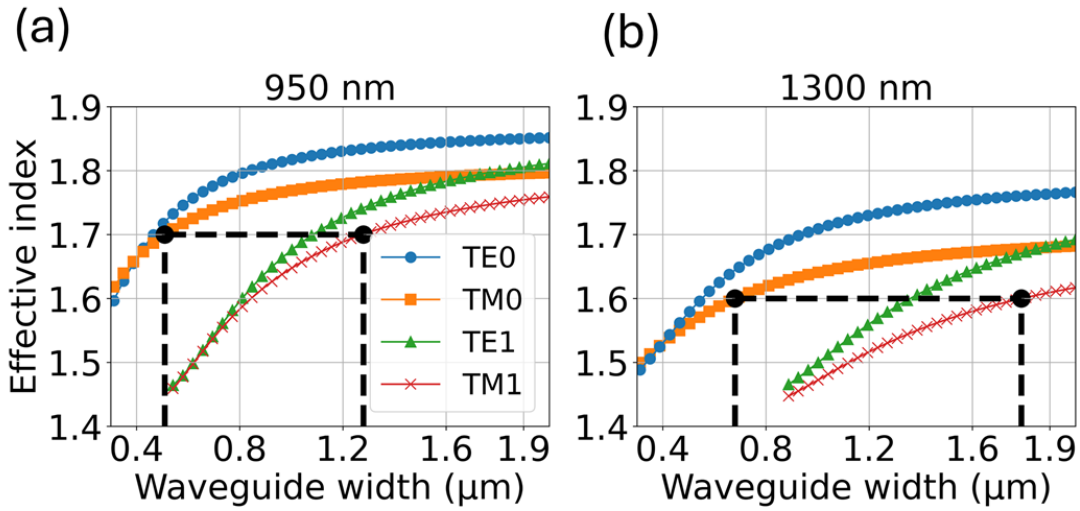

Figure 2.13: Effective indices of an isolated SiN waveguide, calculated on the SiN platform shown in Fig. 2.12, as a function of the waveguide width for the fundamental and first higher-order modes. Results are reported for (a) $\lambda = 950$ nm and (b) $\lambda = 1300$ nm. The black dashed line represents the width where the phase-matching condition is fulfilled.

chosen w_1 lies within the single-mode regime of the first waveguide, in order to avoid undesired conversions between the fundamental and higher-order modes. Considering these constraints and performing interpolation, we determine the appropriate widths as $w_1 = 0.51$ μm and $w_2 = 1.28$ μm for $\lambda = 950$ nm, and $w_1 = 0.68$ μm and $w_2 = 1.79$ μm for $\lambda = 1300$ nm.

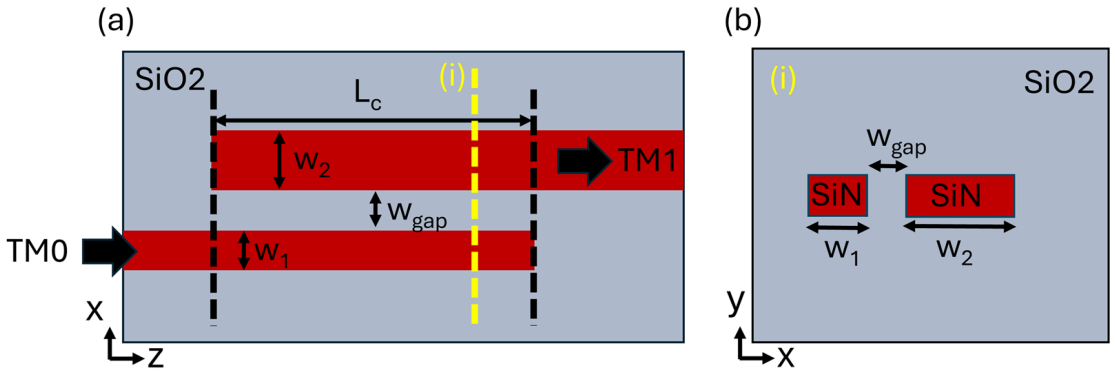

Figure 2.14: Simplified version of the PBS optimized using an eigenmode expansion solver: (a) top view; (b) cross-sectional view at position (i), indicated by the yellow dashed line in (a).

With the widths fixed, the next step is to determine the appropriate gap and coupling length values. To this aim, we use an eigenmode expansion (EME) solver, which we employ to compute the conversion efficiency between TM_0 and TM_1 for different gap values. To perform the simulation, a simplified structure is used, shown in Fig. 2.14 (a) (top view)

and Fig. 2.14 (b) (cross-sectional view). It consists of two coupled waveguides with the previously fixed width values w_1 and w_2, separated by the gap w_{gap}. When using the EME solver, it is necessary to remove the S-bends due to the difficulty of the solver in handling steep angles. Using EME, we calculate the transmission of the TM_0 mode from the input waveguide to the TM_1 mode of the second waveguide as a function of the coupling length (L_c), for different gaps from 300 to 400 nm at $\lambda = 950$ nm and from 500 to 600 nm at $\lambda = 1300$ nm. The results are shown in Fig. 2.15. For both 950 nm and 1300 nm, we observe that the gap primarily affects the coupling length of the device, which is consistent with Eq. (2.33), showing that the coupling length is inversely proportional to the coupling coefficient. To select the gap, and hence the coupling length, two considerations are taken into account: the device should remain compact, and the gap should be sufficiently large to allow for SiO_2 filling during cladding deposition by plasma-enhanced chemical vapor deposition (PECVD).

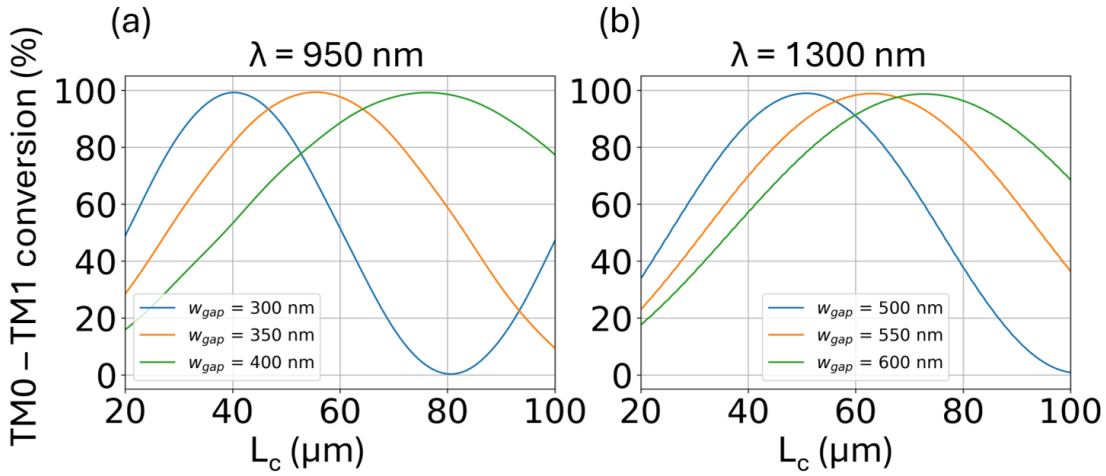

Figure 2.15: TM_0 to TM1 coupling from the lower waveguide to the upper waveguide, as shown in Fig. 2.14 (a), for three different gap widths w_{gap} as a function of the coupling length L_c: (a) $\lambda = 950$ nm; (b) $\lambda = 1300$ nm.

For these reasons, a gap of 350 nm was chosen for the device operating at $\lambda = 950$ nm, since a gap of 300 nm is at the limit for the PECVD technique. At a gap of 400 nm, the device becomes excessively long. In the case of the device operating at $\lambda = 1300$ nm, we decided to use a gap of 550 nm instead of 500 nm which would maintain a coupling length comparable to that of the device operating at $\lambda = 950$ nm while ensuring an additional safety margin for SiO_2 deposition. This also provides useful information in case the device at $\lambda = 950$ nm does not work while the one at $\lambda = 1300$ nm does. This would indicate that the SiO_2 cladding deposition for the device at $\lambda = 950$ nm probably did not fill the gap. The corresponding coupling lengths are $L_c = 55$ μm for 950 nm and $L_c = 62$ μm for 1300 nm. For both wavelengths, the device shows a TM_0 to TM_1 conversion above 99%. In the case of the TE_0 mode, the unwanted coupling to the TE_1 mode is below 6% for both wavelengths, Fig. 2.16.

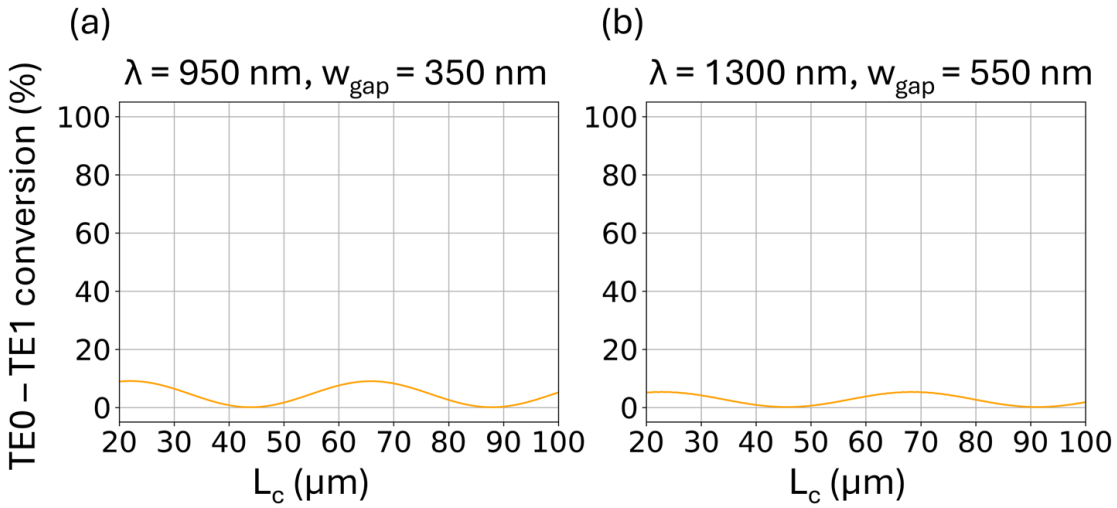

Figure 2.16: TE_0 to TE1 coupling from the lower waveguide to the upper waveguide, as shown in Fig. 2.14 (a), for a fixed gap width w_{gap} = 350 nm as a function of the coupling length L_c: (a) λ = 950 nm; (b) λ = 1300 nm.

We also conducted an analysis to evaluate the impact of fabrication tolerances. In particular, for the fabrication process used in this work, the tolerance on the waveguide width is on the order of ±10 nm.

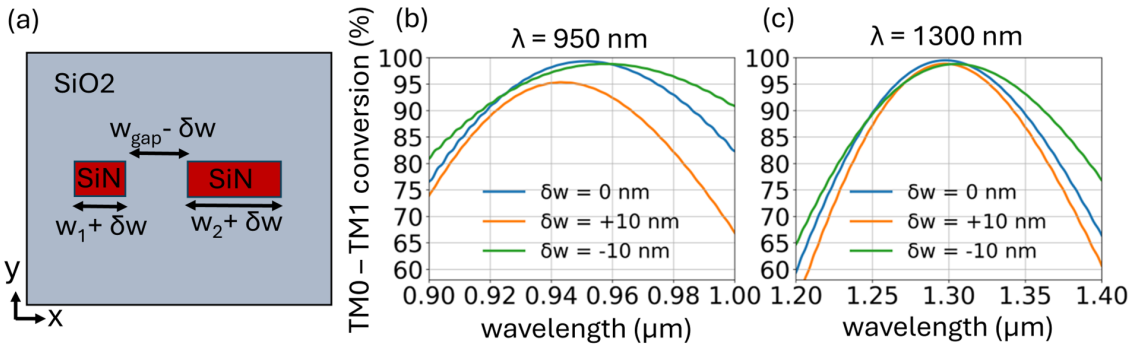

Figure 2.17: Cross-sectional view of the directional coupler with waveguide width deviations (δw) (a). TM_0 to TM_1 coupling as a function of wavelength for three different width deviations (δw): (b) λ = 950 nm and (c) λ = 1300 nm.

We therefore account for these variations in the EME simulations. We performed the same simulations as before, i.e. the TM_0 to TM_1 coupling, but considering waveguides that are either wider or narrower than the intended value. When the waveguides of the directional coupler are wider by 10 nm, the gap is reduced by 10 nm. In the opposite case, when the waveguides are narrower by 10 nm, the gap is increased by 10 nm. This is represented in Fig. 2.17 (a) by the parameter δw. The results of the EME simulation are shown in Fig. 2.17 (b) and (c) for the devices operating around 950 nm and 1300 nm,

respectively. In this case, simulation results are reported as a function of wavelength for different fabrication errors. Deviations in waveguide widths result in a decrease of the maximum conversion efficiency and a shift in the central wavelength. Fabrication errors have a more pronounced impact on the device operating at 950 nm compared to 1300 nm. For $\delta w = +10$ nm at $\lambda = 950$ nm, the conversion efficiency decreases from 99.3% to 95.3%, and the central wavelength is blue-shifted by 7 nm. Although the 4% drop in conversion efficiency is comparatively small, the device is also used to reconvert the TM_1 mode of the central waveguide to the TM_0 mode of the output waveguide, yielding a full transmission of 90.8%, i.e., a 8% penalty compared to the nominal value. At 1300 nm, the conversion from TM_0 to TM_1 decreases only slightly from 99.5% to 98.8%, with a central wavelength shift of 6 nm. The higher robustness of the 1300 nm design is primarily due to the derivative of the mode effective refractive index with respect to waveguide width being larger at 950 nm than at 1300 nm.

2.3 . Tapered coupler: design and simulation

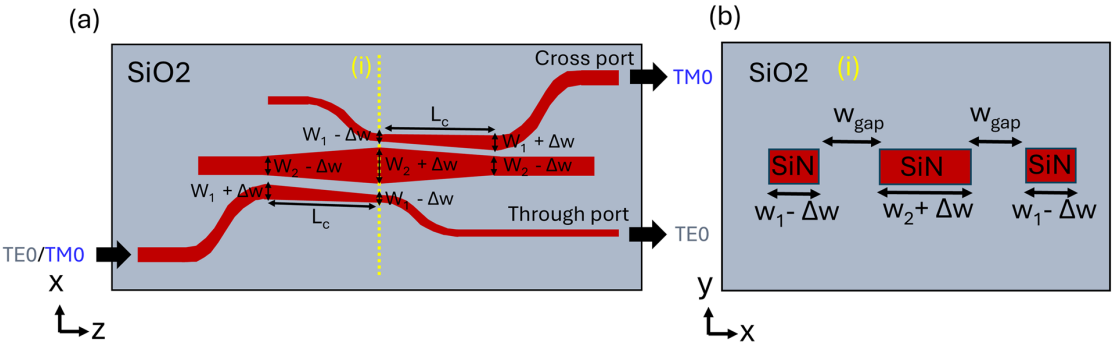

Figure 2.18: Schematic of our second device: a PBS based on a tapered asymmetrical directional coupler. (a) Top view; (b) cross-sectional view indicated by the yellow dashed line at position (i).

To mitigate the impact of waveguide width deviations, we explored the use of linear tapers in the access and central waveguides of the two directional couplers, as shown in Fig. 2.18 (a) and (b), representing respectively the top view and the cross-sectional view at position (i) of the device. In this tapered design, the width of the input waveguide varies from $w_1 + \Delta w$ to $w_1 - \Delta w$, while the central waveguide is tapered from $w_2 - \Delta w$ to $w_2 + \Delta w$, i.e., the gap width remains constant.

Fig. 2.19 shows the evolution of the effective indices of the TM_0 and TM_1 modes at a wavelength of 1300 nm as a function of the position z along the coupling length. Without waveguide width deviation (Fig. 2.19 (a)), the device without tapers (dashed lines) ensures that the two modes have the same effective index along the entire coupler (since the widths of the two waveguides are constant), thereby satisfying the phase-matching condition. However, if the waveguides are both 10 nm wider than the nominal value (Fig. 2.19 (b)), the phase-matching condition is no longer satisfied anywhere in the coupler, and

coupling becomes very inefficient. By tapering the waveguides within the coupler around the two optimal design points w_1 and w_2 (solid lines in Fig. 2.19), the phase-matching condition is satisfied in the ideal case without fabrication imperfections only at the center of the device (Fig. 2.19 (a)). The advantage of this design is that, even when errors in the waveguide widths occur, the phase-matching condition is still satisfied, albeit at a different position z along the coupler (Fig. 2.19 (b)). It should also be noted that in the case without a taper, even if a waveguide width deviation prevents phase matching at a given wavelength, it may still be achieved at another wavelength, although the coupling length will not be optimal.

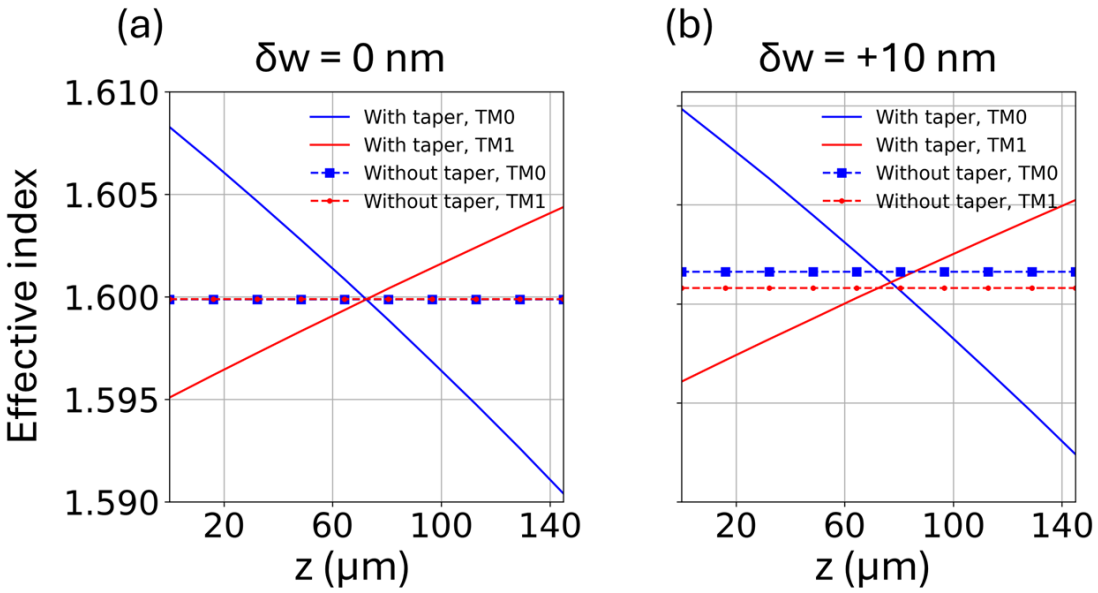

Figure 2.19: Variation of the TM_0 and TM_1 modes' effective indices within the coupling region for the non-tapered and tapered PBS devices operating at a wavelength of 1300 nm: (a) without width deviation, and (b) with a width deviation of $\delta w = +10$ nm.

To verify the stability of the performance of this tapered structure against width deviations, we calculated the TM_0–TM_1 transmission as a function of the coupling length using EME on the structure shown in Fig. 2.20 for several values of Δw, and then repeated the calculations for different waveguide width deviations δw.The results are shown in Figs. 2.21 (a) and 2.22 (a) for 950 nm and 1300 nm, respectively. We also plotted the amount of (unwanted) conversion between the TE_0 and TE_1 modes for 950 nm and 1300 nm, Figs. 2.21 (b) and 2.22 (b), for several values of Δw, and again repeated the calculations for different δw values. We observe that as Δw increases, stability with respect to taper length also increases (a plateau gradually forms), and sensitivity to waveguide width deviation decreases. However, as the conversion stability improves for the TM_0 and TM_1 modes, the amount of coupled power from TE_0 to TE_1 also increases. Therefore, in selecting the parameters, we opted for a compromise between TM stability and TE instability.

We determined the Δw values for 950 nm and 1300 nm as 30 nm and 50 nm, respect-

ively, with corresponding taper lengths of L_c = 121 μm and L_c = 145 μm. The coupling lengths of the tapered device are longer than those of the non-tapered device because phase matching is satisfied only over a limited section of the coupler. The tapered structure exhibits a TM_0–TM_1 conversion of 99 % at both 950 nm and 1300 nm, and a TE_0–TE_1 conversion of 16 % and 9 % at 950 nm and 1300 nm, respectively.

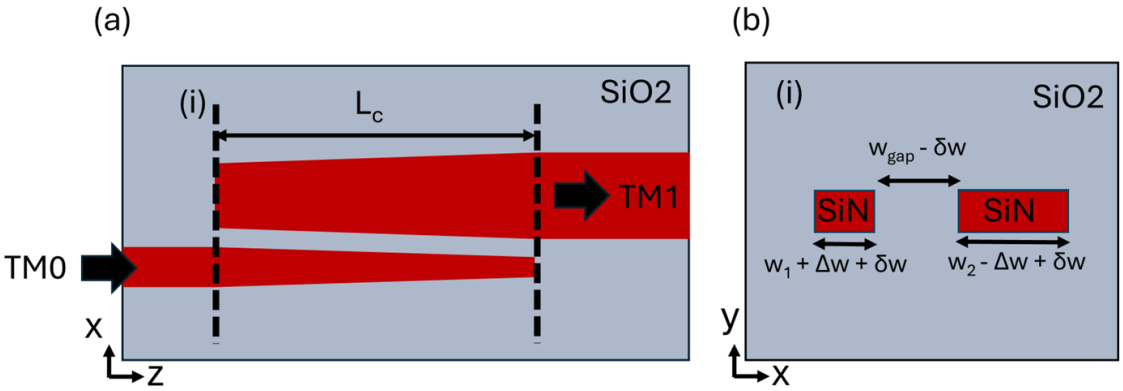

Figure 2.20: Simplified version of the tapered PBS optimized using an eigenmode expansion solver: (a) top view; (b) cross-sectional view at position (i), indicated by the black dashed line in (a).

2.4 . Fabrication and experimental results

Two generations of devices were fabricated and tested using two different setups, as detailed in the two following sections.

2.4.1 . First device generation and fiber-based measurements

The chip layout was prepared including several variations of the nominal device for both straight and tapered designs. These variations targeted both the waveguide widths and the coupling lengths. Specifically, the waveguide widths were adjusted by ±10 nm to account for potential fabrication width deviations. In addition, the coupling lengths were reduced by 5 µm and 10 µm to compensate for overcoupling effects induced by the S-bends, which occur when the input and output waveguides approach and recede from the central waveguide. The S-bend was designed with a radius of 100 μm. In order to de-embed the PBSs performance, reference waveguides without PBSs were included on the chip. The reference waveguide width for the device is chosen as 0.52 μm at λ = 950 nm and 0.71 μm at λ = 1300 nm; for these widths, the waveguide is single mode (see Fig. 2.13).

For input and output optical coupling, edge couplers based on inverse taper geometries were designed, as schematically represented in Fig. 2.23 (a) 3D view and (b) cross-sectional view in (x-z) plane. Lensed fibers with mode field diameters (MFDs) of 2.5 µm and 5 µm at the focal point were considered for operation at 950 nm and 1300 nm, respectively. To best match the fiber and waveguide modes, a finite difference mode solver was used to calculate fiber modes with MFDs equivalent to those of the employed lensed

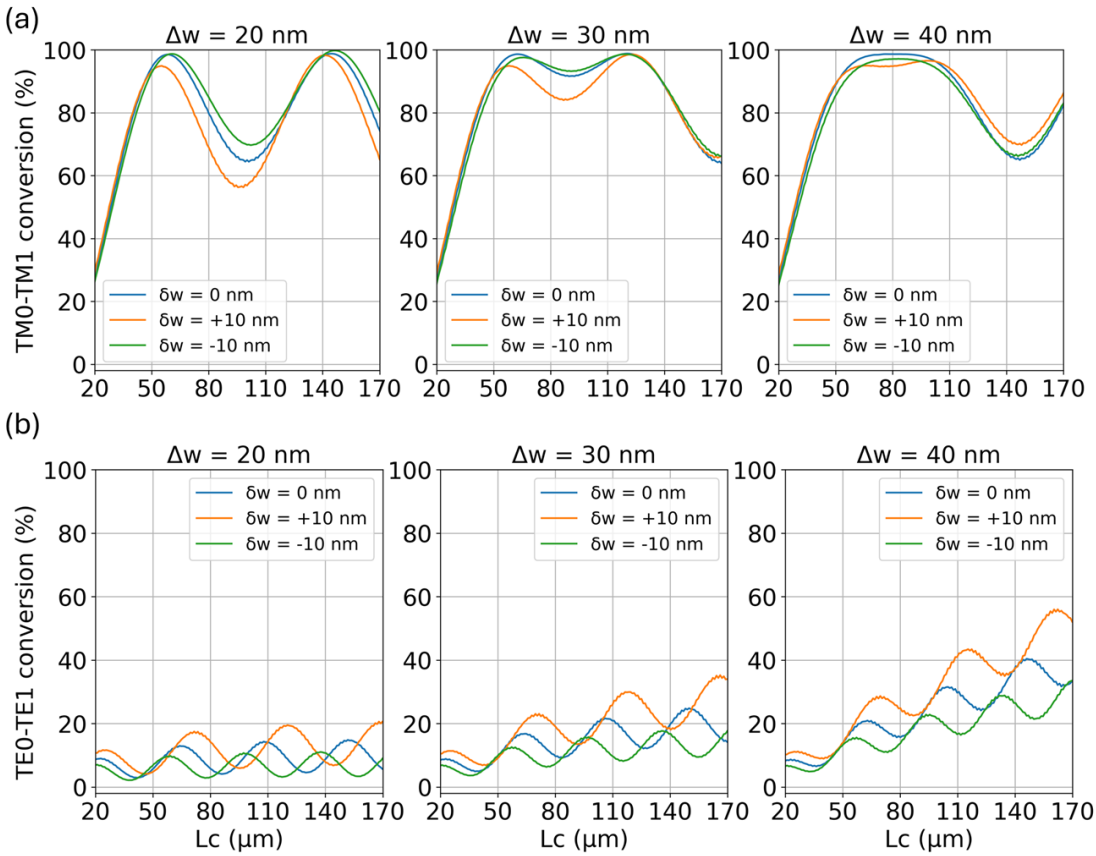

Figure 2.21: Fundamental mode coupling to the first higher-order mode at λ = 950 nm, from the lower tapered waveguide to the upper tapered waveguide shown in Fig. 2.20 (a), as a function of the coupling length for three different waveguide width deviations δw. From left to right, the taper width variation (Δw) increases from 20 nm to 40 nm. (a) TM_0–TM_1 conversion. (b) TE_0–TE_1 spurious conversion.

fibers.

Fig. 2.24 (a) represent the schematic cross-section of an optical fiber, consisting of a core waveguide with refractive index n_{core} and a cladding with refractive index $n_{cladding}$. In Fig. 2.24 (b), the normalized electric field intensity of the fiber fundamental mode (corresponding to Fig. 2.24 (a)) is represented, calculated at a wavelength of 1300 nm using a finite difference eigenmode solver, with a core diameter of 5 μm, core index $n_{core} = 1.48$, and cladding index $n_{cladding} = 1.45$. The transverse section of the normalized electric field intensity of the mode Fig. 2.24 (b) at position (i) is represented in Fig. 2.24 (c). The resulting MFD for these parameters is approximately 5 μm. The same procedure was applied for the wavelength of 950 nm, with a chosen core diameter of 3 μm, a core index $n_{core} = 1.6$, and a cladding index $n_{cladding} = 1.45$, yielding an MFD of 2.5 μm. The coupling efficiency was then calculated using the overlap integral between the fiber mode and the waveguide mode as a function of the waveguide width. Figures 2.25 (a) and (b) show the overlap in-

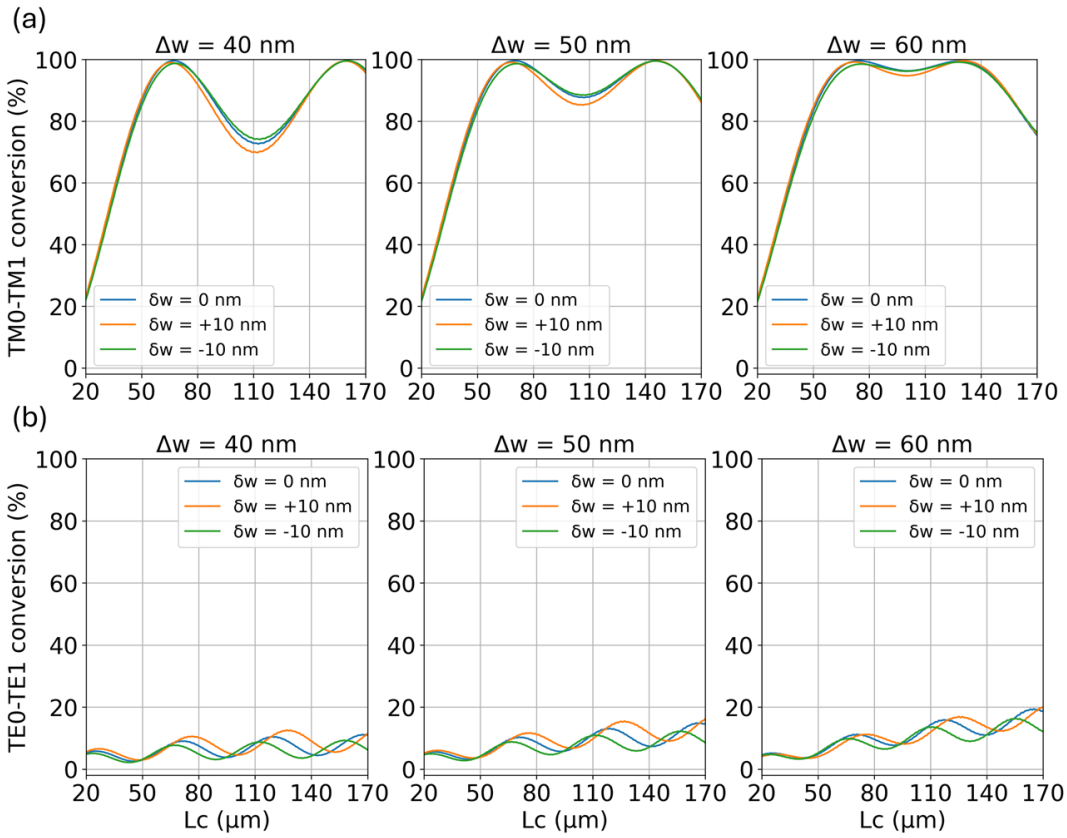

Figure 2.22: Fundamental mode coupling to the first higher-order mode at λ = 1300 nm, from the lower tapered waveguide to the upper tapered waveguide shown in Fig. 2.20 (a), as a function of the coupling length for three different waveguide width deviations δw. From left to right, the taper width variation (Δw) increases from 20 nm to 40 nm. (a) TM_0–TM_1 conversion. (b) TE_0–TE_1 spurious conversion.

tegral between the fiber mode and the edge coupler tip mode for different tip widths for both TE and TM modes, respectively. As expected, reducing the tip width increases the overlap efficiency, since a narrower tip supports a larger mode size.

The transmission performance of the fundamental modes (a) TE_0 and (b) TM_0 for different tip widths of the inverse taper as a function of taper length at λ = 950 nm, obtained using EME simulations, is presented in Fig. 2.26, while the corresponding results at λ = 1300 nm are shown in Fig. 2.27. In contrast to the overlap behavior, it was found that smaller tip widths lead to higher propagation losses within the taper. This is consistent with the fact that a larger mode, produced by a narrower tip, is more susceptible to leakage into the substrate.

Considering these opposing trends, a design compromise was adopted. The tip width was set to 130 nm for the 950 nm device and 180 nm for the 1300 nm device. The corresponding inverse taper lengths were fixed at 150 μm and 170 μm, respectively. Under these conditions, the expected coupling performance is 0.86 dB and 2.23 dB for TE and

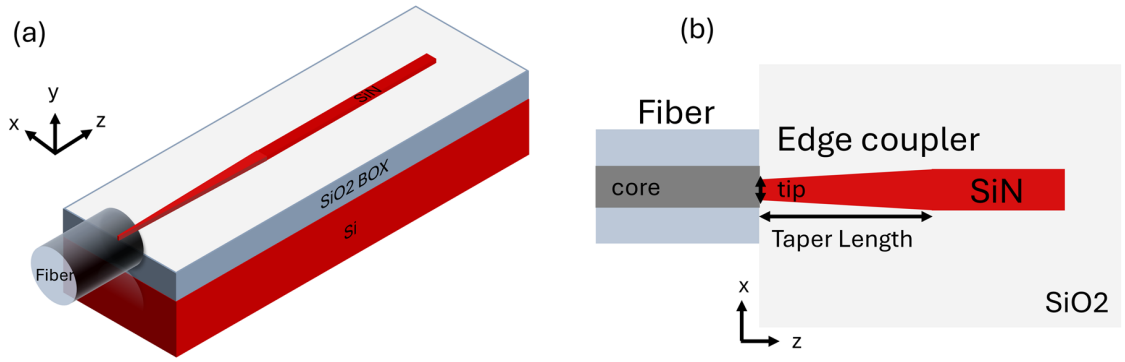

Figure 2.23: Schematic of fiber-to-chip coupling using an edge coupler based on an inverse taper: (a) 3D view; (b) 2D view in the (x–z) plane.

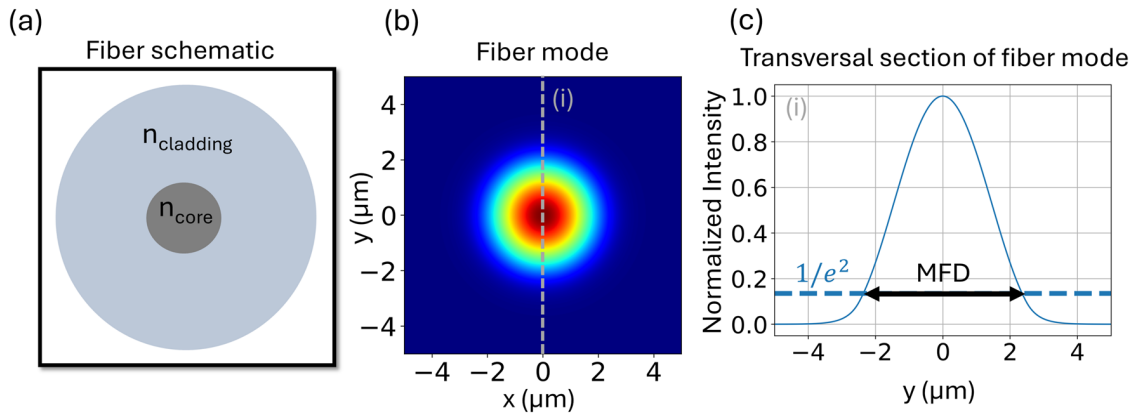

Figure 2.24: (a) Cross-sectional view of an optical fiber. (b) Electric-field intensity of the fundamental mode of the fiber shown schematically in (a), calculated at λ = 1300 nm using a finite eigenmode solver with the following parameters: n_{core} = 1.48, $n_{cladding}$ = 1.45, and a core diameter of 5 μm. (c) Transverse section of the electric field intensity, indicated by the dashed line in (b) at position (i).

TM modes at 950 nm, and 1.25 dB and 1.43 dB for TE and TM modes at 1300 nm.

To test the fabricated chip, we used the setup shown in Fig. 2.28, consisting of a tunable laser, a polarization controller, and a polarizer. The light was injected into the chip using a polarization-maintaining (PM) lensed fiber. At the output of the chip, the light was collected with the same type of PM lensed fiber and measured using a power detector. The polarization was controlled with a polarization controller to align it with that of the polarizer. Different setups were used for measurements at 950 nm and 1300 nm.

To excite the TE_0 mode at the input waveguide, the transmission axis of the polarizer placed before the injection PM fiber is oriented horizontally with respect to the chip (i.e., along the x-direction in Fig. 2.23). In the case of TM_0 mode excitation, the transmission axis of the polarizer is oriented vertically (i.e., along the y-direction in Fig. 2.23).

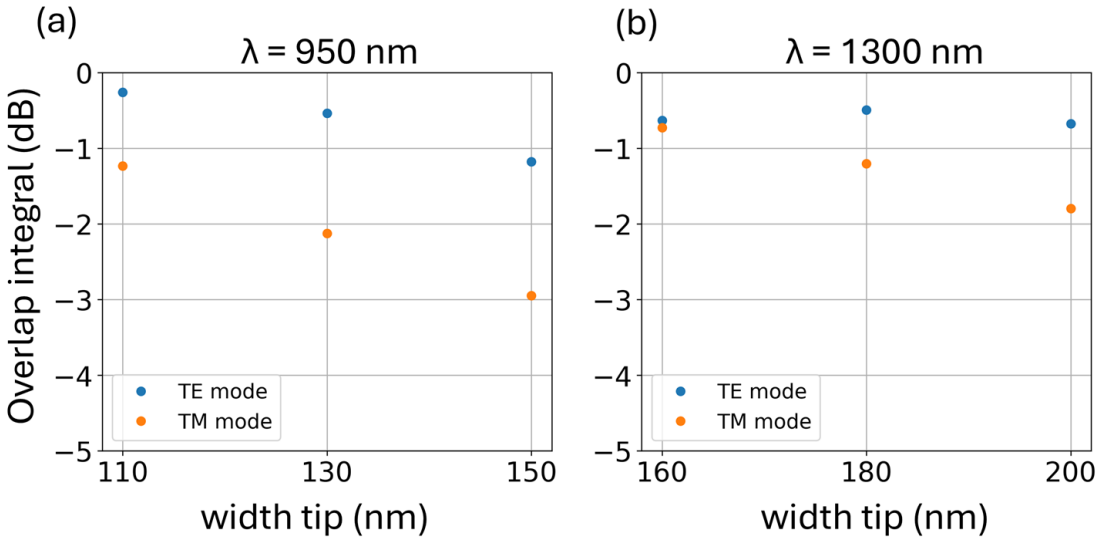

Figure 2.25: Overlap integral calculated between the fundamental TE_0 and TM_0 modes of the fiber and those of the inverse taper edge coupler shown in Fig. 2.23 (b), as a function of the taper tip width: (a) for λ = 950 nm and (b) for λ = 1300 nm.

The experimental measurement data for the reference waveguides are shown in Fig. 2.29. On Fig. 2.29 (a), the transmission of the reference waveguide at 950 nm is plotted as a function of wavelength, and on Fig. 2.29 (b), the corresponding plot is shown for 1300 nm. This reference allows us to account for both coupling losses and propagation losses. At 950 nm, we obtain a transmission of 5 dB and 8 dB for TE and TM, respectively, and at 1300 nm, a transmission of 3.5 dB and 4.3 dB for TE and TM, respectively, which is consistent with the simulation results presented above, considering that coupling losses at both the input and output must be included, in addition to propagation losses, given that the chip length is 0.8 mm.

In Figs. 2.30 and 2.31, we plot the experimental transmission of the straight PBS devices normalized to the reference waveguide (considering the right polarization) at λ = 950 nm and λ = 1300 nm, respectively, as a function of wavelength, for different bias variations on the nominal design. ΔL_c denotes variations of the nominal coupling length, and δw denotes variations of the waveguide widths. The devices were tested by first exciting the TE mode and then the TM mode, measuring the light at the through and cross ports. From left to right, the coupling length is varied, and from top to bottom, the waveguide widths are varied. In the case of the measurements of the devices operating at 950 nm, shown in Fig. 2.30, the designs with δw = 0 nm are not reported, as most of them did not work properly.

Looking at the fabricated devices, we observe that polarization splitting is indeed achieved. For all the devices, when the TE mode is excited, the light is measured at the through port with negligible losses (i.e., < 1 dB). When the TM mode is excited, the light is measured at the cross port with negligible losses around the operating wavelength of

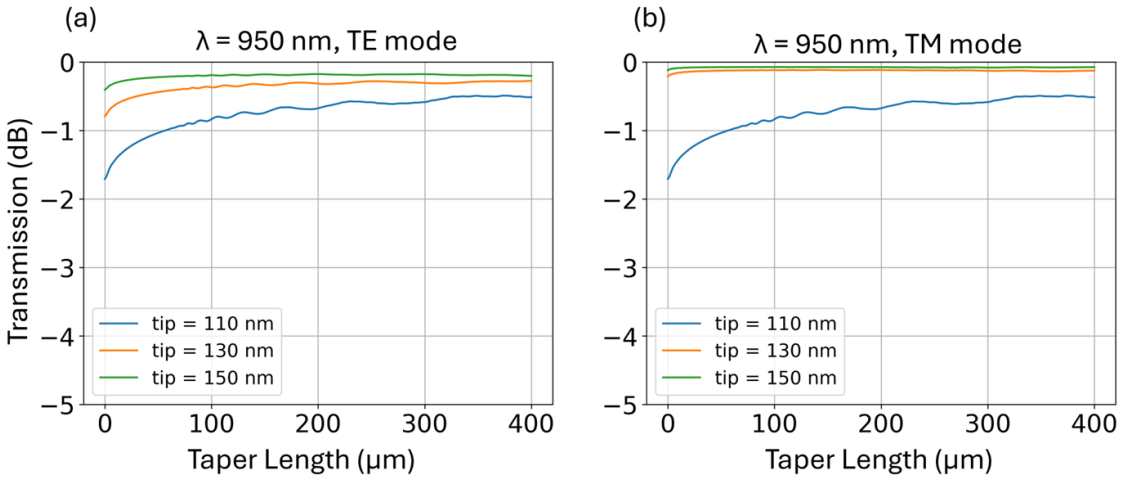

Figure 2.26: Transmission of the inverse taper edge coupler shown in Fig. 2.23 (b) at λ = 950 nm, as a function of the taper length for three different tip widths: (a) TE mode excitation, and (b) TM mode excitation.

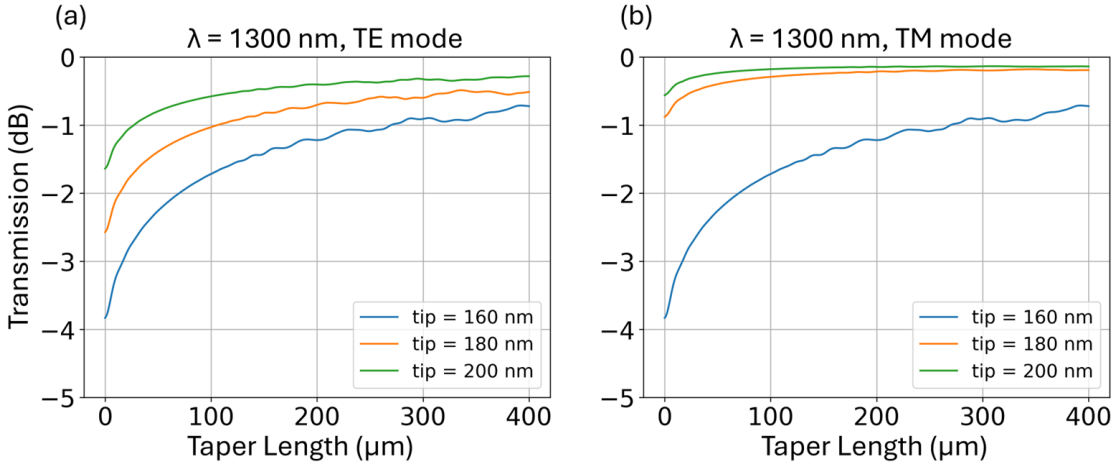

Figure 2.27: Transmission of the inverse taper edge coupler shown in Fig. 2.23 (b) at λ = 1300 nm, as a function of the taper length for three different tip widths: (a) TE mode excitation, and (b) TM mode excitation.

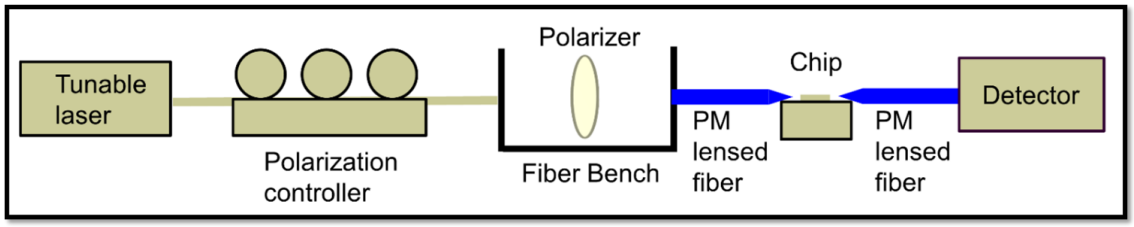

Figure 2.28: Lensed fiber experimental setup for polarization control devices characterization.

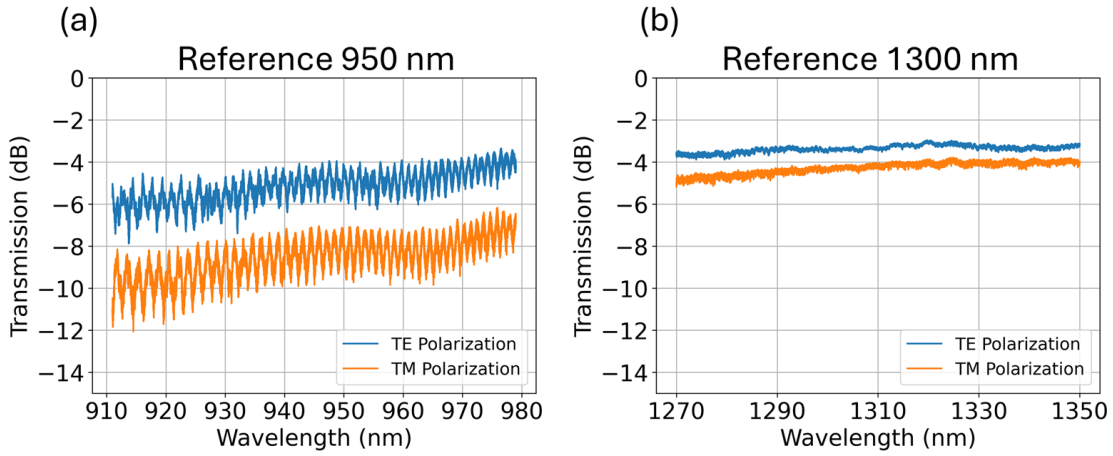

Figure 2.29: Reference waveguide transmission measurements for TE and TM mode excitation: (a) $\lambda = 950$ nm, and (b) $\lambda = 1300$ nm.

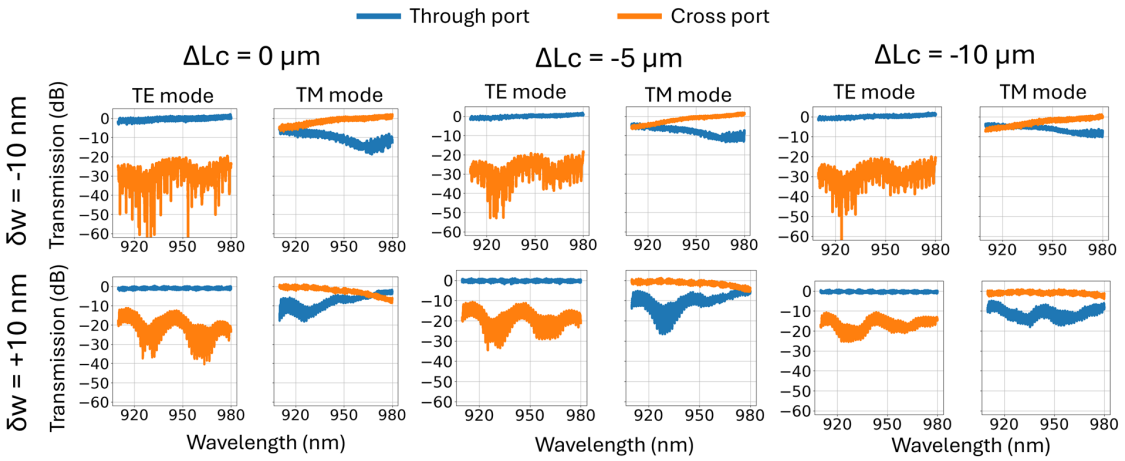

Figure 2.30: Normalized measured transmission spectra of the straight PBS devices at the through and cross ports as a function of wavelength for devices operating around 950 nm, with either TE or TM mode excitation at the input. From top to bottom, the measurements correspond to waveguide width deviations of ±10 nm relative to the nominal design, and from left to right, to coupling length deviations ranging from 0 to –10 μm. Normalization is done using the reference waveguide measurement shown in Fig. 2.29.

the device. The TE mode is only slightly affected by variations in wavelength, coupling length, and waveguide width, as evidenced by the very flat transmission measured at the through port.

However, for the TM mode, we note that the operating wavelength and device performance are quite dependent on both the waveguide width and the coupling length. If the waveguides are wider than designed, as an example in the case of the device at 1300 nm with $\Delta L_c = -5$ μm, an increase of 10 nm in width results in a blueshift of approxim-

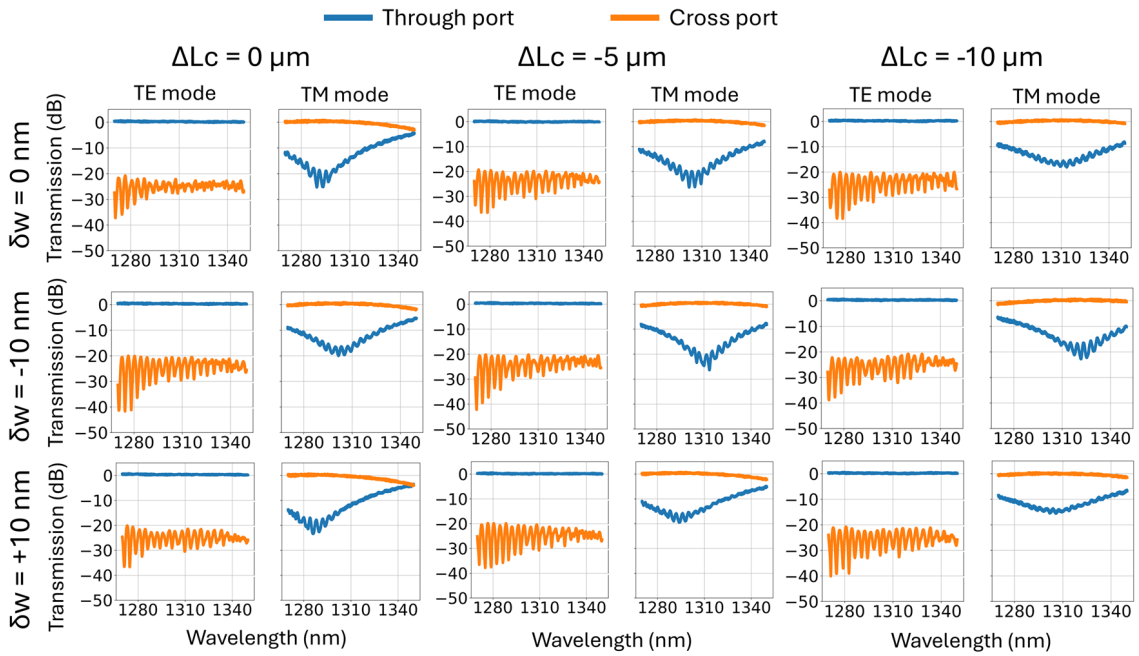

Figure 2.31: Normalized measured transmission spectra of the straight PBS devices at the through and cross ports as a function of wavelength for devices operating around 1300 nm, with either TE or TM mode excitation at the input. From top to bottom, the measurements correspond to waveguide width deviations of ±10 nm relative to the nominal design, and from left to right, to coupling length deviations ranging from 0 to –10 μm. Normalization is done using the reference waveguide measurement shown in Fig. 2.29 (b).

ately 10 nm, whereas a decrease of 10 nm results in a redshift of approximately 10 nm. The same conclusion applies to the coupling length: when it is reduced (for a fixed width), shifts in the operating wavelength are observed.

The polarization extinction ratio PER can be directly observed on the graphs, since they correspond, in dB scale, to the difference between the through and cross ports at a given wavelength. The insertion loss IL corresponds instead to the transmission at either the through or cross port depending on whether the TE or TM mode is injected. Across all devices, whether operating at 950 nm or at 1300 nm, it is important to note that significant oscillations appear in the through port measurements when the TM mode is excited, and in the cross port measurements when the TE mode is excited. These oscillations will be discussed in the following section. However, since the PER values require calculating the difference between the through port and cross port measurements, a moving average was applied to remove the oscillations.

Thus, the best performing device operating at λ = 950 nm is the one with deviations from the nominal design of ΔL_c = –5 μm and δw = +10 nm. This device exhibits a PER higher than 10 dB over the wavelength range from 910 to 980 nm for the TE mode, and from 920 to 950 nm for the TM mode. Over the same respective ranges, the IL remains below 1 dB for both TE and TM modes. The maximum PER is obtained at λ = 930 nm,

reaching 24 dB and 17 dB for the TE and TM modes, respectively.

In the case of devices operating at $\lambda = 1300$ nm, the best performance is obtained for the device with deviations from the nominal design of $\Delta L_c = -5\ \mu$m and $\delta w = -10$ nm. This device exhibits a PER higher than 20 dB over the wavelength range from 1270 to 1350 nm for the TE mode, and a PER higher than 10 dB over the range from 1280 to 1340 nm for the TM mode. Over the same respective ranges, the IL remains below 1 dB. At $\lambda = 1310$ nm, the device achieves a PER of 24 dB and 22 dB for the TE and TM modes, respectively.

The experimental results of tapered PBS devices are shown in Figs. 2.32 and 2.33 for wavelengths of 950 nm and 1300 nm, respectively. The normalized transmission is plotted for TE and TM mode excitation. For both figures, in the top three graphs, from left to right, we show the variation of coupling length at a fixed $\delta w = 0$ nm. In the bottom three graphs, we show the variation of waveguide width at a fixed coupling length variation of $\Delta L_c = -5\ \mu$m.

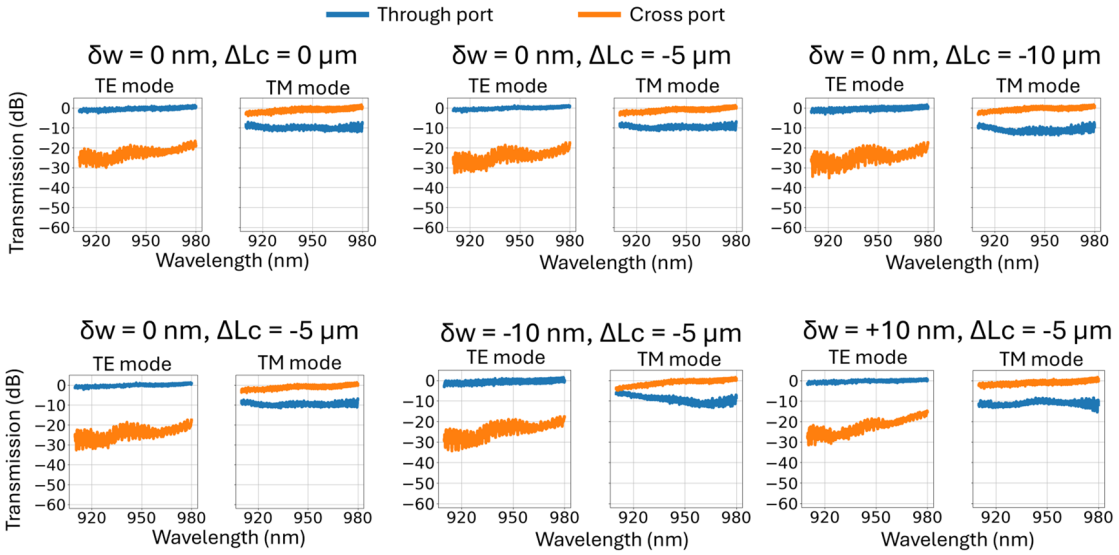

Figure 2.32: Normalized measured transmission spectra of the tapered PBS devices at the through and cross ports as a function of wavelength for devices operating around 950 nm, with either TE or TM mode excitation at the input. The top panels show measurements for coupling length deviations ranging from 0 to –10 μm, with no variation in waveguide width relative to the nominal design. The bottom panels show measurements for waveguide width deviations of ±10 nm relative to the nominal device, for a coupling length deviation of $\Delta L_c = -5\ \mu$m. Normalization is done using the reference waveguide measurement shown in Fig. 2.29 (a).

Overall, the performances of the tapered devices appear significantly more stable with respect to the coupling length and waveguide width deviations. All the tapered devices at 1300 nm (i.e., with all different variations in ΔL_c and δw) exhibit a PER higher than 10 dB over the range from 1270 to 1350 nm for both TE and TM modes, with IL below 1 dB over the same range. For all the tapered devices operating at 950 nm, the PER is higher than 10 dB for both TE and TM modes over the wavelength range from 930 to 980 nm. Over this same range, the IL remains below 1 dB. When comparing the performances of the

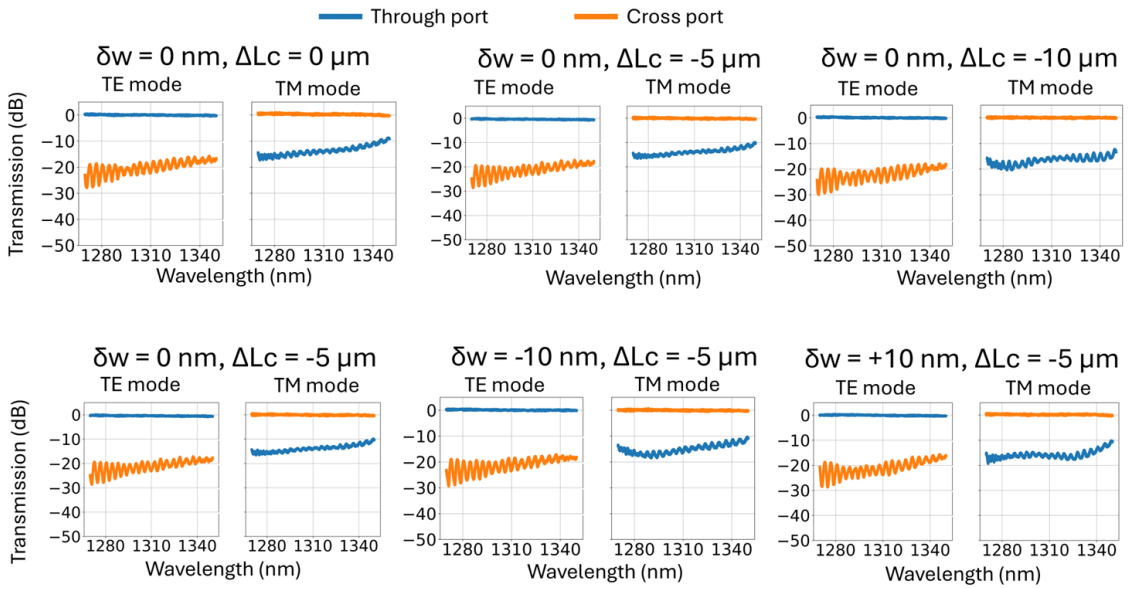

Figure 2.33: Normalized measured transmission spectra of the tapered PBS devices at the through and cross ports as a function of wavelength for devices operating around 1300 nm, with either TE or TM mode excitation at the input. The top panels show measurements for coupling length deviations ranging from 0 to –10 μm, with no variation in waveguide width relative to the nominal design. The bottom panels show measurements for waveguide width deviations of ±10 nm relative to the nominal device, for a coupling length deviation of $\Delta L_c = -5$ μm. Normalization is done using the reference waveguide measurement shown in Fig. 2.29 (b).

devices without taper and with taper, it can be observed that the obtained PER values are reduced, for both TE and TM modes. For the TM mode, this is due to the fact that in the tapered case the phase matching is satisfied only over a limited section of the coupler. For the TE mode, this reduction arises because the presence of the taper facilitates phase matching between the TE_0 and TE_1 modes, particularly at longer wavelengths, thereby increasing the amount of light measured at the cross port.

To emphasize the improved stability of the tapered devices compared to the non-tapered ones, we plotted in Fig. 2.34 the PER of the TM mode as a function of coupling length and waveguide width variations, for three different wavelengths. The PER of the TM mode is shown because it is the one coupled by the directional couplers into the cross port, which makes the device performance for this mode much more dependent on deviations from the nominal design. In the top row (Figs. 2.34 (a) and (b)), the PER_{TM} of devices at 950 nm are shown, and in the bottom row (Figs. 2.34 (c) and (d)), the PER_{TM} of devices at 1300 nm.

For wavelengths between 1300 and 1320 nm in the non-tapered device (Fig. 2.34 (a)), when the waveguide width δw varies, a large disparity in PER is observed on the three wavelengths depending on the fabrication bias. For $\delta w = +10$ nm, the PER varies from 11 dB to 17 dB, while for $\delta w = 0$ nm, the PER varies from 17 dB to 22 dB. In the case of the tapered device, this disparity is almost eliminated: for $\delta w = +10$ nm, the PER is stable

across the three wavelengths at approximately 16 dB. For δw = –10 nm, the PER varies only from 13 dB to 14 dB. This observation also holds true for variations in the coupling length. In Fig. 2.34 (d), for $\Delta L_c = 0$ μm and $\Delta L_c = -5$ μm, the PER disparity across the three wavelengths in the non-tapered devices ranges from 10 to 20 dB and from 16 to 22 dB, respectively. In contrast, for the tapered devices, the PER disparity is limited to 14–15 dB for $\Delta L_c = 0$ μm and 13–14 dB for $\Delta L_c = -5$ μm. The same analysis carried out on the 950 nm devices leads to the same conclusion. Thus, the tapered devices appear much more robust; however, this comes at the cost of a reduced maximum PER.

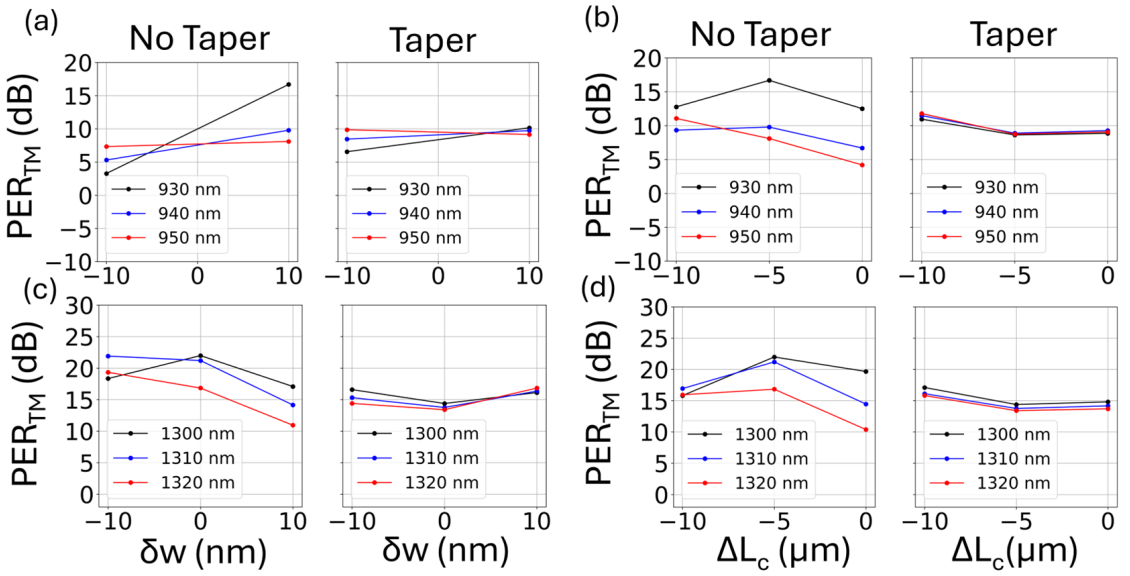

Figure 2.34: PER_{TM} for PBS devices with and without taper, measured at three different wavelengths. Panels (a)–(b) correspond to measurements performed on devices operating around 950 nm, while panels (c)–(d) correspond to those operating around 1300 nm. In panel (a) and (c), the variation of the PER as a function of the waveguide width deviation (δw) is shown for $\Delta L_c = -5$ μm, whereas in (b) and (d), the variation of the PER_{TM} as a function of the coupling length deviation (ΔL_c) is shown for $\delta w = 0$ nm.

2.4.2 . Second device generation and free-space measurements

A first issue that can be observed with the results of the first generation of devices, e.g., in Figs. 2.30 and 2.31, is the appearance of rather evident oscillations in the measured transmission spectra. The origin of these oscillations could be the presence of a Fabry–Perot cavity in the device. This cavity appears in the central waveguide (see Fig. 2.35 (a)). When the TM_0 mode is excited at the input waveguide and converted into the TM_1 mode of the central waveguide, a fraction of the TM_1 mode is not entirely converted back into the TM_0 mode of the upper waveguide and therefore remains in the central waveguide, eventually reaching a discontinuity. The core waveguide ends abruptly and causes a partial reflection. The fraction of the TM_1 mode that is reflected then propagates in the opposite direction, coupling to the TM_0 mode of the upper and lower waveguides in the

backward direction. The remaining fraction of TM_1 in the central waveguide is again reflected by the interface, bouncing back and forth and thus forming a Fabry–Perot cavity. The same phenomenon occurs for the TE mode, in which a fraction of the TE_0 mode from the input waveguide is converted into the TE_1 mode of the central waveguide, thereby leading to the back and forth propagation within the central waveguide.

A second issue was also observed in the characterization of the insertion loss (IL). We used a normalization with the reference waveguides. However, due to the very low losses of the device, accurately characterizing the IL was challenging. For instance, in Fig. 2.36, we show the normalized transmission of the device at 1300 nm with $\Delta L_c = -5$ μm and $\delta w = -10$ nm when the TM mode is injected. The results suggest that the losses would "above zero," which indicates that they are too small to be reliably characterized using only the reference waveguide.

Based on the results obtained with this first generation of devices, a second generation was fabricated. In this new generation, the cavity was suppressed by reducing the width of the central waveguide at its extremities as shown in Fig. 2.35 (b), reaching the cut-off condition for which neither TE nor TM modes can propagate at the wavelengths under study (950 and 1300 nm). As a result, the residual light in the central waveguide can no longer bounce back and forth, as it is fully filtered. Moreover, in this second fabrication run, we also added structures to obtain a more precise measurement of the insertion losses. In particular we included two cascaded versions of the straight PBS, allowing the use of the cut-back method. For this measurement, the chosen devices chosen were the two nominal designs (i.e., $\Delta L_c = 0$ μm and $\delta w = 0$ nm) at $\lambda = 950$ nm and $\lambda = 1300$ nm, each implemented in cascades of 3 and 6 devices (see Fig. 2.37).

The experimental results of this new generation are presented in Fig. 2.38, showing the normalized transmission for TE and TM excitation as a function of wavelength. From left to right, the number of cascaded devices increases. It can be observed that the PBS operate as expected, since the TE and TM modes are measured at the through and cross ports, respectively, with negligible losses (< 1 dB). However, a redshift in the central operating wavelength appears in the data. Specifically, for the TM mode, the device operates at 1330 nm instead of 1300 nm. For the devices designed to operate at 950 nm, the results are not shown because the same redshift observed at 1300 nm also affects them. Nevertheless, our laser at 950 nm has a limited tuning range (910–980 nm), and with this redshift the wavelength at which the device is fully operational is not accessible.

Figure 2.39 shows the SEM image of the coupling region of the nominal PBS device operating at 1300 nm, with a zoom on the directional coupler. The nominal design widths should be $w_1 = 0.68$ μm and $w_2 = 1.79$ μm, with a gap of $w_{\text{gap}} = 0.55$ μm. However, the SEM reveals that $w_1 = 0.66$ μm, $w_2 = 1.77$ μm, and $w_{\text{gap}} = 0.57$ μm, corresponding to a deviation of 20 nm. This deviation is most likely the reason for the redshift of the operating wavelength.

Despite the presence of this redshift, we used this measurements (Fig. 2.38), to determined the insertion loss of the device at 1300 nm via the cut-back method. To this purpose, we took the average measured transmission around the device's operating wave-

length (1330 nm) and performed a linear regression as a function of the number of cascaded devices. Results are shown in Fig. 2.40. The estimated value for IL (corresponding to the slope of the fitting curve) was 0.1 dB for both TE and TM modes, which, to the best of our knowledge, represents a record-low value for a device fabricated in silicon nitride (see comparison table in table 2.1).

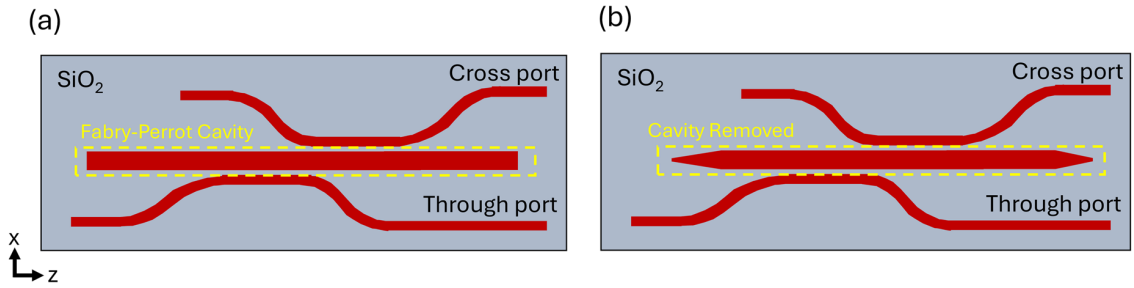

Figure 2.35: (a) PBS device where the central waveguide can act as a Fabry–Perot cavity. (b) Edge widths of the central waveguide are reduced to reach the cut-off limit and suppress the Fabry–Perot cavity.

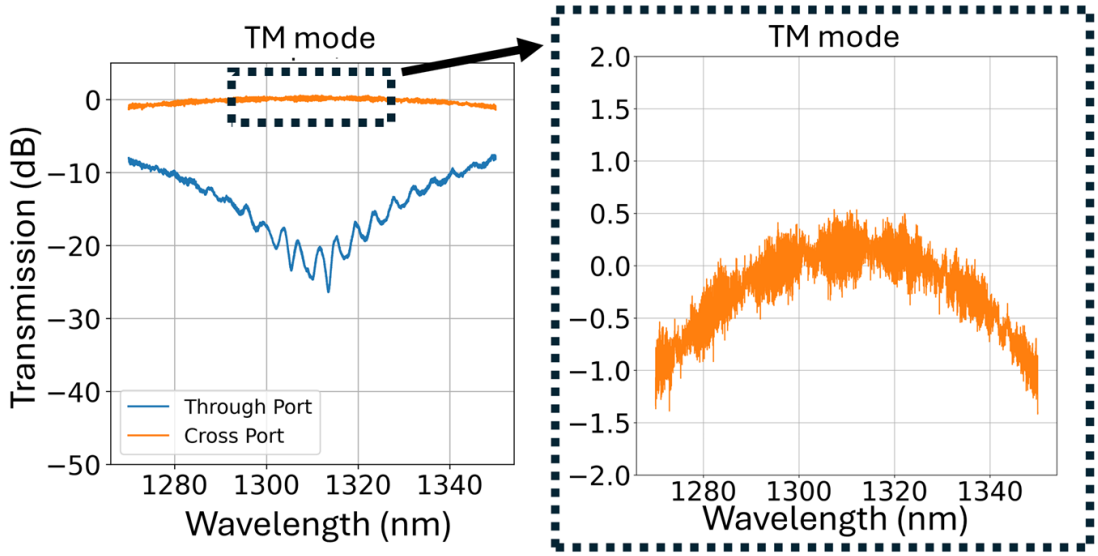

Figure 2.36: Normalized measured transmission at the through and cross ports with TM mode excitation for the device with $\Delta L_c = -5\mu$m and $\delta w = -10$ nm. In the inset, a zoom of the cross-port transmission near the maximum. Normalization is done using the reference waveguide measurement shown in Fig. 2.29 (b).

In the data of Fig. 2.38, we observed that oscillations were still present in the cascaded device. To investigate the origin of these oscillations, and in particular whether they could be linked to the use of PM fibers, we performed a test using the setup shown in Fig. 2.41 (a). In this configuration, the chip and the output fiber from the setup of Fig. 2.28 were

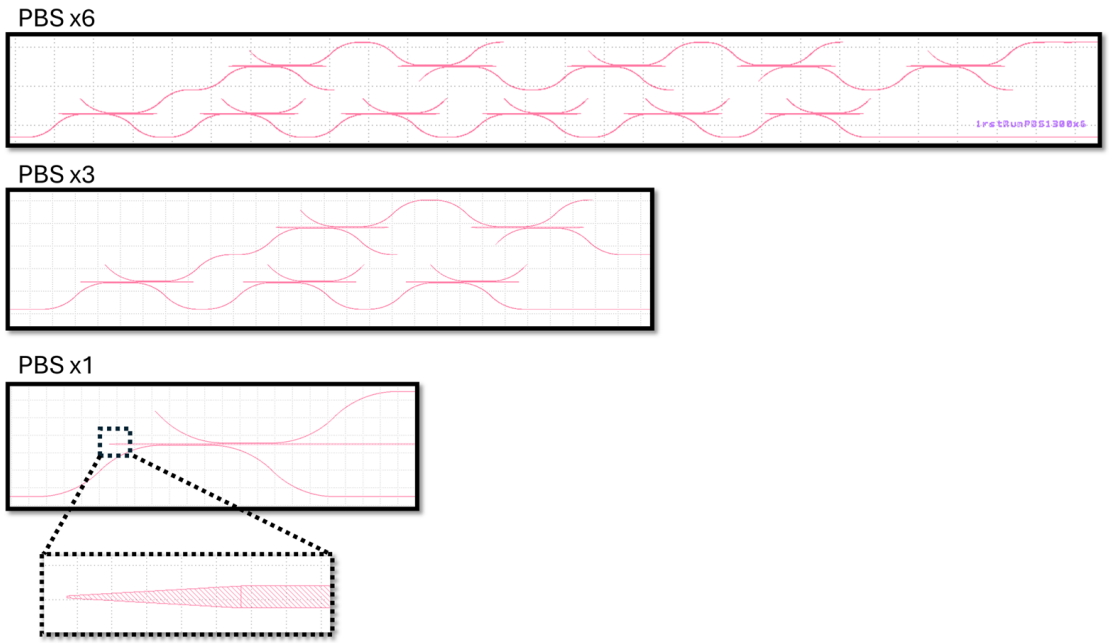

Figure 2.37: GDS layout images of the cascaded PBSs, with a zoom on the central waveguide of the PBS showing the reduced edge width, as illustrated in Fig. 2.35 (b).

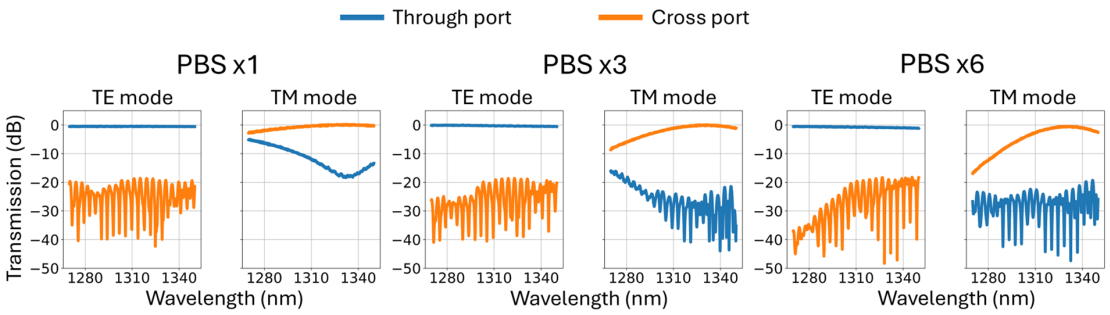

Figure 2.38: Measured transmission of the cascaded PBS devices at the through and cross ports as a function of wavelength for TE and TM mode excitation, with the number of cascaded devices increasing from left to right.

removed and replaced with an objective lens and an output polarizer placed just before the detector. In this way, we could control the input polarization sent into the fiber and measure the output polarization it returned.

The results are shown in Fig. 2.41 (b) for 950 nm and in Fig. 2.41 (c) for 1300 nm setups. The measurement is performed as follows: for TE input, the transmission axis of the polarizer is oriented horizontally (i.e., along the x-direction in Fig. 2.23), and for TM input, the transmission axis is oriented vertically (i.e., along the y-direction in Fig. 2.23). The polarized light is then injected into the fiber, and after exiting, it is collimated by the microscope objective and sent to the polarizer before the detector. This polarizer is oriented

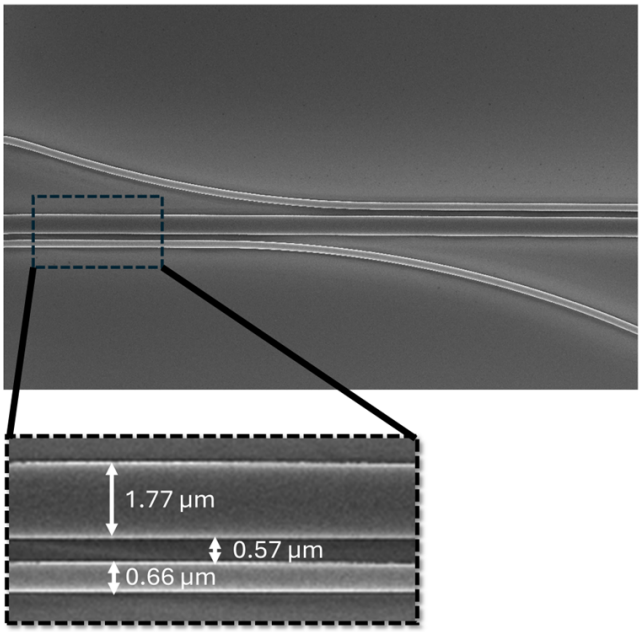

Figure 2.39: SEM image of the nominal PBS device operating at 1300 nm, with a zoom on the coupling region.

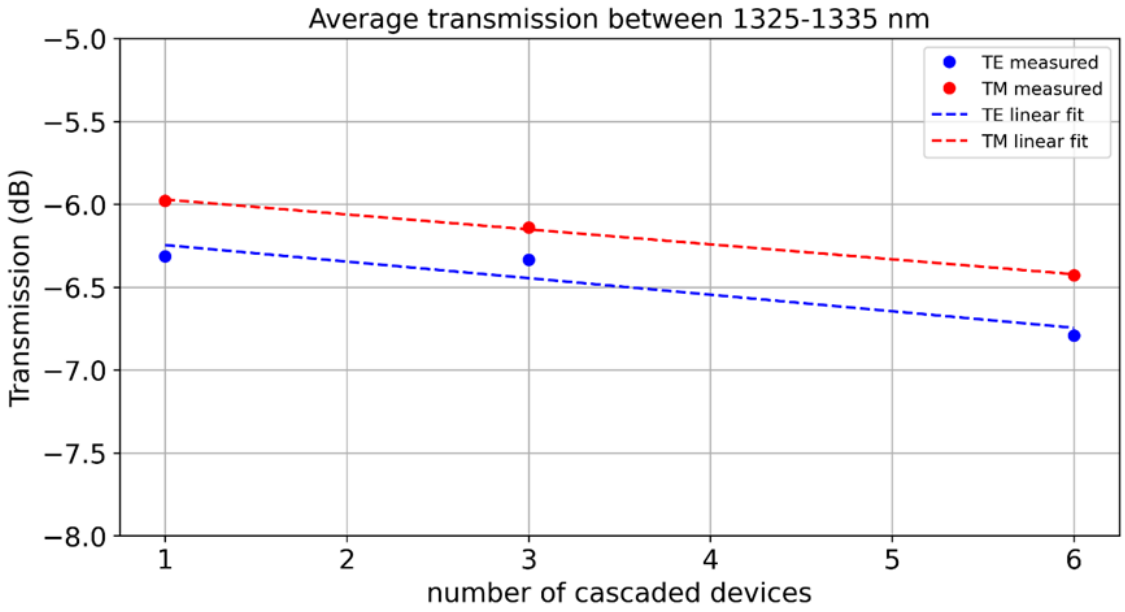

Figure 2.40: Average transmission of the PBS devices between 1325 and 1335 nm, without normalization by the reference waveguide, as a function of the number of cascaded devices. Measured transmissions for TE and TM modes are shown as blue and red dots, respectively, while the dashed lines represent linear fits, with the slope corresponding to the insertion losses.

horizontally when measuring the TE polarization and vertically when measuring the TM polarization. The difference between the TE and TM polarizations measured at the output corresponds to the PER of the setup. In the data shown in Fig. 2.41 (b), we observe that when a TE polarization is launched into the fiber, the fiber still generates a TM component at the output, for which the transmission begins to oscillate below –10 dB. In Fig. 2.41 (c), over the range from 1250 to 1350 nm, the oscillations start to appear below –20 dB. An explanation of these oscillations is provided in Appendix Sec. 7, which describes the working principle of polarization-maintaining fibers.

In order to reliably measure the PER of the PBS devices, this needs to be smaller than that of the setup itself. Over the wavelength range of 910–980 nm, it is not possible to measure the PER of a PBS device above 10 dB without oscillations, and over the range of 1250–1350 nm, it is not possible to measure the PER of a PBS device above 20 dB without oscillations. It should be noted that this condition applies specifically to avoiding oscillations in the measurements, but more generally, if we want to measure the maximum PER of the PBS, it must remain below the maximum PER of the setup. Otherwise, the maximum PER measured during the characterization of the PBS will be limited by the setup rather than by the device itself. To overcome this limitation and enable a more precise characterization of our devices' performance, we built a fully free-space setup, removing the PM fibers and replacing them with microscope objectives. We first tested the setup itself by removing the chip and placing a polarizer at the output (Fig. 2.42 (a)). The experimental measurements for 950 nm and 1300 nm are shown in Fig. 2.42 (b) and Fig. 2.42 (c), respectively. We observe that the oscillations are no longer present and that the maximum measurable PER of the setup is at least 30 dB for 950 nm and 40 dB for 1300 nm.

A comparison of the measurements of the same device using the fiber and free-space setups is shown in Fig. 2.43 (b) at 950 nm and (c) at 1300 nm. These correspond to the device with $\Delta L_c = -5$ μm and $\delta w = +10$ nm at 950 nm, and the device with $\Delta L_c = -5$ μm and $\delta w = -10$ nm at 1300 nm, with TM polarization injected. A clear improvement is observed: the oscillations caused by the fiber have been completely eliminated, enabling the maximum device performance to be measured reliably.

Figs. 2.44 present the experimental results of the devices at 950 nm and 1300 nm, respectively, showing the normalized transmission as a function of wavelength for different width biases in both straight and tapered designs. The straight devices are shown in the top row ((a), (b), (e), (f)), and the tapered ones in the bottom row ((c), (d), (g), (h)). The legend indicates the waveguide width deviations, with circular markers representing measurements at the through port and square markers representing measurements at the cross port. We chose to present in Figs. 2.44 only the measurements of the devices exhibiting the highest PER for each waveguide width bias (δw). In the case of the straight devices operating at 950 nm, the best measured PER corresponds to $\Delta L_c = 0$ μm when $\delta w = 0$ nm, to $\Delta L_c = -5$ μm when $\delta w = -10$ nm, and again to $\Delta L_c = 0$ μm when $\delta w = +10$ nm. At 1300 nm, the best PER measurements are shown, obtained with $\delta w = 0$ nm and $\delta w = -10$ nm, both corresponding to $\Delta L_c = -5$ μm. For the tapered devices, only the con-

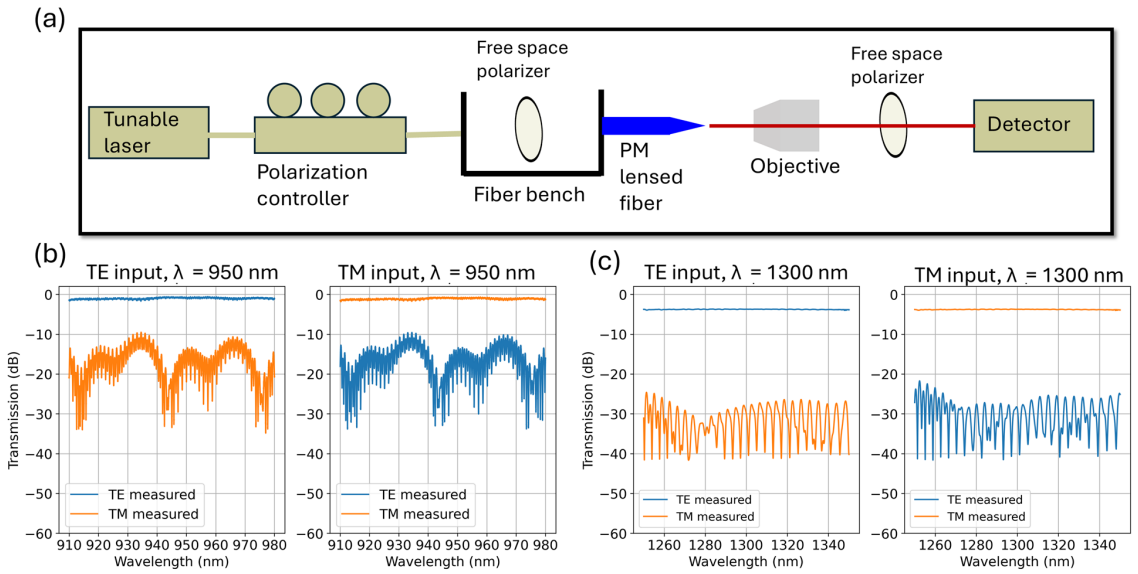

Figure 2.41: (a) Experimental setup used for the characterization of the PM fiber. (b) Measured transmission at $\lambda = 950$ nm when either TE- or TM-polarized light is injected into the fiber in setup (a). The input polarization is set by a polarizer, and the output is analyzed with a second polarizer oriented to measure either the TE or TM polarization emerging from the fiber. (c) Same measurement as in (b), but at $\lambda = 1300$ nm.

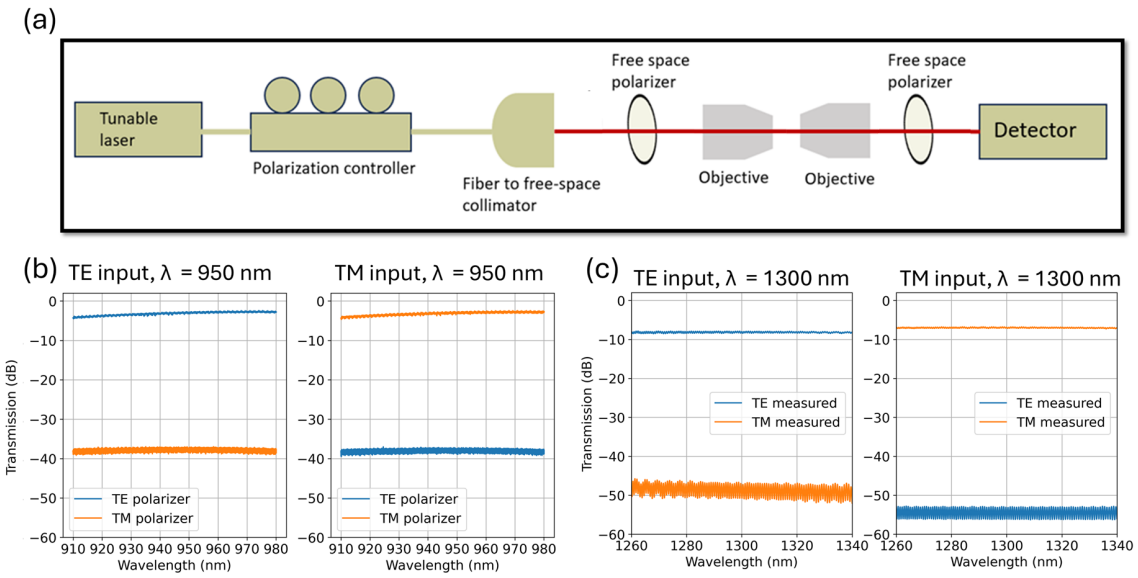

Figure 2.42: (a) Experimental free-space setup. (b) Measured transmission at $\lambda = 950$ nm when either TE- or TM-polarized light is injected and measured in setup (a). The initial polarization after the fiber to free-space collimator is set by a polarizer, and the output is analyzed with a second polarizer oriented to measure either TE or TM polarization. (c) Same measurement as in (b), but at $\lambda = 1300$ nm.

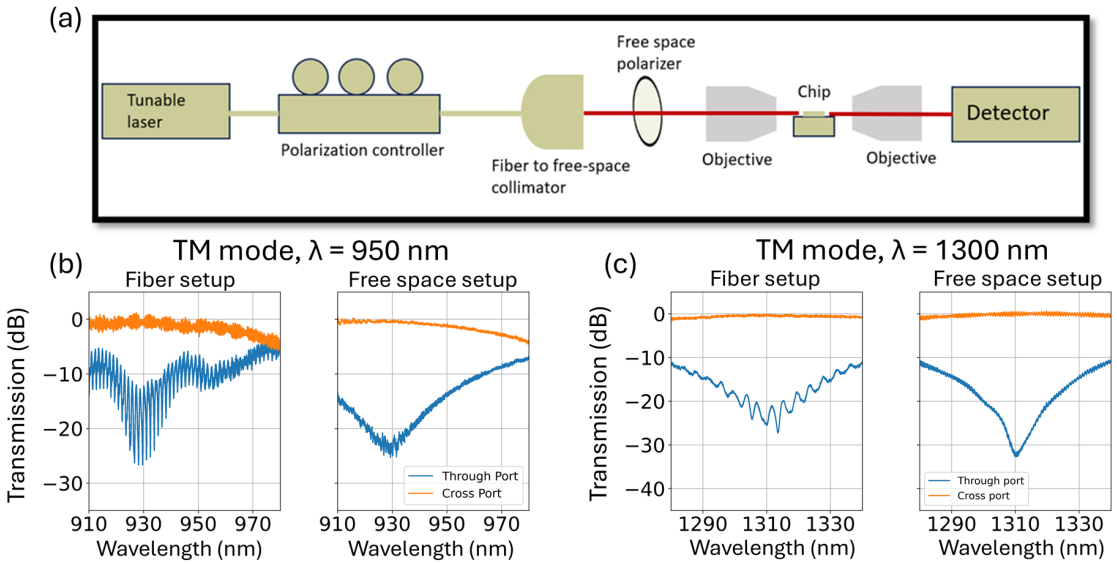

Figure 2.43: (a) Experimental free-space setup, same as in Fig. 2.42 (a), but with the chip placed between the two microscope objectives. (b) Normalized measured transmission at the cross port of the same PBS device using either the fiber setup shown in Fig. 2.41 or the free-space setup in panel (a) for $\lambda = 950$ nm, with TM mode excitation into the PBS device having $\Delta L_c = -5$ μm and $\delta w = +10$ nm. (c) Same as (b) but at $\lambda = 1300$ nm for the PBS device with $\Delta L_c = -5$ μm and $\delta w = -10$ nm.

figuration with $\Delta L_c = -5$ μm is reported across the three width biases, since this was the only version fabricated on the chip presenting the width biases.

The best-performing straight device at 950 nm is with $\Delta L_c = -5$ μm and $\delta w = +10$ nm and achieves a PER of 29 dB for TE and 24 dB for TM at 930 nm. At 1300 nm, it is with $\Delta L_c = -5$ μm and $\delta w = -10$ nm. It reaches a PER of 32 dB for both TE and TM at 1310 nm. In the tapered versions for both wavelength, performance was relatively uniform regardless of waveguide width deviation. This stability, however, comes at the cost of a reduced maximum PER. Around 950 nm, the maximum PER for TM polarization drops to 16 dB with tapering, compared to 24 dB without tapering. Similarly, at 1300 nm, tapering yields a maximum PER of 22 dB, while the straight devices reach 32 dB. On the other hand, the bandwidth is significantly larger in the tapered devices. For example, with $\delta w = 0$ nm or $\delta w = +10$ nm, the PER_{TM} remains above 10 dB throughout the entire 910 nm – 980 nm range. In comparison, the straight device with $\delta w = +10$ nm shows a PER decrease from 24 dB to below 10 dB over the much narrower 930–960 nm interval. For the straight devices with $\delta w = -10$ nm and $\delta w = 0$ nm, the PER drops from 16 dB to below 10 dB within 940–965 nm. At 1300 nm, all tapered devices maintain PER_{TM} above 10 dB across 1250–1350 nm, whereas the straight versions with $\delta w = -10$ nm and $\delta w = 0$ nm only sustain this level over 1275–1335 nm. For the TE mode, the straight devices exhibit a stable PER above 25 dB across the full wavelength ranges (910–980 nm and 1250–1350 nm), independent of width biases. In contrast, the tapered versions maintain PER above 15 dB over the same

ranges. As already discussed in the previous section (sec. 2.4.1) , this reduction arises from the fact that the taper favors phase matching for TE modes at these wavelengths.

Finally, looking at table. 2.1, which presents the performances of various SiN-based and SOI-based PBS devices reported in the literature, the device proposed here exhibits the lowest insertion loss among SiN PBS devices reported to date while maintaining state of the art performance in terms of PER and operational bandwidth. Moreover, the closest comparable work (Fig. 2.9 (a) from [51] using ADC PBS) shows higher losses, most likely due to abrupt transitions within the coupling region, which are prone to inducing additional losses. In our case, the very low insertion losses were achieved thanks to the use of smooth transitions between the input waveguide and the first coupler, as well as between the second coupler and the cross output waveguide, implemented with very gradual S-bends. This approach effectively minimized abrupt geometry changes (see the difference between the PBS geometry Fig. 2.10 in our work and Fig. 2.9 from reference [51]).

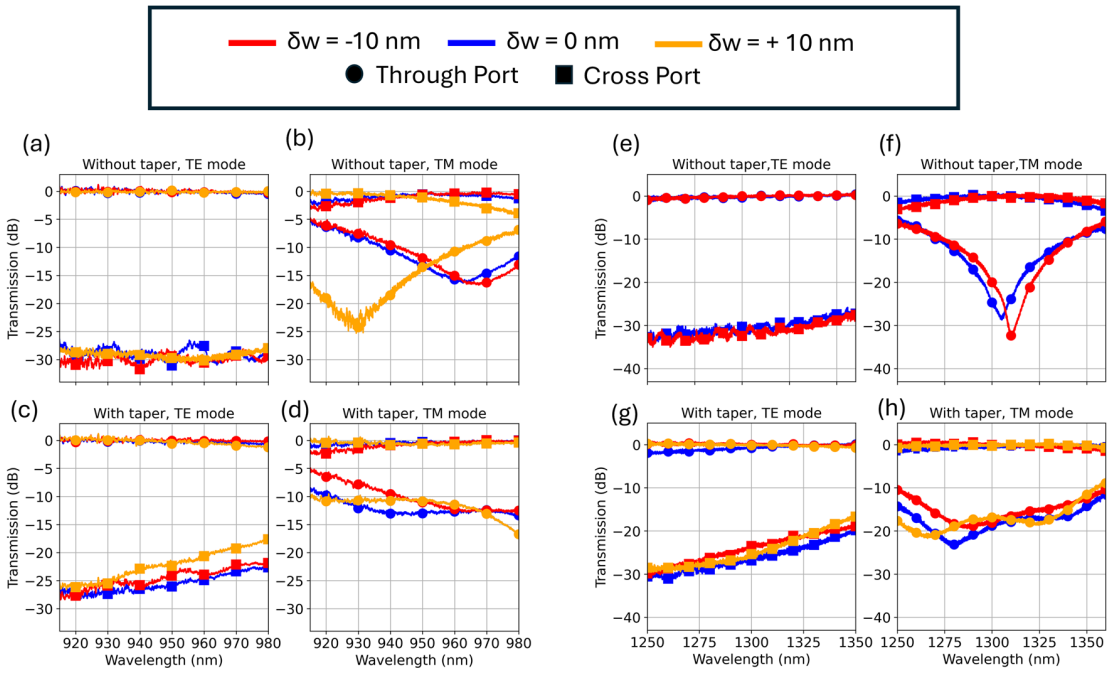

Figure 2.44: Normalized measured transmission spectra of the PBS devices using the free-space setup shown in Fig. 2.43 (a). Circle and square symbols represent measurements at the through and cross ports, respectively. Measurements were performed for different waveguide width deviations (δw). Panels (a)–(d) correspond to the measure of the device operating around $\lambda = 950$ nm, and panels (e)–(h) to those of the device operating around $\lambda = 1300$ nm. Panels (a), (b), (e), and (f) show the measurements of the straight version of the PBS, while panels (c), (d), (g), and (h) show those of the tapered version.

3 - Polarization beam splitter based on sub-wavelength metamaterials

3.1 . Theory and state of the art

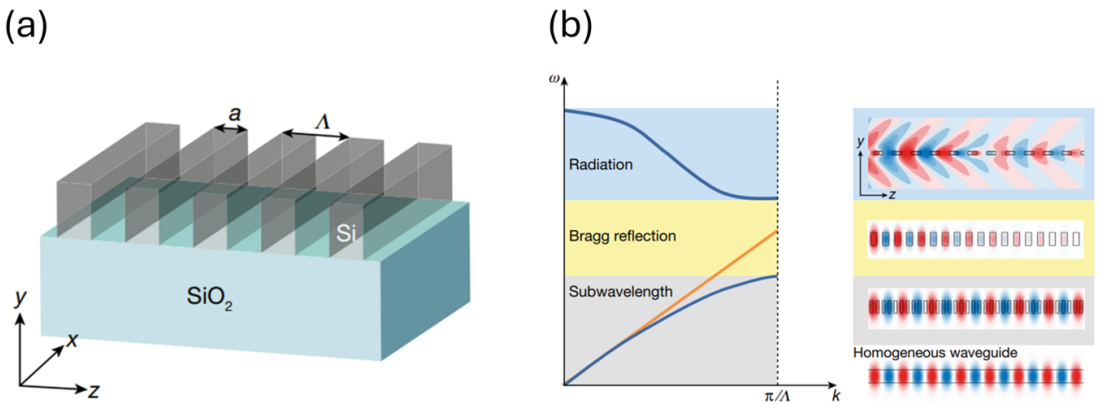

Figure 3.1: Light propagation through a periodic dieletric structure. (a) Schematic of periodically etched silicon-on-insulator structure. (b) Dispersion diagram (left) and corresponding electric field profiles (right) for a periodic waveguide along the direction of propagation (z-direction in a). The three possible regimes appear from bottom to top: subwavelength-guided wave propagation, Bragg reflection and radiation. The orange line represents the optical dispersion curve for a homogeneous medium with an equivalent core refractive index. In the right panel, the electric field's positive, negative, and zero values are shown in blue, red, and white, respectively. The black rectangles represent silicon segments. Figure and caption are adapted from [61]

Sub-wavelength (SWG) metamaterials consists of periodic or semi-periodic dielectric structures characterized by a period smaller than half of the wavelength. Under this condition, diffraction effects are suppressed and the structure behaves as a homogeneous anisotropic medium in which light propagates with an equivalent effective refractive index [61, 64, 65]. An example of a periodically etched silicon-on-insulator structure is show in Fig. 3.1 (a) from [61]. The fundamental and original theory behind SWG metamaterials is based on the stratified model developed by Rytov [61, 66]. The model consider a periodic structure of dielectric slabs with alternating refractive indices n_1 and n_2. Rytov demonstrated that if the period is smaller than half of the wavelength, the structure is equivalent to a uniaxial crystal with the following equivalent refractive indices:

$$n_{\parallel}^2 = \frac{a}{\Lambda} n_1^2 + (1 - \frac{a}{\Lambda}) n_2^2 \tag{3.1}$$

$$n_{\perp}^{-2} = \frac{a}{\Lambda} n_1^{-2} + (1 - \frac{a}{\Lambda}) n_2^{-2} \tag{3.2}$$

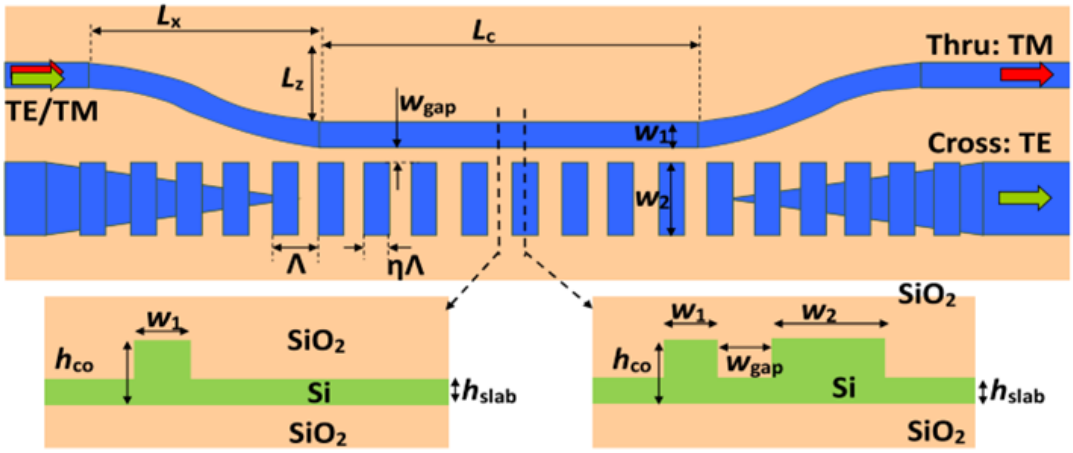

Figure 3.2: PBS based on SWG directional coupler from [62]

In such periodic structures, photonic crystal theory [61, 64] identifies three distinct regimes: the SWG regime, the Bragg regime, and the radiation regime (see Fig. 3.1 (b)). In the Bragg regime, there are no propagating modes: the guided mode entering the structure decays exponentially due to complete reflection. In the radiation regime, the structure behaves as a diffraction grating, radiating the optical power from the waveguide into free space, both upwards and downwards. In the SWG regime, the waveguide supports Bloch–Floquet modes. These three regimes are defined with respect to the Bragg limit period, i.e.,

$$\Lambda_{Bragg} = \frac{\lambda_0}{2n_{eff}} \tag{3.3}$$

where λ_0 is the operating wavelength and n_{eff} is the effective index of the guided mode. When the period of the structure is smaller than Λ_{Bragg}, the device operates in the SWG regime; when it equals Λ_{Bragg}, it operates in the Bragg regime; and when it is larger than Λ_{Bragg}, it operates in the diffraction regime.

In the SWG regime, a waveguide built using such periodic material arrangement is then optically equivalent to a waveguide based on an homogeneous material with an equivalent refractive index that can be engineered by controlling the filling factor (or duty cycle) and the period of the grating (see Eqs. 3.1–3.2). However, the accuracy of this homogenization process done using Rytov models decreases as the period of the structure approaches the Bragg limit. To address this, new models have been recently developed: the analytical laminar model and the slab model, the latter also accounting for the waveguide thickness. Both provide improved accuracy near the Bragg limit, with the slab model remaining particularly effective even close to the Bragg regime [65].

In the SOI platform, the flexibility enabled by silicon SWG structures led to breakthrough advances in many areas of silicon photonics [65], including broadband, ultra-efficient fiber-to-chip couplers [67–70], polarization independent optical antenna [71], high

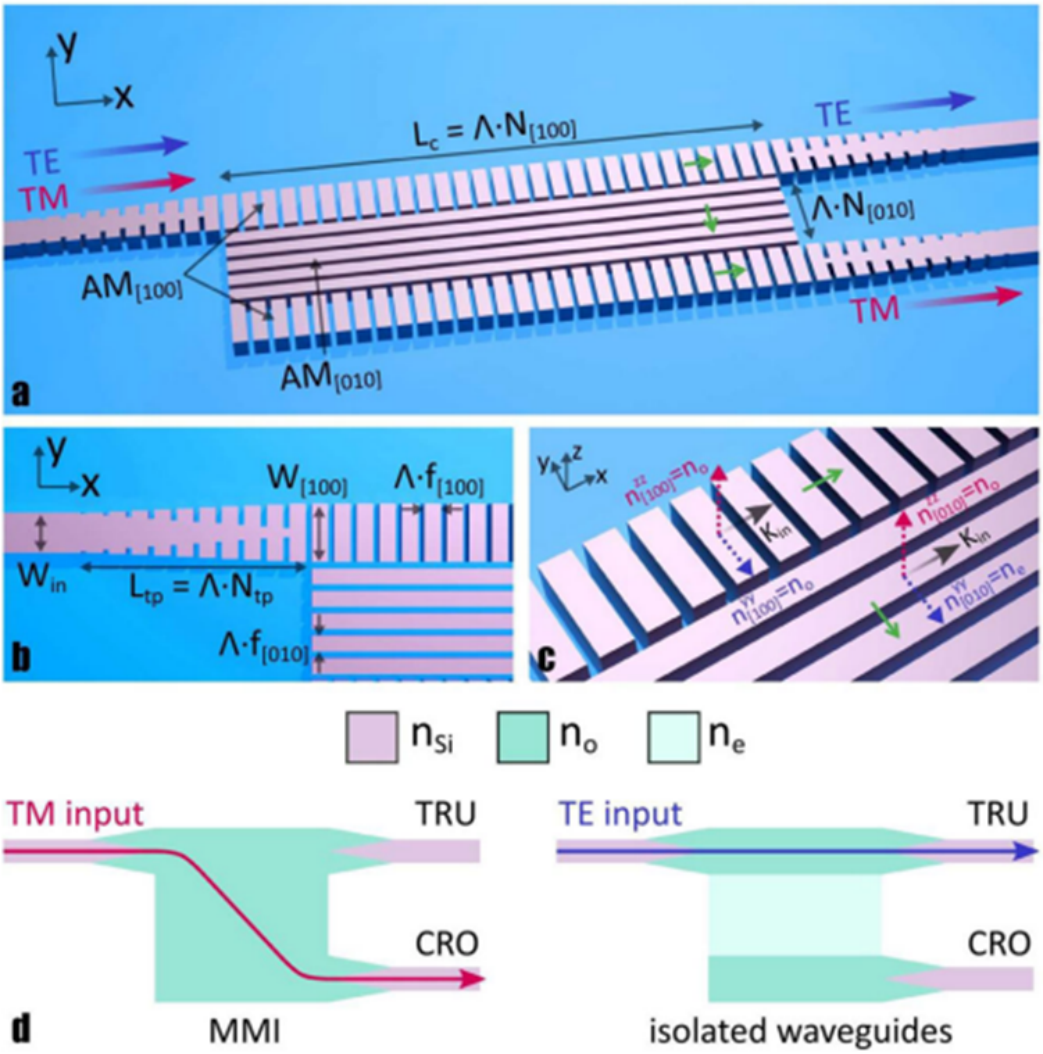

Figure 3.3: PBS based on SWG MMI from [63]

- performance integrated filters [72–74] , power and polarization splitters covering multiple optical communication bands [75–79] and evanescent field control [80]. However, in the case of the SiN platform, the use of SWG structures has only recently started to gain traction. As examples, SWG waveguides have been employed to realize high-efficiency fiber-to-chip grating couplers [81], high-performance ring resonators [82], broadband power splitters [83], and ultra-low-loss optical delay lines [84].

In the specific context of polarization control, a few notable works on the SOI platform have been demonstrated using SWG structures to improve device performance. In Fig. 3.2, the device exploits an SWG structure to separate the TE_0 and TM_0 modes in a 340-nm-thick SOI platform using a directional coupler. The TE_0 mode in the upper solid-core waveguide is directly coupled to the TE_0 mode of the SWG waveguide under phase-

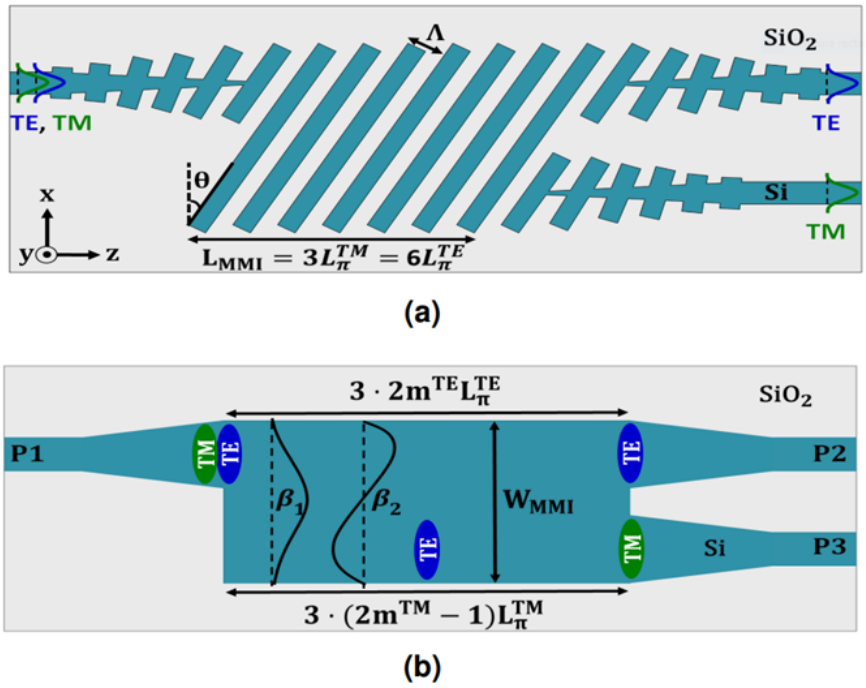

Figure 3.4: PBS based on tilted SWG MMI from [44]

matching conditions. The TM_0 mode, on the other hand, remains in the solid-core waveguide since the phase-matching condition is not satisfied for it. Compared to the PBS device based on a directional coupler shown in Fig. 2.9 (b), presented in Sec. 2.1.3 and realized in a 220-nm-thick SOI, the birefringence of a 340-nm-thick waveguide is smaller than that of a 220-nm-thick one, and therefore direct separation cannot be achieved without significant crosstalk. The use of the SWG structure here provides much stronger birefringence than a regular 340-nm-thick SOI solid waveguide, enabling high polarization selectivity and thus realizing a PBS while minimizing potential coupling crosstalk. The device exhibits a PER higher than 20 dB over the wavelength range from 1530 to 1560 nm for the TE mode, and higher than 20 dB from 1520 to 1600 nm for the TM mode. The IL varies between 0.2 and 2.6 dB over the 1520–1600 nm range for the TE mode, while for the TM mode, it varies between 0.3 and 2.8 dB. The maximum PER obtained is 30 dB and 25 dB for the TE and TM modes, respectively, at $\lambda = 1540$ nm. The device has a total length of 22 μm.

In the device shown in Fig. 2.3 (a), presented in Sec. 2.1.1, we saw that the main limitation of using MMIs as PBSs is their large required length, due to the fact that $L_\pi^{TE} \approx L_\pi^{TM}$. In [42], by employing a quasi-state imaging method, the PBS length was reduced to 1034 μm, but this came at the cost of higher losses, and the device length remains large. In the PBS shown in Fig. 3.3 from [63], the use of SWGs addresses this issue by leveraging the fact that in anisotropic materials, the behavior of light waves depends on the orientation of

the electric field components with respect to the optical axis of the material (in Fig. 3.3, the optical axis is indicated by the green arrow in panel (c)). By changing the orientation of the anisotropic SWG structure, it is therefore possible to control light propagation. University of Málaga [65] showed that the fundamental TE mode is strongly affected by changes in the orientation with respect to the optical axis, while the TM mode is virtually unaffected. Thus, when the TE mode is excited in the input waveguide of Fig. 3.3, the difference in the orientation of the structure causes the light to behave as if it were propagating in an isolated waveguide. In contrast, when the TM mode is excited, it is not affected by the orientation difference, and the structure acts as an MMI exclusively for it. This makes it possible to separate the TM mode without affecting the TE mode, achieving a compact MMI footprint since only one mode needs to be separated. The device exhibits a PER higher than 20 dB over the wavelength range from 1460 to 1670 nm for both TE and TM modes, with IL below 1 dB across the same range. The total device length is only 12.25 μm, compared to the 1034 μm of the device in Fig. 2.3 (a).

In Fig. 3.4 from [44], another approach is presented to address the MMI length limitation. As discussed earlier, one of the distinctive features of SWG structures is that their orientation has very little effect on the TM mode but strongly affects the TE mode. By tilting the SWG, the equivalent refractive index experienced by the TE mode is substantially modified, while the TM mode remains largely unaffected. The idea is therefore to tilt the SWG in order to reduce the effective index of the TE mode while keeping that of the TM mode largely unchanged, thereby increasing the difference between L_π^{TE} and L_π^{TM}. This enables polarization separation to be achieved with small integer values of m^{TE} and m^{TM}. Simulation results indicate that the device exhibits a PER above 20 dB over the wavelength range of 1500–1700 nm for the TE mode, and above 20 dB over 1500–1640 nm for the TM mode. The IL remains below 1 dB over 1520–1670 nm for both modes. A maximum PER of 47 dB for TE and 38 dB for TM is obtained at λ = 1600 nm and λ = 1560 nm, respectively. This is achieved with a total length of only 92 μm, compared once again to the 1034 μm required by the design in Fig. 2.3 (a).

Regarding the SiN platform, very little work has been carried out so far to explore the use of SWG structures in polarization control devices. This is the topic of the rest of this chapter which presents the design and experimental demonstration of a PBS based on a fully etched SWG waveguide.

3.2 . Subwavelength asymmetric directional coupler

The schematic of the PBS we demonstrated is shown in Fig. 3.5 (a) top view and (b) cross-sectional view. The main difference from the previously designed structure in Fig. 2.10 is that we have replaced the central waveguide with an SWG structure. This has two main advantages. The first one is that it allow reducing the device length by up to a factor of two. Indeed, in the case of an SWG structure, it is possible to directly couple the TE_0 mode of the conventional waveguide to the TE_0 mode of the SWG waveguide, while the TM_0 mode remains uncoupled. This was discussed in the presentation of the design in

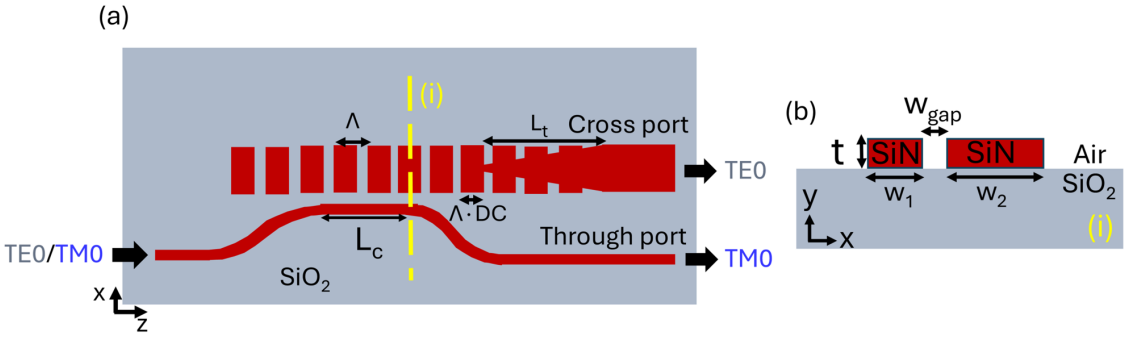

Figure 3.5: Schematic of our device, consisting of a PBS based on an SWG directional coupler. (a) Top view in the x–z plane. (b) Cross-sectional view in the x–y plane.

Fig. 3.2, in Section 3.1. For the SiN platform, this represents a significant advantage, since it enables the separation of TE and TM modes without the need for an intermediate higher-order mode. The second advantage arises from the fact that, unlike most devices reported in the literature, we chose to design this one without an upper cladding. Working without cladding is required for certain applications, such as the deposition of quantum dots [23] on silicon nitride waveguides, as discussed in Sec. 1. Removing the cladding, however, also has the effect of making the TM modes extremely weakly guided in SWG waveguides which can be used to improve the device performance as discussed below.

The operating principle of the device is as follows: the TE_0 mode of the input waveguide is coupled to the TE_0 mode of the SWG waveguide under the phase-matching condition $n_{eff}^{TE_0}(w_1) = n_{eff_{SWG}}^{TE_0}(w_2)$. The SWG waveguide is then converted into a conventional waveguide using an inverse taper. At the same time, the SWG waveguide does not support a guided TM_0 mode, so the mode remains in the input waveguide and exits through the through port. Any spurious TM coupling is also quickly leaked out of the SWG waveguide, improving the device PER.

This device was fabricated in the C2N cleanroom on the SiN platform with a 400 nm core thickness and a 3 μm SiO_2 buried oxide. The refractive index of SiN is 2.03 at $\lambda = 950$ nm and 2.02 at $\lambda = 1300$ nm. To design the PBS, certain parameters of the SWG waveguide are constrained by both physics and fabrication. First, the period must be smaller than the Bragg period in order to operate in the SWG regime, i.e., $\Lambda < \Lambda_{Bragg} = \lambda_0/2n_{eff}$, where n_{eff} is the effective index of the 3D Bloch–Floquet mode of the SWG structure. This condition imposes fabrication limits. Since $n_{eff} < n_{SiN}$, for $\lambda = 950$ nm this results in $\Lambda < 950/4 \approx 240$ nm. This value is slightly overestimated because we assumed the effective index to be equal to $n_{SiN} \approx 2$, the index of the core material. Setting the period to $\Lambda = 240$ nm and considering a minimum fabricable feature size of about 120 nm, which is close to the limit tested in the cleanroom for 400-nm-thick SiN devices, fixes the duty cycle to 50% for $\lambda = 950$ nm. Applying the same procedure for $\lambda = 1300$ nm, we obtain $\Lambda < 1300/4 = 325$ nm. In this case, we choose $\Lambda = 300$ nm to ensure that the SWG mode propagates. The duty cycle can then be selected between 40% and 60%. We opted for a duty cycle of 60% which

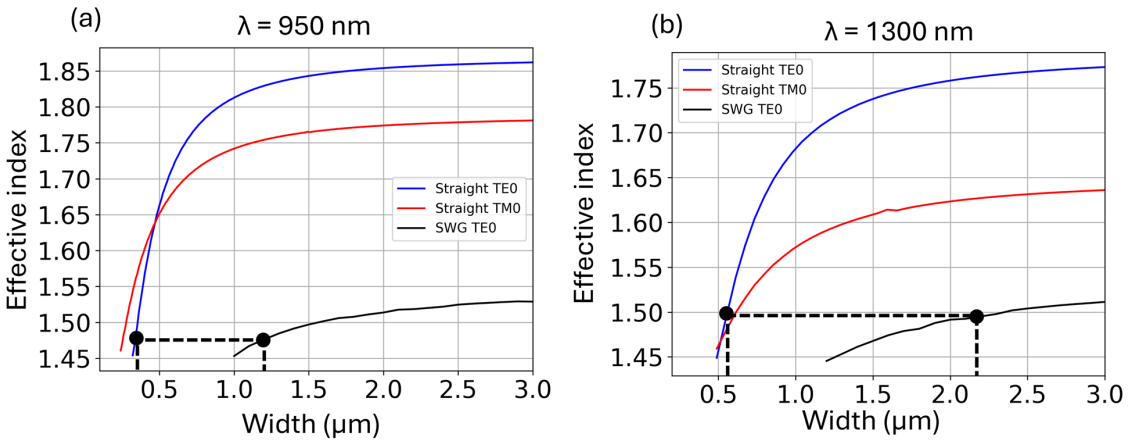

Figure 3.6: Effective indices of the TE_0 and TM_0 modes for a isolated solid core waveguide as a function of waveguide width, represented by blue and red lines, respectively. The black line shows the effective index of the TE_0 mode of a isolated SWG waveguide as a function of waveguide width, calculated using the MPB library. The dashed lines indicate the selected widths where the phase-matching condition is fulfilled. (a) at λ = 950 nm, calculated using the SWG parameters shown in Fig. 3.2 (a) with Λ = 240 nm and DC = 50%. (b) at λ = 1300 nm, calculated with Λ = 300 nm and DC = 60%.

provides a larger effective index for a given waveguide width. This ensures a sufficiently high effective index to facilitate mode conversion during the transition between the solid waveguide and the SWG waveguide.

Once the values of period and duty cycle for the SWG waveguide have been chosen, we need to calculate the effective index of the TE_0 modes of the waveguide. For this, we cannot use the previously employed finite-difference eigenmode solver, since it is a 2D solver and SWG waveguide modes are 3D Bloch–Floquet modes. Instead, we use the MPB library from MEEP [85], which is a frequency domains eigenmode solvers of Maxwell's equations for periodic dielectric structures. Figure 3.6 shows the effective indices of the TE_0 and TM_0 modes in the solid core, as well as the TE_0 mode in SWG waveguides, for (a) λ = 950 nm and (b) λ = 1300 nm. To select the appropriate widths (w_1 and w_2 in Fig. 3.5 (b)), we impose the following conditions: the TE_0 and TM_0 modes of the solid core waveguide must propagate, and the SWG waveguide width must not be too large to avoid strong mode confinement and weak coupling. For λ = 950 nm, we choose w_1 = 0.34 μm and w_2 = 1.2 μm corresponding to $n_{eff}^{TE_0}$ = 1.48. For λ = 1300 nm, we choose w_1 = 0.55 μm and w_2 = 2.2 μm corresponding to $n_{eff}^{TE_0}$ = 1.5.

Similarly to the PBS design presented in Sec. 2.2.2, we use an EME solver to calculate the conversion from the TE_0 mode of the solid-core waveguide to the TE_0 mode of the SWG waveguide, and to determine the optimal coupling length and gap. Since the eigenmode solver requires 2D modes, the SWG geometry is approximated as an equivalent homogeneous material with an equivalent refractive index, as discussed in Sec. 3.1. The geometry used in the EME simulations is shown in Fig. 3.7: (a) top view and (b) cross-

sectional view. It is important to note that approximating the SWG waveguide with an equivalent homogeneous material assumes an isolated SWG waveguide which is not the case here. The coupling lengths obtained in this way are hence approximate due to the underlying assumptions. These values will serve primarily as an initial estimate and will later be refined using a few 3D FDTD simulations.

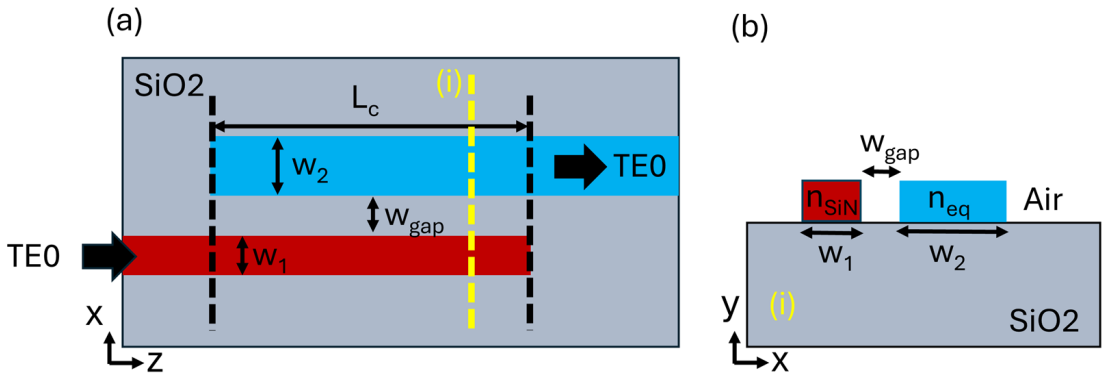

Figure 3.7: Schematic of a simplified version of the PBS based on an SWG directional coupler, optimized using EME simulations by modeling the SWG geometry as a homogeneous material with an equivalent refractive index (n_{eq}). (a) Top view. (b) Cross-sectional view at position (i), indicated by the yellow dashed line in (a).

In Fig.3.8 (a), we show the cross-sectional view of the isolated SWG geometry approximated as an equivalent homogeneous material. The waveguide width w_2 has already been fixed in order to satisfy the phase matching condition, namely $w_2 = 1.2\ \mu$m for 950 nm and $w_2 = 2.2\ \mu$m for 1300 nm. Fig. 3.8 (b) and (c) show the calculated effective index of the TE_0 mode as a function of the equivalent material index for $\lambda = 950$ nm and $\lambda = 1300$ nm, respectively. From these graphs, we extract the equivalent material indices that yield the effective index values (i.e., $n_{eff}^{TE_0} = 1.48$ at 950 nm and $n_{eff}^{TE_0} = 1.50$ at 1300 nm), which satisfy the phase-matching condition: $n_{eq} = 1.64$ and $n_{eq} = 1.705$ for 950 nm and 1300 nm, respectively. These values, together with the previously determined w_1 and w_2 for 950 nm and 1300 nm, are then used in the EME simulations. In this work, we fix the gap at 400 nm, although it could be further reduced to make the device even more compact.

The results of the simulations are presented in Fig. 3.9 which reports on the conversion from TE_0 in the solid waveguide to TE_0 in the SWG waveguide as a function of coupling length for both $\lambda = 950$ nm and $\lambda = 1300$ nm. We obtain $L_c = 70\ \mu$m and $L_c = 105\ \mu$m for the two wavelengths, respectively.

We then take the optimized device parameters for the $\lambda = 950$ nm device and run a 3D FDTD simulation, whose results are reported in Fig. 3.10. In this simulation, the TE_0 mode is launched into the top waveguide (see Fig. 3.10 (a)), which is adiabatically converted into an SWG waveguide using an inverse taper. The bottom waveguide of width w_1 approaches progressively via an S-bend, and the TE_0 mode couples from the SWG waveguide to the solid waveguide. The solid bottom waveguide is then separated from the SWG waveguide via another S-bend. The main difference compared to Fig. 3.5 is that the

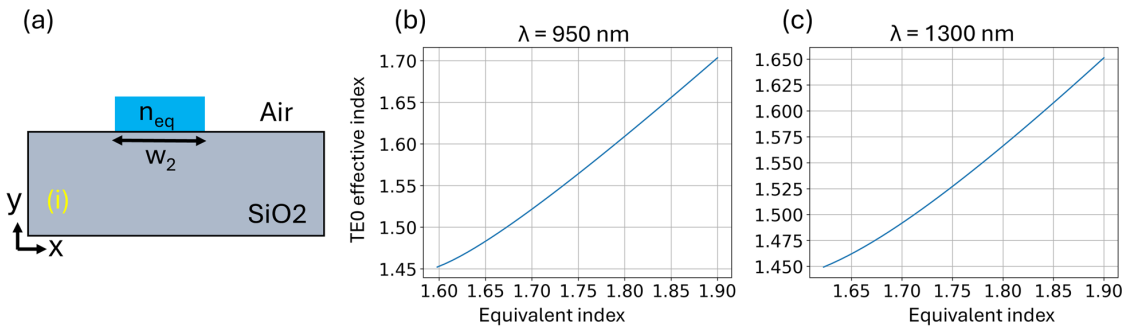

Figure 3.8: (a) Cross-sectional view of an isolated SWG waveguide modeled as a homogeneous material. (b–c) Effective index of the TE_0 mode of the structure shown in (a) as a function of the equivalent material index. The waveguide width is fixed: (b) $w_2 = 1.2$ μm for $\lambda = 950$ nm, and (c) $w_2 = 2.2$ μm for $\lambda = 1300$ nm.

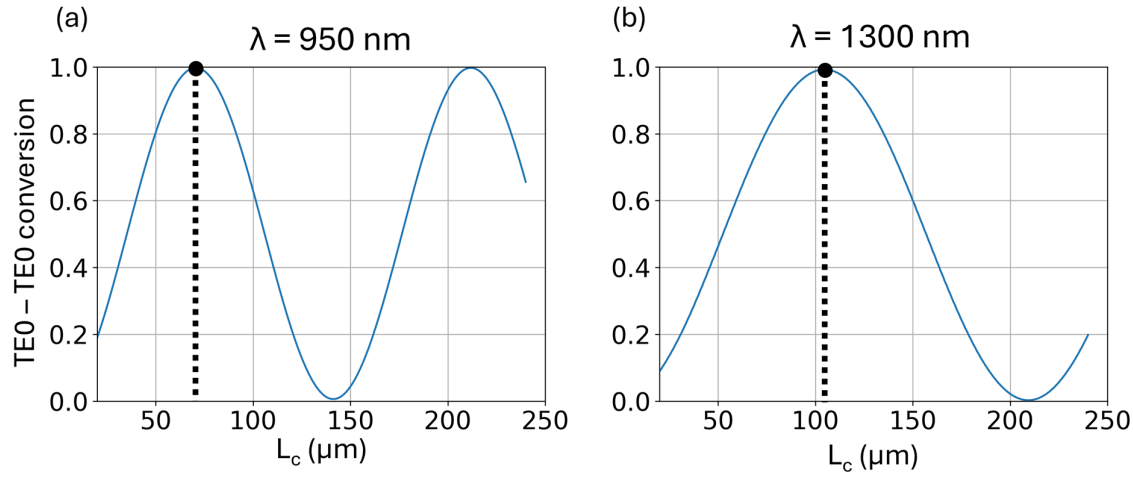

Figure 3.9: TE_0–TE_0 coupling from the solid core waveguide to the SWG waveguide as a function of the coupling length. The simulation was performed using EME with the SWG geometry approximated as a homogeneous material having an equivalent refractive index as illustrated in Fig. 3.7. (a) At $\lambda = 950$ nm. (b) At $\lambda = 1300$ nm.

input source is now placed in the top waveguide with the SWG transition added. However, this is irrelevant since the device operates symmetrically: if TE_0 is injected into the bottom waveguide, it exits from the top waveguide, and vice versa. The simulation was set up in this configuration to avoid including the entire S-bends, thereby drastically reducing the simulation time. Fig. 3.10 (b) shows the electric field distribution obtained when launching the TE_0 mode into the top waveguide, using the optimized parameters extracted from the EME analysis. If the device operates as intended, light should be fully transferred from the SWG waveguide to the solid bottom waveguide, with only negligible power measured at the top output waveguide, where the power monitor is located in Fig. 3.10 (a). The field distribution was calculated over the wavelength range 910–980 nm. However, only the

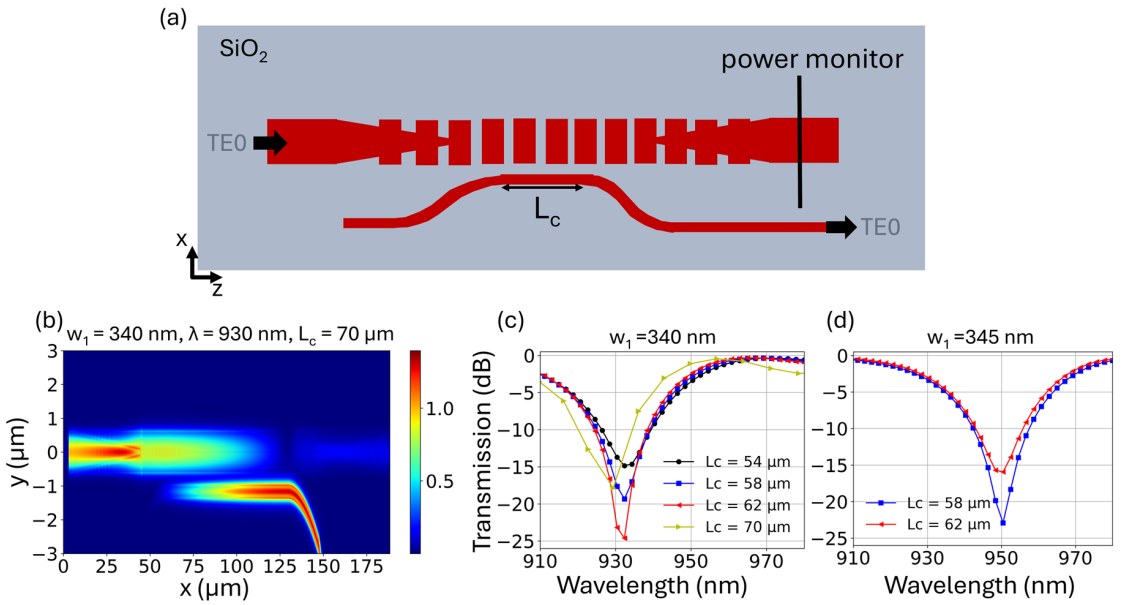

Figure 3.10: (a) Schematic of the PBS based on an SWG directional coupler, with the TE_0 mode excited in the top waveguide instead of the bottom one, as previously shown in Fig. 3.5. (b) Electric-field amplitude calculated using 3D FDTD simulations for the structure schematically represented in panel (a). The field distribution is shown for λ = 930 nm, with w_1 = 340 nm and L_c = 70 μm. (c) Simulated transmission with the fundamental TE mode excitation measured at the power monitor indicated in panel (a), calculated for different coupling lengths at w_1 = 340 nm. (d) Same as in (c), but with w_1 = 345 nm.

result at 930 nm is shown, as this wavelength yields the maximum field amplitude in the solid bottom waveguide, measured at x = 128 μm (dark red). At the same position, the top waveguide exhibits a minimum field amplitude (dark blue). Beyond x = 128 μm, the field amplitude in the top waveguide begins to increase again, as indicated by the progressive brightening of the blue color. This resurgence is attributed to over-coupling induced by the presence of the S-bends.

To determine the optimal design, the transmission at the power monitor in Fig. 3.10 (a) was calculated for different coupling lengths, starting from 54 μm and progressively increasing the length. The results is shown in Fig. 3.10 (c) and indicate that the transmission decreases gradually, reaching a value as low as –25 dB at L_c = 62 μm. However, the operating wavelength is blue-shifted by about 20 nm compared to the 950 nm target. To correct this, the width of the bottom solid waveguide was slightly increased to w_1 = 345 nm while keeping the coupling length fixed at 62 μm. The resulting transmission spectrum is shown in Fig. 3.10 (d). Increasing the bottom waveguide width by 5 nm shifts the operating wavelength back to 950 nm but reduces the conversion efficiency (compare Fig. 3.10 (c) and (d) red curve). Subsequently, reducing the coupling length to L_c = 58 μm lowered the transmission, leading to the final device parameters of w_1 = 345 nm and L_c = 58 μm.

For λ = 1300 nm, we used the same approach, relying on 3D FDTD simulations to fine-

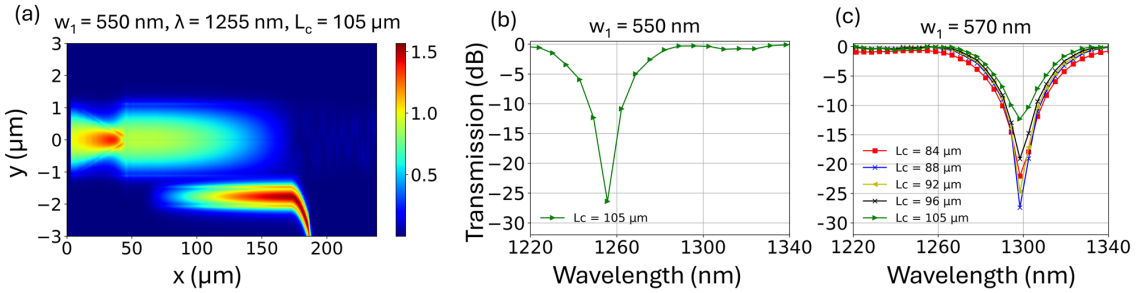

Figure 3.11: (a) Electric-field amplitude calculated using 3D FDTD simulations for the structure schematically represented in Fig. 3.10 (a). The field distribution is shown for $\lambda = 1255$ nm, with $w_1 = 550$ nm and $L_c = 105$ μm. (b) Simulated transmission with fundamental TE-mode excitation, measured at the power monitor indicated in Fig. 3.10 (a), calculated for $L_c = 105$ μm and $w_1 = 550$ nm. (c) Same as in (b), but for different coupling lengths and with $w_1 = 570$ nm.

tune the PBS parameters. Fig. 3.11 (a) shows the electric field distribution obtained when launching the TE_0 mode into the top waveguide, as before. However, in this case, the EME-optimized parameters result in an almost complete coupling at a wavelength of 1255 nm. This is evident when looking at positions beyond $x = 175$ μm in Fig. 3.11 (a), where most of the field amplitude resides in the solid bottom waveguide (dark red), while the amplitude in the top waveguide remains weak, as indicated by the limited variation in the dark blue intensity once the bottom waveguide has been separated by the S-bend. This observation is further confirmed by the transmission measured at the power monitor in Fig. 3.11 (b), where a transmission of –26 dB is obtained. Since the device is already optimal at 1255 nm, the coupling length does not require further adjustment. Instead, only the bottom waveguide width is increased to 570 nm to red-shift the operating wavelength to 1300 nm. The coupling length is then fine-tuned following the same procedure as for the 950 nm device: the transmission is computed while progressively increasing the coupling length, starting from $L_c = 84$ μm, as shown in Fig. 3.11 (c). The minimum transmission was thus obtained for $w_1 = 570$ nm and $L_c = 88$ μm.

Once the PBS design parameters have been selected, the final step is to design the transition from the solid waveguide to the SWG waveguide. This is achieved using a taper that gradually narrows the solid-core waveguide while progressively introducing the SWG geometry, thereby ensuring a smooth transition. To determine the optimal transition length (L_t), we performed 3D FDTD simulations using the structure illustrated in Fig. 3.12 (a). In this configuration, the TE_0 mode of the solid waveguide is converted into the TE_0 mode of the SWG, propagates through the SWG waveguide, and is then converted back into the TE_0 mode of a solid waveguide using another transition. The TM_0 mode, on the other hand, is filtered out by the transition, as the SWG structure without cladding does not support a propagating TM_0 mode. Figures 3.12 (b)-(c) show the transmission at the power monitor as a function of wavelength for different transition lengths, with TE or TM mode excitation: (b) for $\lambda = 950$ nm and (c) for $\lambda = 1300$ nm. At $\lambda = 950$ nm and $\lambda = 1300$ nm , shorter transition lengths cause slightly higher losses, but these remain limited (<1 dB).

For the first fabrication run, we chose $L_t = 40\ \mu$m, although as shown in the simulations, $L_t = 10\ \mu$m would work equally well. For $L_t = 40\ \mu$m, the expected losses at 950 nm are 0.25 dB and 10 dB for the TE and TM modes, respectively. At 1300 nm, the expected losses are 0.1 dB for TE and 8 dB for TM modes.

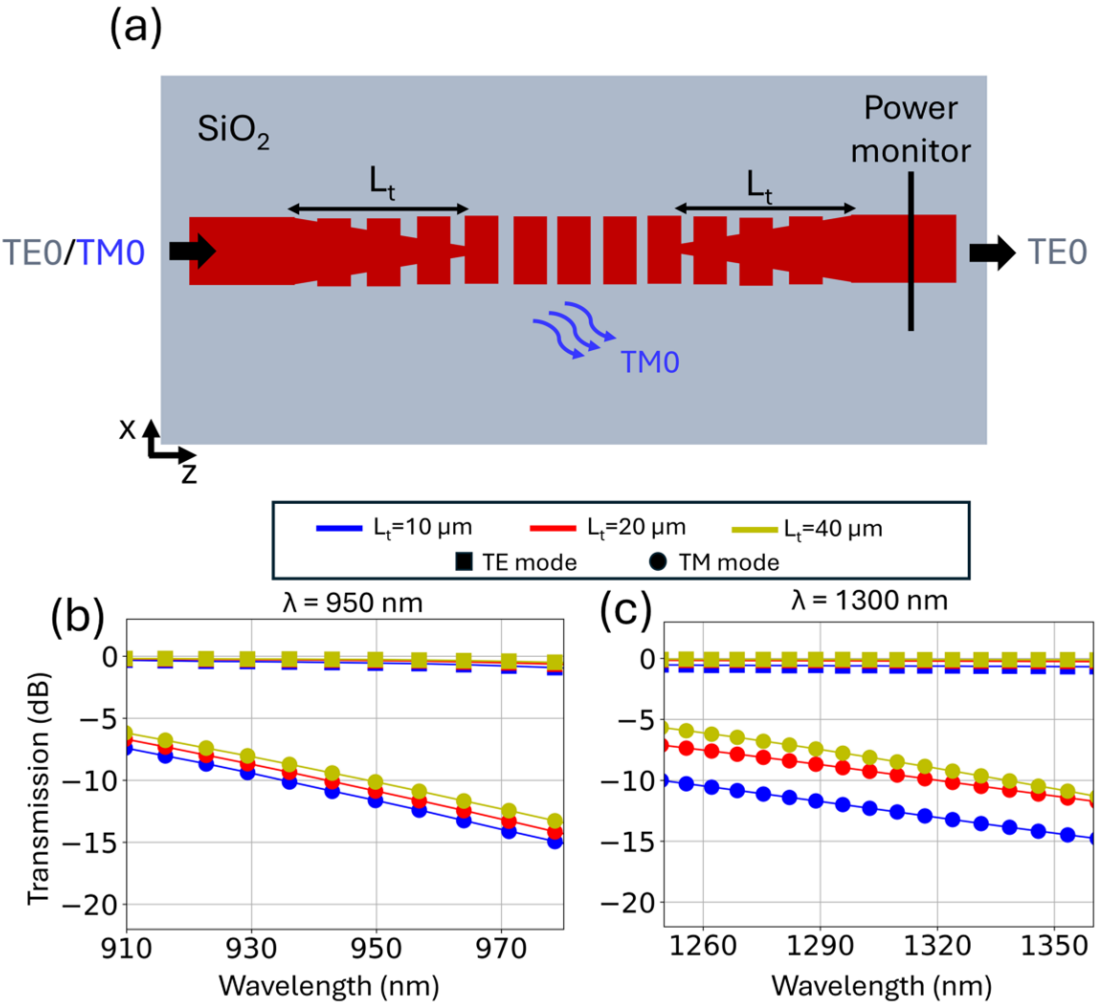

Figure 3.12: (a) Schematic of the solid to SWG waveguide transition. (b–c) Transmission calculated at the power monitor indicated in panel (a) using 3D FDTD simulations, as a function of wavelength for different transition lengths (L_t). Square markers represent TE mode excitation, while circular markers correspond to TM mode excitation. (b) Results for $\lambda = 950$ nm. (c) Results for $\lambda = 1300$ nm.

3.3 . Fabrication and experimental results

Two chips were fabricated separately: one containing devices operating at 950 nm and the other containing devices operating at 1300 nm. For both sets of devices, the experimental setup shown in Fig. 2.42, discussed in Sec. 2.4.2, was used. In Figure 3.13 optical (a) and electron microscope images (b) of one of the 1300 nm PBS devices is showed, corresponding to the nominal design. In Fig. 3.13 (c), which provides a zoom of the coupling region, the SWG geometry can be observed. It is already noticeable that the width of the upper waveguide deviates significantly from the optimized design, with a measured value of about 602 nm compared to the 570 nm target.

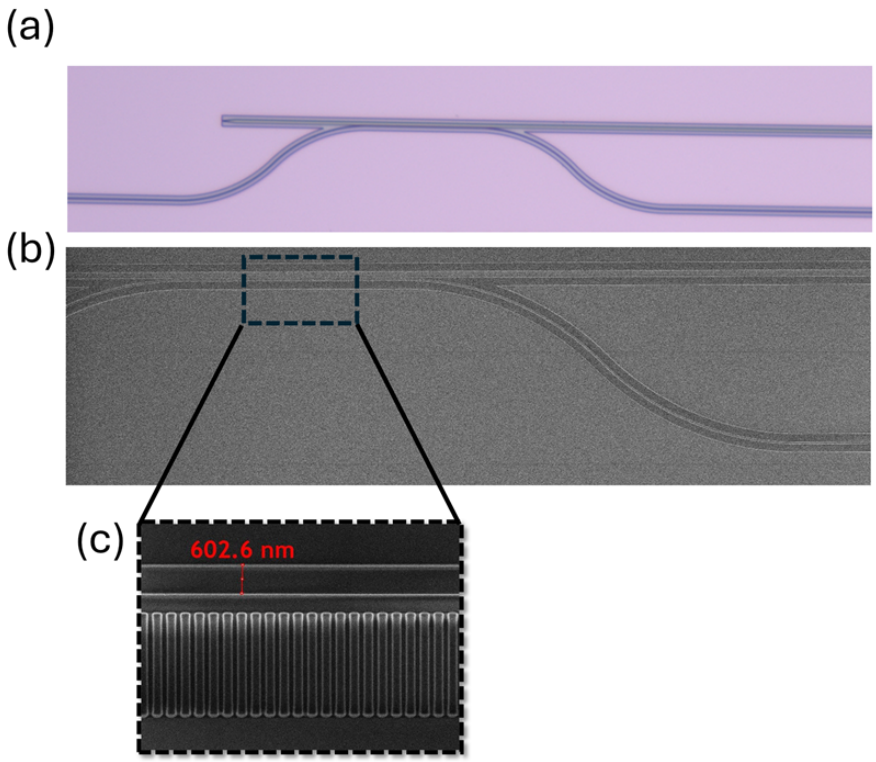

Figure 3.13: Image of the fabricated nominal SWG PBS designed for operation around 1300 nm. (a) Optical microscope image and (b) SEM image of the fabricated PBS. (c) SEM image zoom on the SWG region.

To characterize the losses associated with the transition from solid core to SWG waveguide, reference waveguides with and without SWG sections were included. The results are shown in Fig. 3.14 and represent the transmission of the SWG reference normalized to the reference without SWG, for both TE and TM excitation. As predicted by simulation, the TE_0 mode propagates with very little loss of less than 0.5 dB at 950 nm and 1300 nm while the TM_0 mode is filtered out by the SWG transition (see Fig. 3.14), with losses of 12.5 dB at 950 nm and 11 dB at 1300 nm. The SWG structure therefore acts as a TM filter, allowing only the TE mode to propagate. It is important to note that these results were obtained

for a specific SWG waveguide length; increasing the propagation length would cause the TM mode to leak out even further.

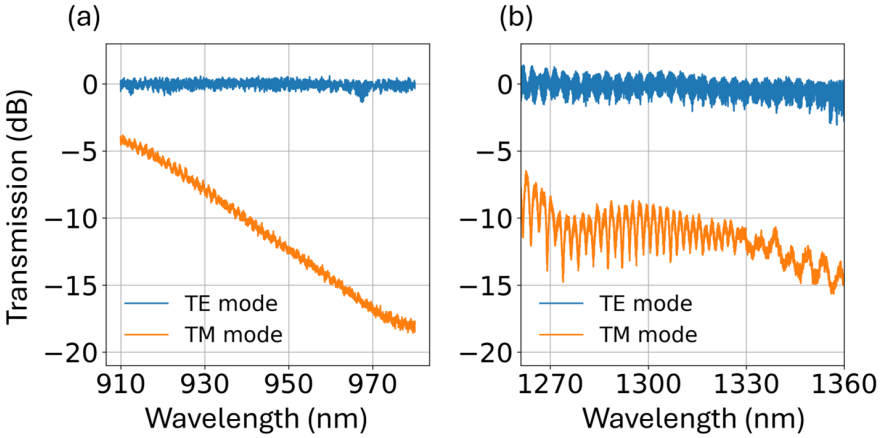

Figure 3.14: Measured transmission of the SWG reference waveguide, normalized to the transmission of the reference waveguide without SWG. (a) At λ = 950 nm. (b) At λ = 1300 nm.

Figure 3.15 shows the measured performance of the fabricated PBS devices at 950 nm. The plots show the transmissions at the through and cross ports (Fig. 3.15) when either TE or TM is excited. In the case where polarization separation occurs, TE should be measured at the cross port and TM at the through port. From left to right, the coupling length was varied from -5 μm to +5 μm. This range is narrower than that used for regular PBSs (Sec. 2) due to the optimization process: previously, for the structure of Fig. 2.10, optimization relied solely on EME simulations, which do not handle steep angles well. Therefore, the S-bends were removed from the simulation. With 3D FDTD, removing the S-bends was not necessary and they were included in the simulations. We hence chose to vary only the coupling length. From the experimental results, we observe that for all

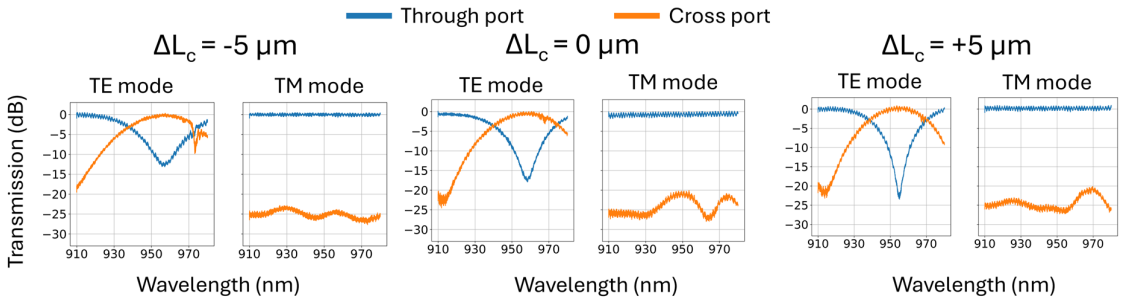

Figure 3.15: Normalized measured transmission of the SWG PBS at the through and cross ports as a function of wavelength for devices operating around a wavelength of 950 nm, with either the TE or TM mode excited at the input. The panels show measurements for coupling length deviations from -5 μm to -5 μm. Normalization is done using the reference waveguide, with respect to the polarization.

three coupling lengths, polarization separation occurs with good performance. Among

the three devices, the best one appears to be the one with a coupling length increased by 5 µm. This discrepancy with simulations is very likely due to a slight fabrication error on the waveguide widths, as can be observed from the fact that the operating wavelength is slightly red-shifted of about 5 nm. A slight re-optimization of the coupling length thus allowed reaching the maximum performance.

For the best device, we obtained a maximum PER of 24 dB for TE and 26 dB for TM at λ = 955 nm. The insertion losses are below 1 dB for both TE and TM modes. For the first device we fabricated, the cascaded ADC PBS presented in Sec. 2.2, we measured insertion losses below 1 dB and maximum PER values of 29 dB and 24 dB for TE and TM, respectively, at λ = 930 nm . This device had a total length of 110 μm. Given that the device length of the SWG PBS is 63 μm, the use of an SWG waveguide enabled an approximately twofold reduction in length while maintaining comparable performance in terms of PER and insertion loss

Figure 3.16 shows the performance measurements for the device operating around a wavelength of 1300 nm. Also in this case, from left to right, the coupling length was varied from -5 μm to +5 μm, with TE and TM injected successively. For two out of three devices, we observe that TE exits at the cross port and TM remains at the through port. However, the operating wavelength is red-shifted by about 50 nm. This red shift can be explained from the SEM image Fig. 3.13, where as we already discussed the waveguide width deviate from the optimal value of about 30 nm. Nevertheless, despite this fabrication error, the PBS operates well at a wavelength of 1355 nm, yielding, for ΔL_c = 0 μm, a PER of 20 dB for TE and 30 dB for TM at 1355 nm. To benchmark this device, we compare it with our previous design the ADC based PBS presented in Sec. 2.2. That device exhibited insertion losses below 1 dB and a maximum PER of 32 dB for both TE and TM at λ = 1310 nm, for a total device length of 124 μm. In comparison, the SWG PBS achieves similar insertion losses. For TM, the PER remains comparable to the ADC PBS, whereas for TE it is slightly degraded due to the fabrication imperfections discussed above. Nevertheless, the SWG design has a clear advantage in terms of footprint: its total length is 88 μm, corresponding to a reduction of about 30 %.

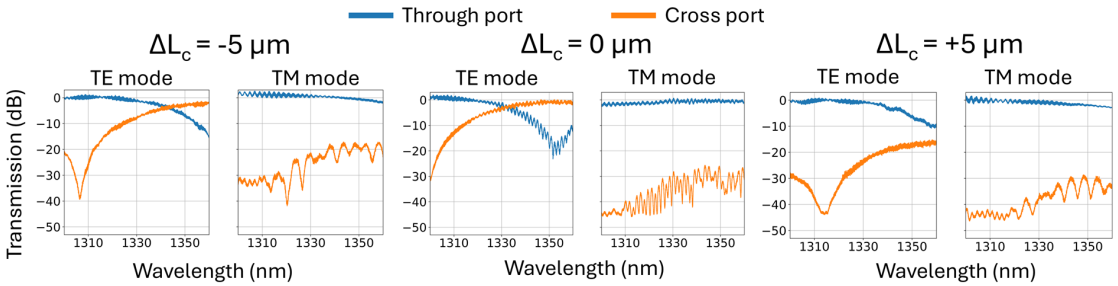

Figure 3.16: Normalized measured transmission of the SWG PBS devices at the through and cross ports as a function of wavelength for devices operating around 1300 nm, with either the TE or TM mode excited at the input. The panel show measurements for coupling length deviations from -5 to -5 μm. Normalization is done using the reference waveguide, with respect to the polarization.

Comparing our devices with those listed in Table 2.1, the fabricated SWG PBSs have

lengths of 63 μm and 88 μm for operation at 950 nm and 1300 nm, respectively. The smallest device reported in the table 2.1 is the angled MMI with a length of 80 μm [56], which is still longer than our 950 nm device and comparable to our 1300 nm device, with relatively similar maximum PER performance. Moreover, the device could be made even more compact by reducing the gap, which would further reduce the coupling length. As shown in Fig. 3.12, the SWG transition could also be made more compact while maintaining low losses.

4 - Polarization splitter-rotator

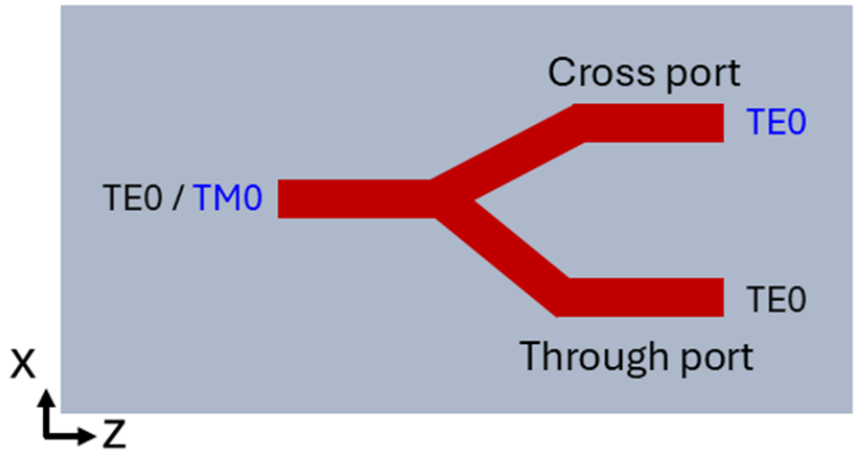

Figure 4.1: Schematic illustrating the operating principle of a polarization splitter and rotator

Another key function in polarization control is the ability to separate two orthogonally polarized modes while at the same time rotate the polarization of one of the two, for example by converting a TM polarization into a TE polarization and routing it to a different port as illustrated in Fig. 4.1. To evaluate the performance of a such structure, the commonly used metrics are the polarization conversion loss (PCL), the insertion loss (IL), and the polarization extinction ratio (PER), defined as follows:

$$PER_{TE} = 10 \cdot \log_{10} \frac{T_{through}^{TE-TE}}{T_{cross}^{TE-TE}}, \tag{4.1}$$

$$PER_{TM} = 10 \cdot \log_{10} \frac{T_{cross}^{TM-TE}}{T_{through}^{TM-TM}}, \tag{4.2}$$

$$IL = -10 \cdot \log_{10} T_{through}^{TE-TE}, \tag{4.3}$$

$$PCL = -10 \cdot \log_{10} T_{cross}^{TM-TE}. \tag{4.4}$$

Here, $T_{through}^{TE-TE}$ and T_{cross}^{TE-TE} represent the transmission of the TE mode measured at the through and cross ports, respectively, when the TE mode is excited at the input. Similarly, $T_{through}^{TM-TM}$ represent the transmission of the TM mode measured at the through port when the TM mode is excited at the input, and T_{cross}^{TM-TE} the transmission of the converted TM mode as TE mode measured at the cross port. Thus, the PER, as in the case of a PBS, measures the effectiveness of polarization separation. The PCL quantifies the efficiency of TM to be converted as TE mode: a PCL equal to zero indicates that the TM mode is fully

converted into the TE mode. Finally, the insertion loss measures the losses of introduced by the PSR device.

Three main approaches are often used to realize such a device: evanescent coupling, rotating the polarization axes of a waveguide and mode evolution. These approaches are briefly discussed in the next section, mostly using the SOI platform as a reference since it is the one most commonly used for the development of integrated polarization rotators and splitter-rotators. An example in the SiN platform is discussed for the mode evolution technique.

4.1 . Theory and state of the art

4.1.1 . Directional couplers

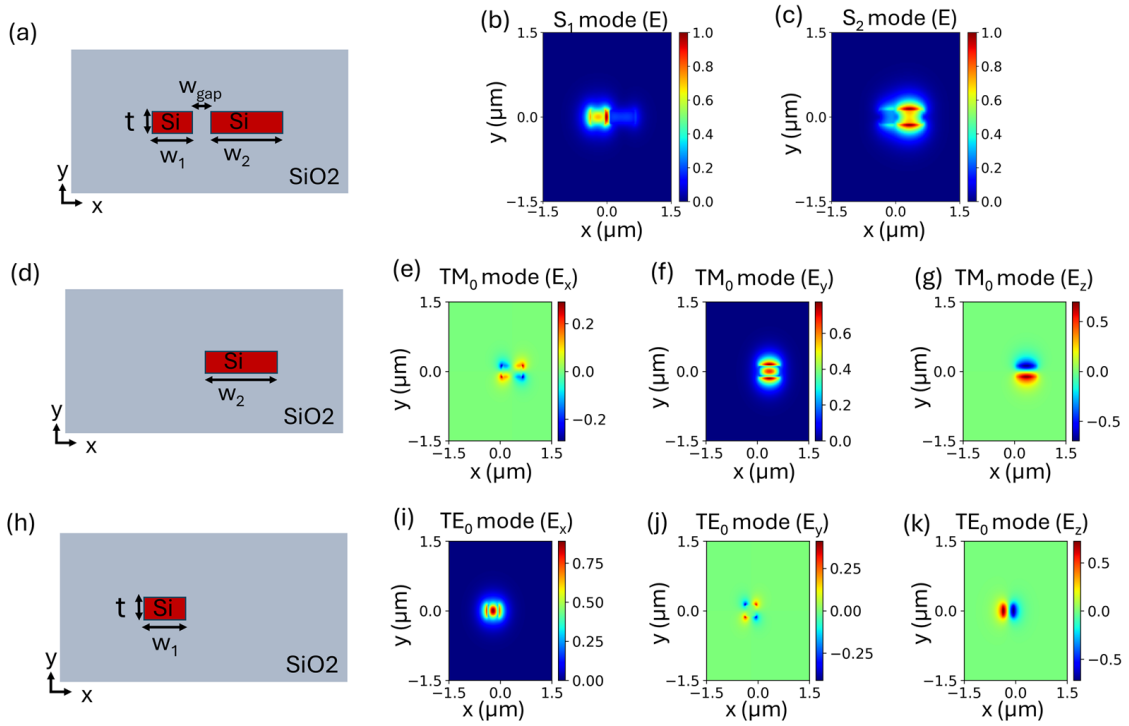

Figure 4.2: (a) Cross-sectional view of a directional coupler with silicon core waveguide surrounded by SiO_2 cladding. The waveguide widths w_1 and w_2 are chosen to satisfy the phase-matching condition between the TE_0 and TM_0 modes. (b) Electric-field amplitude of the supermode most closely corresponding to the TE_0 mode of the isolated waveguide with width w_1. (c) Same as (b), but for the supermode most closely corresponding to the TM_0 mode of the isolated waveguide with width w_2. (d) Isolated waveguide with width w_2 from panel (a). (e–g) Electric-field components of the TM_0 mode of the isolated waveguide shown in panel (d). (h) Isolated waveguide with width w_1 from panel (a). (i–k) Electric-field components of the TE_0 mode of the isolated waveguide shown in panel (h).

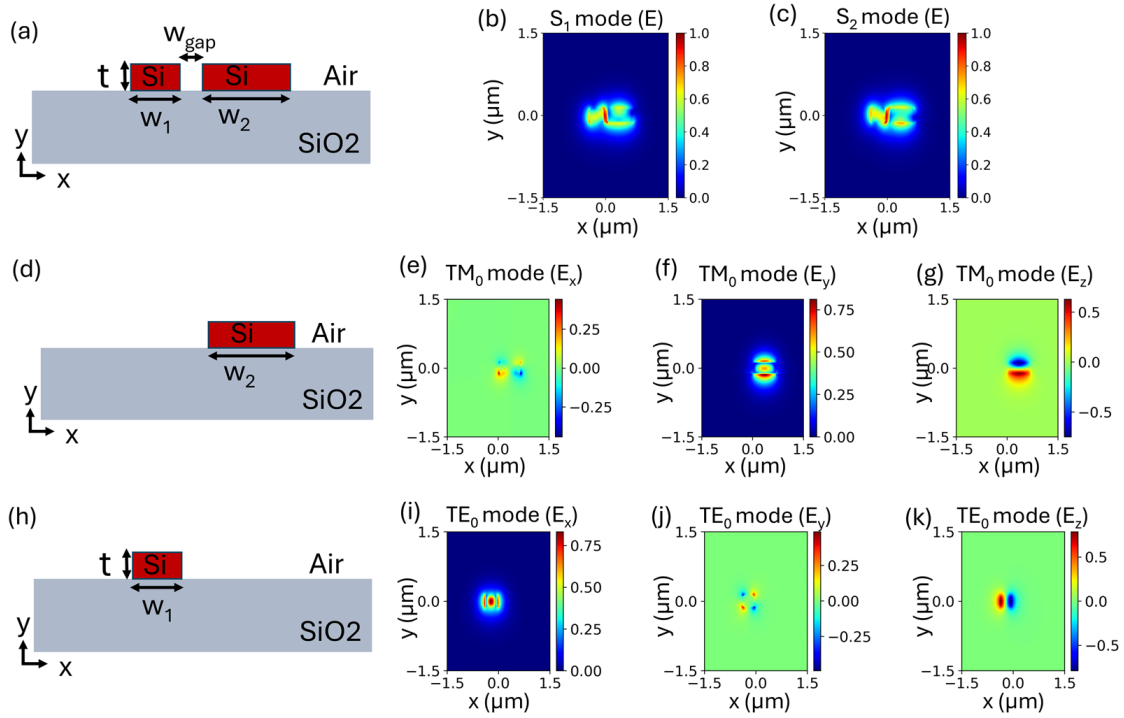

Figure 4.3: (a) Cross-sectional view of a directional coupler with a silicon core waveguide, air cladding, and SiO_2 buried oxide. The waveguide widths w_1 and w_2 are chosen to satisfy the phase-matching condition between the TE_0 and TM_0 modes. (b) Electric field amplitude of the symmetric supermode resulting from the interaction between the TE_0 and TM_0 modes of the isolated waveguides. (c) Same as (b), but showing the electric field amplitude of the antisymmetric supermode. (d) Isolated waveguide with width w_2 from panel (a). (e–g) Electric-field components of the TM_0 mode of the isolated waveguide shown in panel (d). (h) Isolated waveguide with width w_1 from panel (a). (i–k) Electric-field components of the TE_0 mode of the isolated waveguide shown in panel (h).

The first approach for simultaneous polarization splitting and rotation relies on directional couplers. The theory of directional couplers has already been detailed in Sec. 2.2.1. In particular, we noted that the main condition for efficient mode conversion is the fulfillment of the phase-matching condition. One might therefore assume that by simply satisfying the phase-matching condition between the TE0 and TM0 modes (i.e., $n_{eff}^{TE_0} = n_{eff}^{TM_0}$), it would be sufficient to achieve mode conversion from TM_0 to TE_0 using a directional coupler which is not however the case. Using a finite-difference eigenmode solver, we computed the electric field of the modes of a directional coupler cross-section. The cross-section is shown in Fig. 4.2 (a), and the electric field amplitudes of the two supermodes are presented in Fig. 4.2 (c) and (d). These modes were calculated on an SOI platform with a thickness $t = 250$ nm and upper silica cladding. The waveguide widths w_1 and w_2 were chosen to be 330 nm and 600 nm, respectively, in order to satisfy the phase-matching condition between the TE_0 and TM_0 modes (i.e., $n_{eff}^{TE_0}(w_1) = n_{eff}^{TM_0}(w_2)$). The gap was set

to 100 nm. It can be observed that the modes are close to the TE_0 and TM_0 modes of an isolated waveguide, as indicated by the absence of interaction between the two modes across the two waveguides. The slight interaction observed on the right side of Fig. 4.2 (b) corresponds to a weak coupling between the TE_0 mode of the waveguide with width w_1 and the TE_1 mode of the waveguide with width w_2. In Fig. 4.2 (c), a slight interaction is also visible between the waveguides, but it corresponds to a weak coupling between the TM_0 modes of the two waveguides. However, no symmetric and antisymmetric supermodes involving TE_0 and TM_0 are observed, which are necessary to realize a PSR using the directional coupler approach.

The absence of interaction can be explained by examining the isolated waveguide modes. In Fig. 4.2 (d), the isolated waveguide with width w_2 is shown, while panels (e–g) present the electric field components (E_x, E_y, E_z) of the TM0 mode calculated for this waveguide. Similarly, the isolated waveguide with width w_1 is presented in Fig. 4.2 (h), while panels (i–k) present the electric field components (E_x, E_y, E_z) of the TE_0 mode. The symmetry of the structure imposes the electric field components of TE-like and TM-like modes to have well-defined parities, which can be opposite. Under horizontal mirror symmetry (mirror plane parallel to the x-axis, centered in the middle of the core waveguide in Fig. 4.2 (d) and (h)), the E_x and E_z components are odd (see Fig. 4.2 (e) and (g)), while E_y (see Fig. 4.2 (f)) is even for the TM_0 mode. Conversely, for the TE_0 mode, E_x and E_z are even (see Fig. 4.2 (i) and (k)), while E_y (see Fig. 4.2 (j)) is odd. However, for coupling to occur between TE and TM modes, the coupling coefficient in Eq. 2.14 must be nonzero. Nevertheless, due to the opposite parity of the electric field components of the TE_0 and TM_0 modes, their scalar product in Eq. 2.14 (with $A \equiv E_{TE_0}$ and $B \equiv E_{TM_0}$ in Eq. 2.14) becomes an odd function in the vertical (y) direction. Since the integrand is odd over a symmetric interval ($-\infty$ to $+\infty$), the coupling coefficient becomes zero.

In order for an interaction to occur between the TE_0 and TM_0 modes in a directional coupler, it is necessary to break the symmetry. One method to achieve this is by removing the SiO_2 cladding, as shown in Fig. 4.3 (a). The electric fields of the modes of the two-waveguide system are then recalculated, and their amplitudes are displayed in Fig. 4.3 (b) and 4.3 (c). These modes were computed on the same SOI platform as before; however, with the SiO_2 cladding removed, the width w_2 was kept constant, while the width w_1 was slightly readjusted to 333 nm in order to restore the phase-matching condition that was lost when removing the SiO_2 cladding. From the field distributions, it can be clearly seen that an interaction between the TE_0 and TM_0 modes of the isolated waveguides now takes place when considering the coupled system. As before, Figs. 4.3 (d–g) and 4.3 (h–k) show the isolated structures as well as the electric field components for the TM_0 and TE_0 modes, respectively. By breaking the symmetry, the TE and TM mode electric field components under horizontal mirror symmetry are no longer even (see Figs. 4.3 (f), (i) and (k)) or odd (see Figs. 4.3 (e), (g) and (j)), allowing the coupling integral to be nonzero, since the integrand in Eq. 2.14 is no longer an odd function in the y-direction. It then becomes possible to convert the TM_0 mode into the TE_0 mode (and vice versa) through coupling. This approach is straightforward, but the coupling strength is highly sensitive to the index

contrast between the core and the cladding, which must be sufficiently large to maximize the coupling coefficient. In Figs. 4.3 (e) and (j), it can be observed that there is still a significant non-dominant polarization component: E_x for the TM_0 mode and E_y for the TE_0 mode, respectively. This distinctive effect arises from the high index contrast and greatly enhances cross-polarization coupling [86, 87].

A directional coupler was used on an SOI platform to implement the PSR shown in Fig. 4.4, as reported in [88]. Under phase-matching conditions, the TM_0 mode of the input waveguide is converted into the TE_0 mode of the second waveguide. Conversely, since the phase-matching condition is not satisfied for the TE_0 mode, it remains confined to the same waveguide. The device exhibits a PCL below 1 dB over the 1520–1570 nm range, with a minimum of 0.6 dB. The IL remains below 1 dB across the 1480–1580 nm wavelength range. As for the PER, it is higher than 10 dB over 1480–1580 nm for the TE mode, and over 1530–1580 nm for the TM mode. The device has a total length of 37 μm.

The device in Fig. 4.5 from [89] is similar to that in Fig. 4.4. The concept is equivalent, except that here the TE_0 mode is converted and separated into the TM_0 mode, and a linear taper is added on one of the waveguides to improve tolerance against width deviations, in the same way as in our work on tapered PBSs (Sec. 2.3). This device exhibits a PCL below 1 dB over the 1530–1560 nm range, while insertion loss and PER values are not reported in the article. A tolerance study has been made and show that with a 14 nm variation in waveguide width the PCL remains below 1 dB at λ = 1540 nm. The device length is 140 μm.

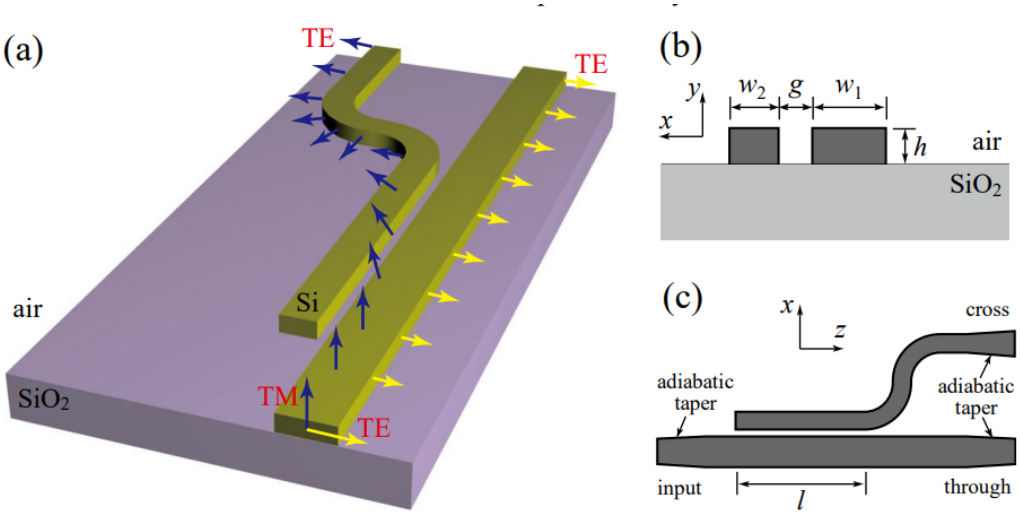

Figure 4.4: PSR based on DC from [88]

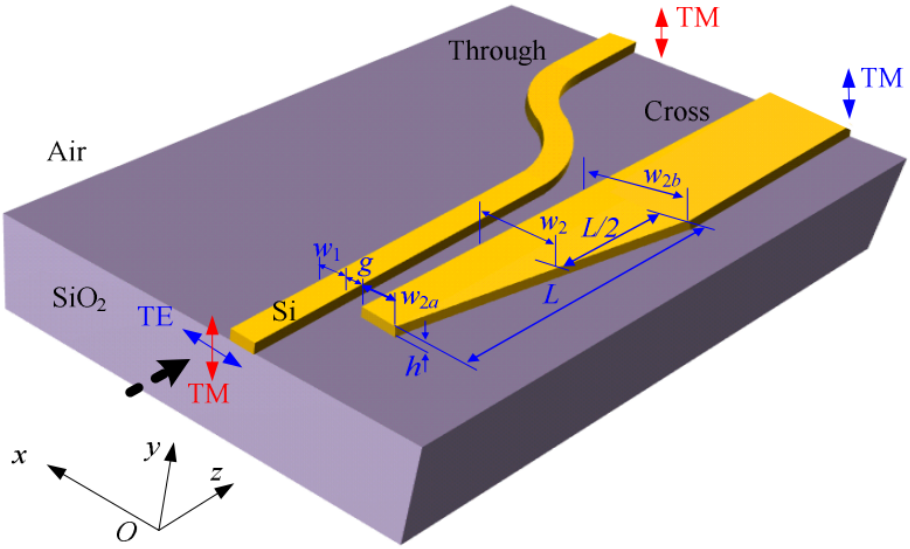

Figure 4.5: PSR based on a tapered DC from [89]

4.1.2 . Asymmetric waveguide

The second possible approach to achieve polarization splitting and rotation consists in cascading two devices to first perform polarization splitting (e.g., using one of the PBSs described in Sec. 2.1) and then rotating the polarization states on one of the two outputs. The rotation can be achieved exploiting for example an asymmetric waveguide. As explained previously, when the refractive index profile of the waveguide is symmetric, it is not possible to convert a fundamental TE mode into a fundamental TM mode because the parity of the fields components causes the overlap integral to vanish. It is therefore necessary to break the structural symmetry in order to obtain hybrid eigenmodes with different effective indices.

Figure 4.6 (a) shows the cross-section of a symmetric waveguide, while panels (b), (c), and (d) show, respectively, the field amplitude and the real part of its E_x and E_y components for the TE_0 mode. These modes were calculated on a SOI platform with a thickness (t) of 300 nm and a waveguide width (w) of 500 nm. By comparing the two electric field components, it is evident that the E_x intensity is stronger than E_y, forcing the mode to be oriented along the x-axis, as illustrated in Fig. 4.6 (a). Figure 4.6 (e) shows the cross-section of an L-shaped waveguide. In this case, the modes can be oriented at an angle θ, and are referred to as hybrid modes (H_1 and H_2 in Fig. 4.6 (e)). The orientation of a mode is defined by the following relation [87]:

$$\tan(\theta) = \frac{\iint n^2 E_x^2 dxdy}{\iint n^2 E_y^2 dxdy}. \tag{4.5}$$

Here, n represents the refractive index of the waveguide cross-section, and E_x and E_y are the transverse components of the electric field. In Fig 4.6 (f), (g) and (h) the amplitude and

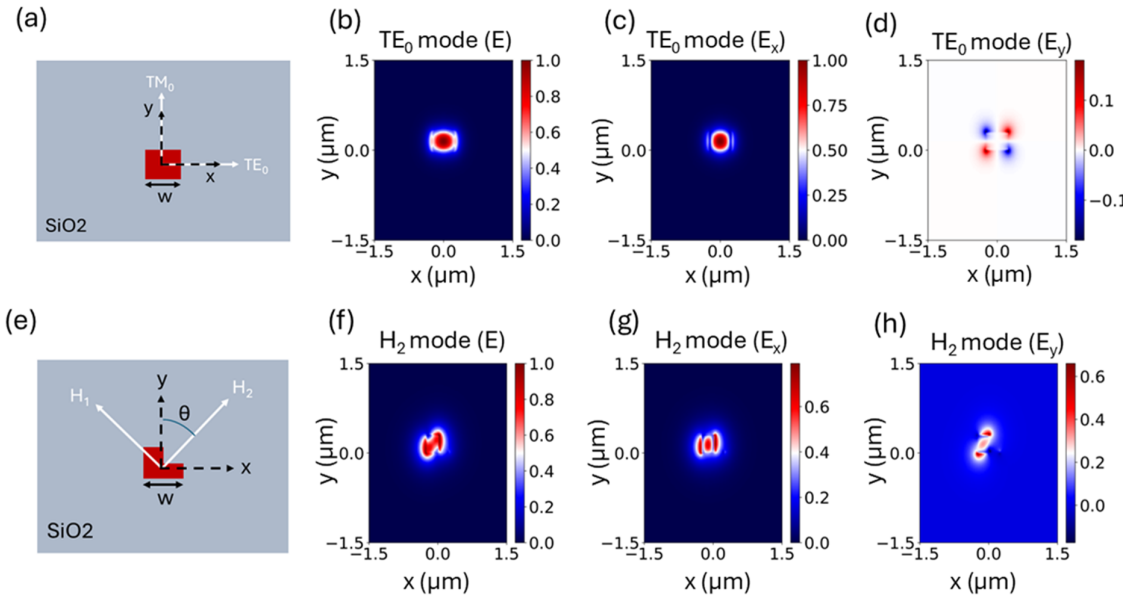

Figure 4.6: (a) Cross-sectional view of a rectangular silicon waveguide, showing that the fundamental TE_0 and TM_0 modes are oriented along the *x*- and *y*-axes, respectively. (b) Electric field amplitude of the TE_0 mode. (c) Electric field component of the TE_0 mode along the *x*-axis. (d) Electric field component of the TE_0 mode along the *y*-axis. (e) Cross-sectional view of an L-shaped silicon waveguide, resulting in the existence of hybrid modes H_1 and H_2, with a modal orientation θ for H_2. (f) Electric field amplitude of the hybrid mode H_2. (g) Electric field component of the H_2 mode along the *x*-axis. (h) Electric field component of the H_2 mode along the *y*-axis.

the real part of the electric field components of the hybrid mode H_2 in the asymmetric waveguide is showed. In this case, the intensities of the E_x and E_y components are relatively similar, resulting in a mode orientation of $\theta = +45^\circ$. These mode was also calculated on the same platform as Fig. 4.6 (a) with the same width (w) of 500 nm, but with an etched section width of 270 nm and an etch depth of 230 nm.

The structure used to perform the rotation is shown in Fig. 4.7 (a) (top view) and in (b)–(d) cross-sections of the device. The TE_0 mode is excited in the first section and propagates until it reaches the second section (position (i) in Fig. 4.7 (a)), where it is projected onto the basis of hybrid modes (H_1 and H_2). At the beginning of this section, the field is mostly oriented along the *x* direction; however, both hybrid modes (H_1 and H_2) propagate with different phase velocities. After a certain length L_π, a phase shift of π is accumulated between the two hybrid modes, resulting in the field being oriented along the *y* direction. This field is then projected onto the TM_0 mode of the third section (position (iii) in Fig. 4.7 (a)), thereby achieving the polarization rotation.

This concept was experimentally demonstrated on a SOI platform in [90], where the schematic of their device is shown in Fig. 4.8 (b). The device operates exactly as described above. Its length is only 3 μm but it exhibits relatively high PCL and IL of about 4 dB over the 1540–1600 nm wavelength range. This is due to modal mismatch during the projection

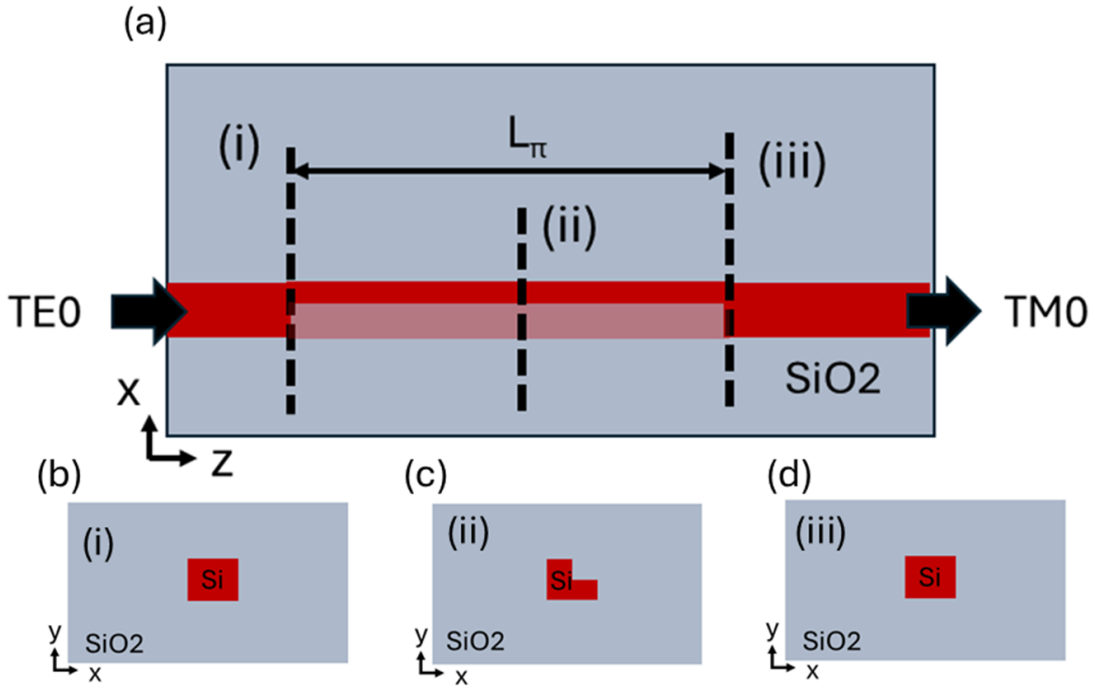

Figure 4.7: (a) Schematic of a polarization rotator based on a partially etched asymmetric waveguide. (b) Cross-sectional view at position (i). (c) Cross-sectional view at position (ii). (d) Cross-sectional view at position (iii).

of the TE_0 and TM_0 modes onto the hybrid modes. To mitigate this issue, in [90], they also proposed the device shown in Fig. 4.8 (c), which incorporates a taper to smooth the transition and reduce the losses. This tapered version achieves improved performance, with PCL and IL reduced to about 2.5 dB. The device is slightly longer 4.2 μm.

4.1.3 . Mode evolution technique

As in the previous approach, also in this case the two functionalities of polarization conversion and splitting are implemented separately. Instead of using an asymmetric waveguide, polarization rotation can be achieved via mode evolution, which, similarly to the two previous approaches, also relies on the presence of hybrid modes. Under certain conditions, it is possible to achieve conversion from the TM_0 mode to the TE_1 mode using a taper, as illustrated in Fig. 4.9 (a).

To detail the principle of the mode-evolution technique, we used a finite-difference eigenmode solver to calculate the effective indices of the TE_0, TM_0, and TE_1 modes as a function of the waveguide width on a 260-nm-thick SOI platform without an upper SiO_2 cladding. The results are shown in Fig. 4.9 (b). Near the phase-matching point between the TM_0 and TE_1 modes, an anti-crossing region appears, which is the hallmark of mode hybridization. To illustrate this effect, three waveguide widths were selected at positions (i), (ii), and (iii) in Fig. 4.9 (a), corresponding respectively to regions before, within, and after

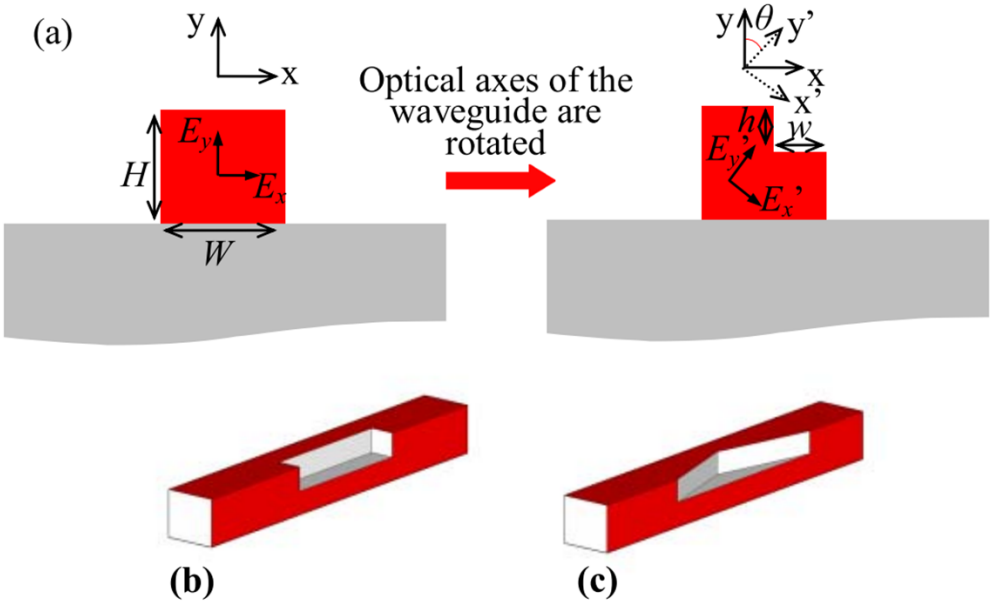

Figure 4.8: Polarization rotator based on a partially etch asymmetric waveguide from [90]

the anti-crossing in Fig. 4.9 (b). The corresponding cross-sections are shown in Figs. 4.9 (c–e), with the electric field amplitudes of the respective modes. The chosen widths are $w_1 = 630$ nm, $w_2 = 723$ nm, and $w_3 = 850$ nm. From the field profiles, we can observe that at width w_1, before the anti-crossing, the mode is essentially TM_0 (Fig. 4.9 (c)); at width w_3, after the anti-crossing, it becomes essentially TE_1 (Fig. 4.9 (e)); while at width w_2, within the anti-crossing, it corresponds to a mode hybridization between TM_0 and TE_1 (Fig. 4.9 (d)).

The polarization rotation realized by the device in Fig. 4.9 (a) thus proceeds as follows. As the waveguide width gradually increases along the propagation direction (z), the effective index of the mode also increases. Initially, before the anti-crossing region, the mode is essentially TM_0 with an orientation angle θ (see Eq. 4.5) close to 0°. Approaching the anti-crossing, the orientation evolves progressively, reaching 45° in the hybrid region (Fig. 4.9 (d)). As the waveguide width continues to increase beyond the hybridization region, the mode orientation evolves from 45° toward 90°, becoming essentially TE_1 (Fig. 4.9 (e)). This gradual mode transformation along the taper is the reason for the name “mode evolution". The conversion from the input TM_0 mode to the output TE_1 mode can be complete, if the taper is sufficiently long to ensure an adiabatic transition. It is worth noting that, in the effective index graphic Fig. 4.9 (b), the modes in the anti-crossing region are no longer purely TM_0 or TE_1 but hybrids of the two. We chose to label modes with orientation angles below 45° as TM_0 and those above 45° as TE_1, even though they are hybrid in nature. This choice, which is also commonly found in the literature (see [17] and [91]),

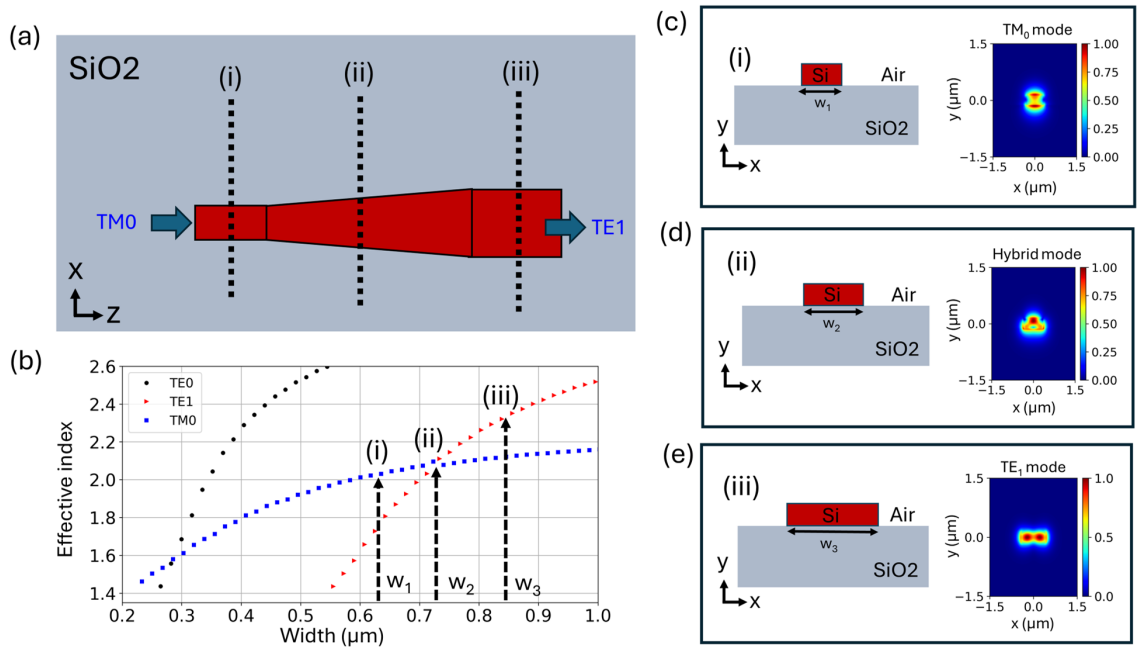

Figure 4.9: (a) Schematic top view of a taper implementing the mode evolution technique, converting the TM_0 mode into the TE_1 mode. (b) Effective indices of the TE_0, TM_0, and TE_1 modes calculated as a function of the waveguide width for an SOI platform with air cladding and SiO_2 buried oxide. An anti-crossing region between the TM_0 and TE_1 modes appears when the width approaches the phase-matching condition. Three width values, w_1, w_2, and w_3, are selected, corresponding respectively to positions before, within, and after the anti-crossing region. These widths correspond to the taper positions indicated by the black dashed lines in panel (a) at positions (i), (ii), and (iii). (c) Cross-sectional view of the waveguide with width w_1 at position (i) in panel (a), before the anti-crossing region shown in panel (b), alongside the electric field amplitude of the TM_0 mode calculated for this waveguide. (d) Same as (c), but at position (ii) in panel (a), within the anti-crossing region in panel (b), showing the electric field amplitude of a hybrid mode formed by the hybridization between the TM_0 and TE_1 modes. (e) Same as (c–d), but at position (iii) in panel (a), after the anti-crossing region in panel (b), showing the electric field amplitude of the TE_1 mode.

is made to simplify the plots.

If we now look at Fig. 4.9 (b), around the phase-matching point between the TE_0 and TM_0 modes, no anti-crossing region appears. This indicates the absence of hybridization between the TE_0 and TM_0 modes and therefore the impossibility of conversion. The reason why TM_0 can be converted into TE_1 (and vice versa) while it cannot be converted into TE_0 (and vice versa) is again linked to the parity of the electric field components. Figure 4.10 shows the electric field components for the TM_0 ((a)–(c)), TE_1 ((d)–(f)), and TE_0 ((g)–(i)) modes of the waveguide represented in Fig. 4.9 (e). The absence of the SiO_2 cladding breaks the horizontal mirror symmetry (mirror plane along the x-axis) but preserves the vertical mirror symmetry (mirror plane along the y-axis). The E_y and E_z components of the TM_0 and TE_1 modes are even with respect to the vertical mirror symmetry (see Figs. 4.10 (b),(c) and (e),(f)), while the E_x components are odd (see Figs. 4.10 (a) and (d)).

Since they share the same parity and the horizontal mirror symmetry has been broken (air cladding), the overlap between the TM_0 and TE_1 modes can be nonzero. In contrast, with respect to the vertical mirror symmetry, the E_y and E_z components of the TE_0 mode are odd (see Fig. 4.10 (h) and (i)), while the E_x component is even (see Fig. 4.10 (g)). Since the TE_0 and TM_0 electric field components have opposite parity with respect to the vertical mirror symmetry, their modal overlap cancels out.

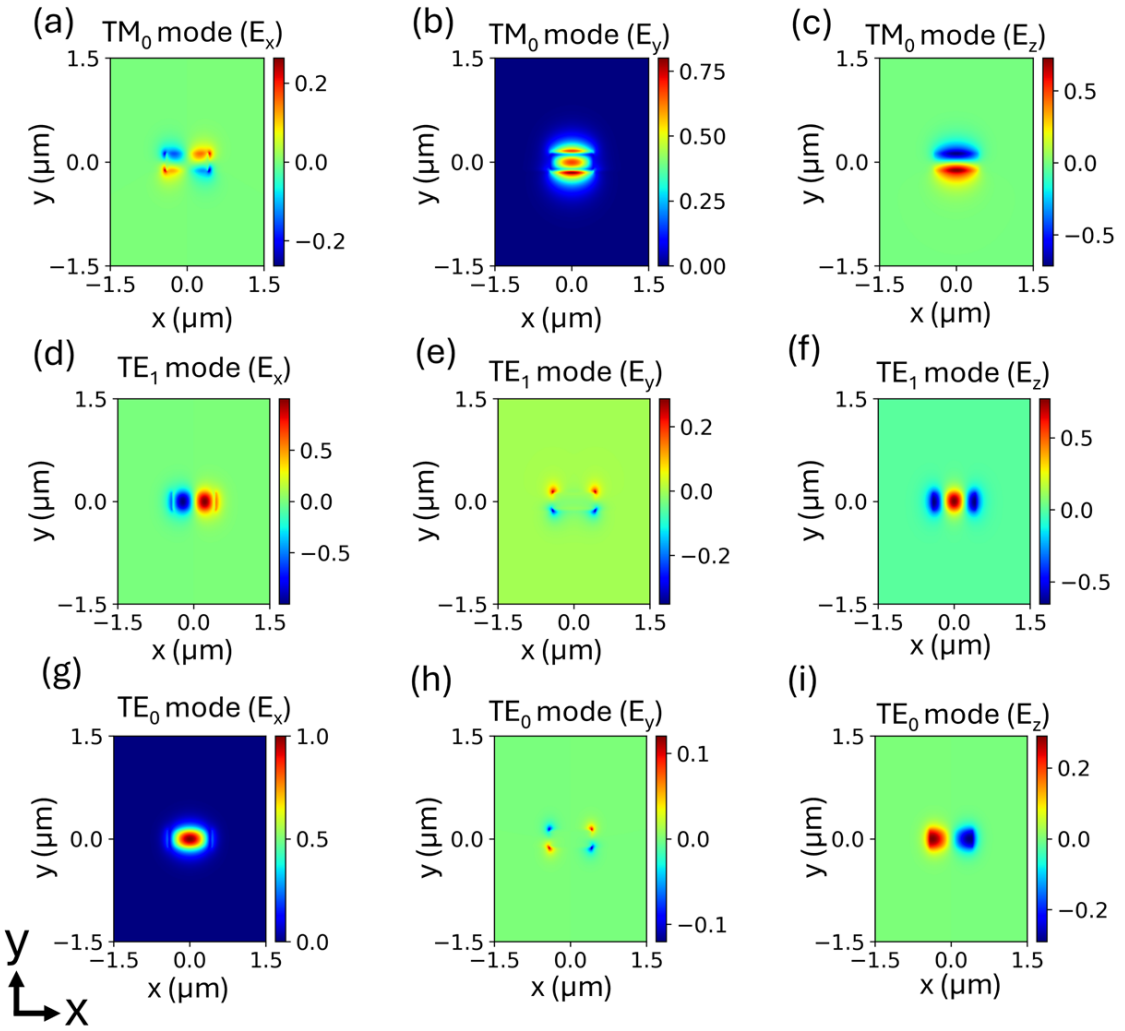

Figure 4.10: Electric field components calculated using a finite-difference eigenmode solver for the waveguide shown in Fig. 4.9 (e). (a–c) Electric field components of the TM_0 mode. (d–f) Electric field components of the TE_1 mode. (g–i) Electric field components of the TE_0 mode.

As an example, a PSR based on mode-evolution method was demonstrated in [92], with the corresponding structure shown in Fig. 4.11 (a). The device consists of two stages. The first stage performs polarization rotation via mode evolution: a taper is employed to progressively convert the TM_0 mode into the TE_1 mode. To break the symmetry, the SiO_2 upper cladding was removed. Once the TE_1 mode is generated at the output of the rota-

tion stage, the second stage begins. This consists of an asymmetric directional coupler, which transfers the TE_1 mode into the TE_0 mode of an adjacent waveguide. Conversely, when the TE_0 mode is injected into the input waveguide, it remains unaffected by the taper and continues to propagate to the coupling region, where it also remains uncoupled since no phase-matching condition is satisfied. Consequently, two output ports carrying the same polarization are obtained. The MMI section is employed only as a filter to further improve performance. The PSR was fabricated on an SOI platform without an SiO_2 upper cladding. It exhibits a PCL of about 1.5 dB over the wavelength range of 1520–1620 nm and an insertion loss (IL) of less than 0.5 dB at 1560 nm. The PER exceeds 20 dB for the TE mode in the 1520–1600 nm wavelength range, and remains above 10 dB for the TM mode within the same wavelength range. The overall device length is 70 μm.

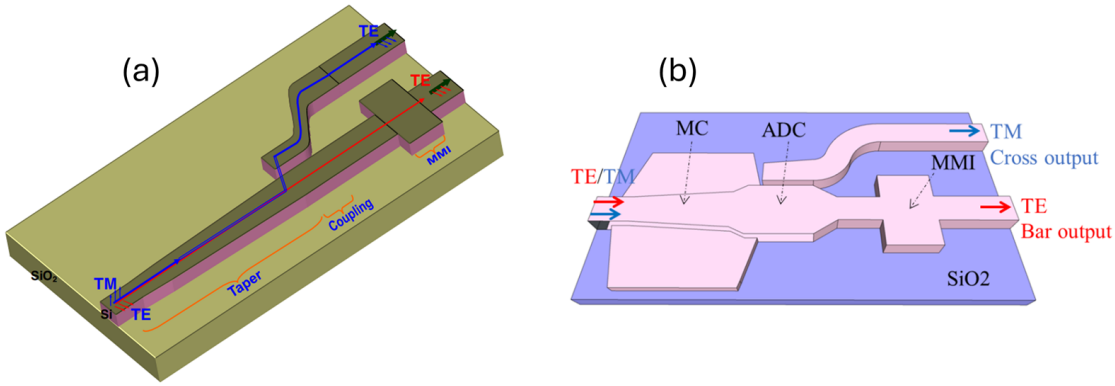

Figure 4.11: PSR devices based on mode evolution approach: (a) from [92] and (b) from [93].

Fig. 4.11 (b) shows a PSR realized using the mode-evolution technique in [93]. Its operating principle is identical to that of the device in Fig. 4.11 (a), but it is realized on a SiN platform. Moreover, to break the symmetry while preserving the SiO_2 cladding, a rib waveguide structure was employed. The PSR achieves a PER of 20 dB over the 1545–1570 nm wavelength range, with both PCL and IL estimated at approximately 1.5 dB. The total device length is about 2900 μm. In terms of PCL, IL, and PER, the two devices in Fig. 4.11 (a) and (b) are comparable; however, their lengths differ significantly due to the much lower index contrast of SiN platform compared to SOI platform.

Table 4.1 summarizes the performance of several PSR devices implemented on SOI and SiN platforms using the various methods presented in this section.

Table 4.1: Performance comparison of several polarization splitter–rotators fabricated on SOI and SiN platforms.

	Wavelength range (nm)	Min PCL (dB)	Max PER (dB)	Size (µm)	$BW_{PCL<1dB}$ (nm)	platform	Asymmetry	Ref
SWG ADC	1500 - 1600	0.13	TE>30-TM:15	50	100	SOI	Air cladding	[94]
ADC	1480 - 1580	0.6	TE:22-TM>30	37	50	SOI	Air cladding	[88]
Tapered ADC	1520 - 1620	0.1	-	140	30	SOI	Air cladding	[89]
Bi-level taper and ADC	1530 - 1565	0.3	TE:40-TM:15	170	25	SOI	Dual etch	[91]
Asymmetric silicon waveguide	1520 - 1620	> 3	-	3	-	SOI	Dual etch	[90]
Dual-etching tapered ADC	1510 - 1580	0.18	TE:26-TM:34	28	45	SOI	Dual etch and Air cladding	[95]
Taper, ADC and MMI	1520 - 1620	1.5	TE:30-TM:28	70	-	SOI	Air cladding	[92]
Bi-level taper, ADC and MMI	1530 - 1565	~ 1.5	TE:24-TM:24	2900	-	SiN	Dual etch	[93]
Bi-level taper and tapered ADC	1260 - 1360	~ 1.5	TE:26-TM:26	750	-	SiN	Dual etch	[96]
Taper and ADC	1240 - 1280	~ 2	TE:28-TM:20	685	-	SiN	Air cladding	Our work

4.2 . Design and simulation

We implemented a polarization splitter-rotator exploiting the mode evolution technique and operating at a wavelength around 1300 nm. Mode evolution was chosen for two main reasons. First, the low index contrast of SiN makes PSRs based on directional couplers challenging. Second, although asymmetrical waveguides could in principle be used, the transition losses and the strong sensitivity to the partial etch depth motivated us to adopt the mode evolution approach instead. In our design, we propose to work without SiO_2 cladding, thereby introducing the necessary asymmetry. These devices were fabricated in the C2N cleanroom on the same SiN platform described in Sec. 1. The schematic of the device is shown in Fig. 4.12. The device operates as follows: when the TM_0 mode is excited into the input waveguide, it is converted into a TE_1 mode by a adiabatic taper. This TE_1 mode is then coupled to the TE_0 mode of the upper waveguide via a directional coupler, allowing a TE_0 to exit through the cross port. Conversely, in the case of TE_0 injection, since no hybridization occurs, the mode is not affected by the taper nor efficiently coupled by the directional coupler due to the lack of phase matching and can hence exit unaffected by the through port.

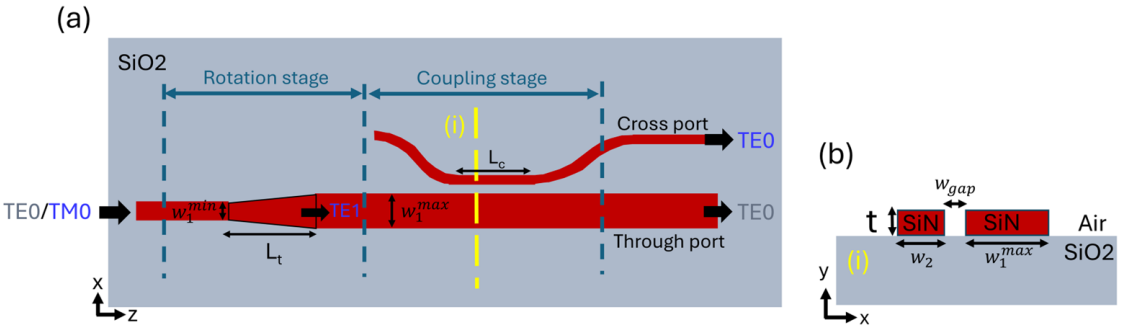

Figure 4.12: Schematic of our device, consisting of a PSR based on the mode evolution technique combined with an ADC. (a) Top view. (b) Cross-sectional view at the position (i) indicated by the yellow dashed line.

To begin designing such a structure, it is first necessary to calculate the effective in-

dex of the modes in the absence of cladding. The effective indices of the TE_0, TE_1 and TM_0 modes as a function of waveguide width are shown in Fig. 4.13 (a). A zoom of the hybridization region is shown in Fig. 4.13 (b), where hybrid TM_0 / TE_1 modes exist. To achieve polarization rotation, the first step is to choose two waveguide widths, w_1^{min} and w_1^{max}, over which the transition will occur. The only condition is that these widths must be selected around the hybridization region such that the guided mode at w_1^{min} is a quasi-TM mode (not hybridized) and at w_1^{max} is a quasi-TE mode. We therefore chose the widths $w_1^{min} = 1.55\ \mu$m and $w_1^{max} = 1.75\ \mu$m, and then verified that the modal orientations using Eq. 4.5 were close to 0° and 90°, respectively.

To determine the taper length required for this rotation, an EME simulation was performed. Figure 4.14 (a) shows the conversion efficiency from TM_0 to TE_1 as a function of the taper length and using the chosen minimum and maximum waveguide widths. It can be observed that a relatively long taper (~ 500 μm) is required for complete rotation, due to the low index contrast between SiN and air. Based on this analysis, the taper length was fixed at 505 μm, achieving an expected conversion performance of 99%. With the rotation section defined, the next step is to design the coupling section.

As previously done for the PBS devices (see Sec. 2.2.2), the appropriate waveguide widths are selected to satisfy the phase-matching condition ($n_{eff}^{TE_1}(w_1^{max}) = n_{eff}^{TE_0}(w_2)$). Since w_1^{max} is already fixed by the rotation section, the value of $w_2 = 0.83\ \mu$m is directly determined by fulfilling the phase-matching condition. The final design parameters, namely the coupling length and the gap, are then chose using EME simulations. Figure 4.14 (b) shows the conversion efficiency of the TE_1 mode of the bottom waveguide to the TE_0 mode of the upper waveguide with width w_2 as a function of the coupling length L_c. Based on this analysis, the coupling length was set to $L_c = 180\ \mu$m for a gap $w_{gap} = 0.4\ \mu$m.

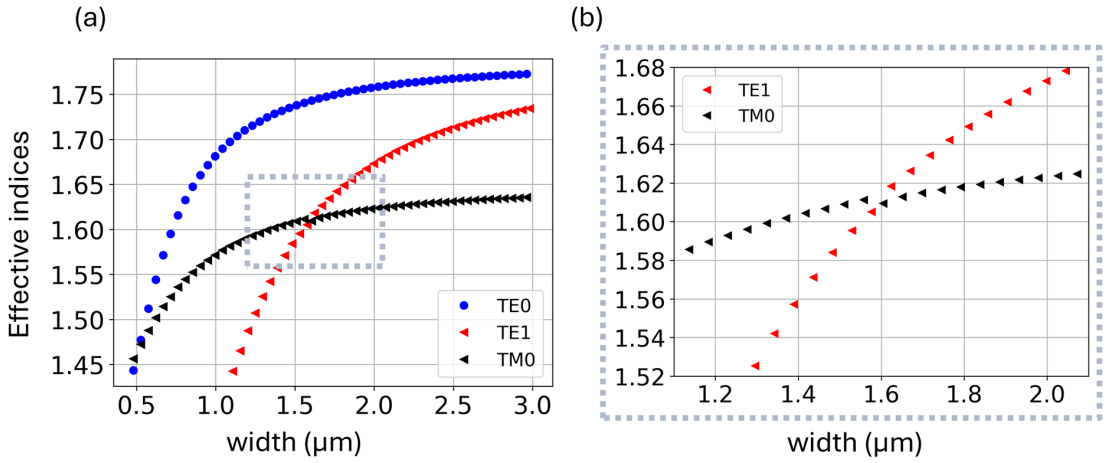

Figure 4.13: (a) Effective indices of an isolated SiN waveguide, calculated on the SiN platform with air cladding and SiO_2 buried oxide, as a function of waveguide width for the TE_0, TE_1, and TM_0 modes. (b) Zoom-in of the anti-crossing region between the TE_1 and TM_0 modes.

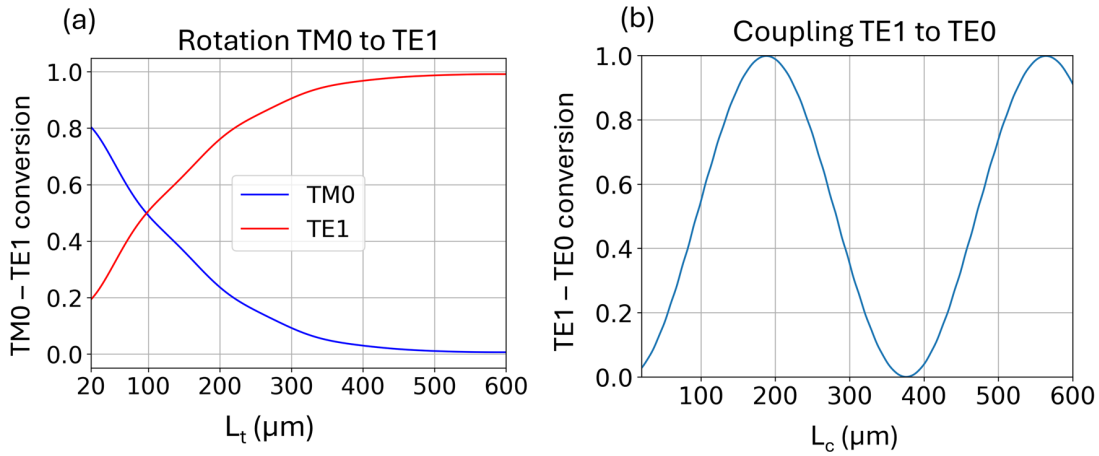

Figure 4.14: (a) TM_0-to-TE_1 mode conversion calculated using EME as a function of the taper length (L_t) in the rotation stage shown in Fig. 4.12 (a). (b) TE_1-to-TE_0 coupling efficiency calculated using EME as a function of the coupling length (L_c) in the coupling stage shown in Fig. 4.12 (a).

4.3 . Fabrication and experimental results

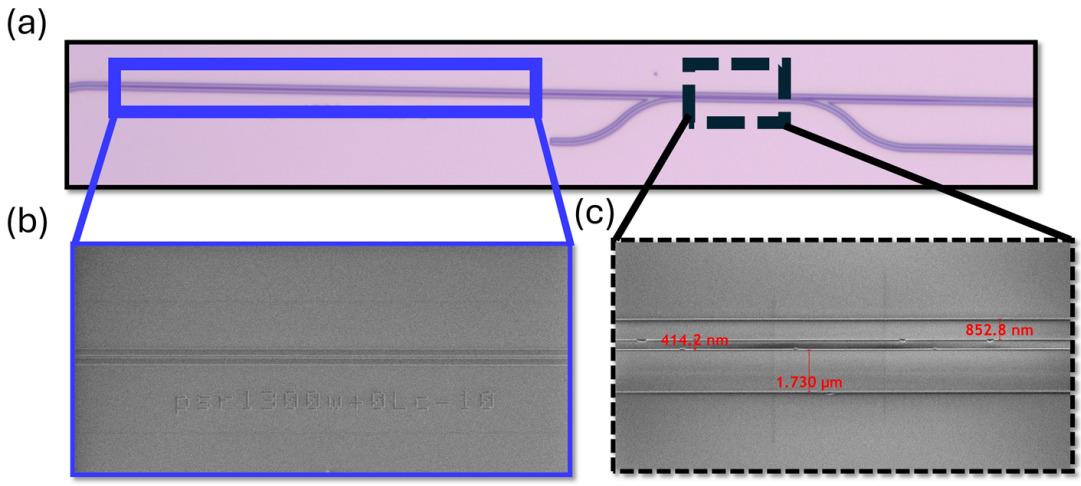

Figure 4.15: Images of the fabricated PSR. (a) Optical microscope image of the complete device. (b) SEM image focused on the rotation stage. (c) SEM close-up of the coupling stage.

Figure 4.15 (a) shows a microscope image of one of the fabricated PSRs while Figs. 4.15 (b) and (c) present SEM images of the rotation and coupling sections, respectively. In the coupling section, the upper waveguide has a width of $w_1^{max} \approx 850$ nm (20 nm larger than the designed value), whereas the lower waveguide has a width of $w_2^{max} \approx 1730$ nm (20 nm smaller than designed). Typically, fabrication errors are expected be uniform, but for this device the deviations had opposite sign, indicating an unusual issue during fabrication.

To characterize the fabricated device, the setup shown in Fig. 4.16 was used. Com-

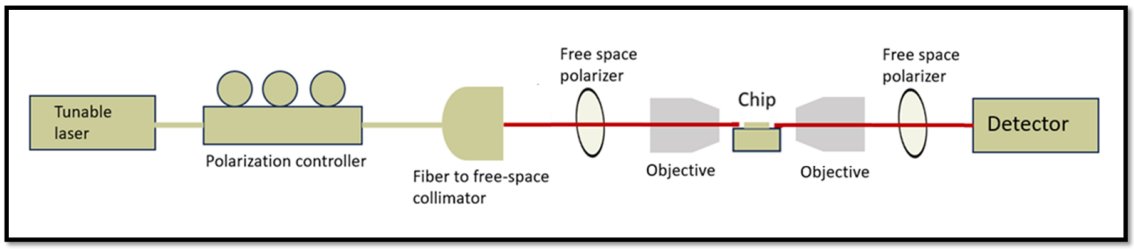

Figure 4.16: Experimental free-space setup for PSR characterization

pared to the setup shown in Fig. 2.42 and used for the characterization of the PBSs, in this case an output polarizer was added to more precisely analyze the polarization state of the output light.

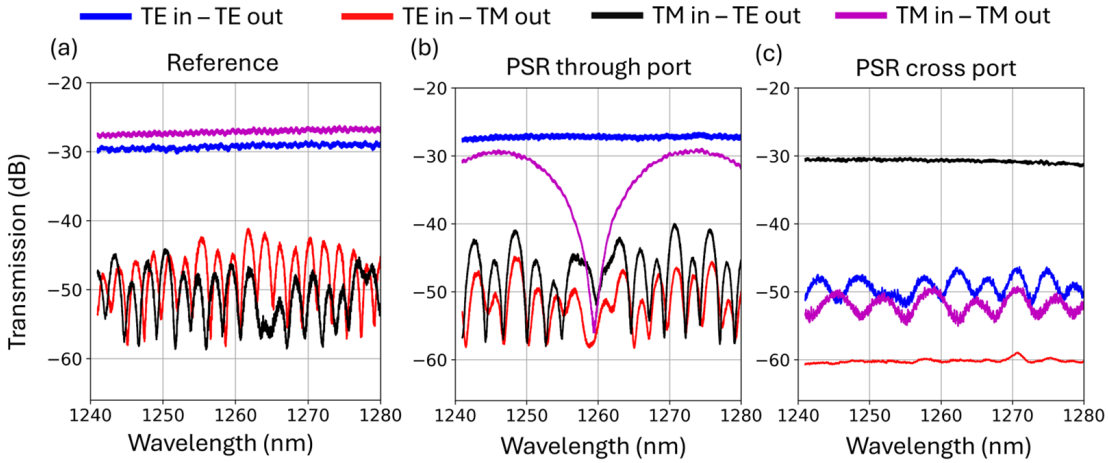

Figure 4.17: (a) Reference waveguide measurements for TE and TM mode excitation, controlled by the orientation of the input polarizer placed after the collimator in the setup shown in Fig. 4.16, and analyzed as TE or TM by the output polarizer positioned before the detector. (b) Same measurement principle applied to the PSR device at the through port. (c) Same measurement principle applied to the PSR device at the cross port.

Figure 4.17 (a) shows the measurements of a reference waveguide without the PSR, obtained using the setup in Fig. 4.16. The blue curve corresponds to the transmission when both the input polarizer (just before the chip) and the output polarizer (after the chip) are oriented horizontally (x-axis in Fig. 4.12 (b)). Note that in each case the polarization controller before the collimator is used to align the initial laser polarization with that of the input polarizer. The red curve corresponds to transmission with the input polarizer oriented horizontally and the output vertically; the black curve to vertical and horizontal; and finally, the magenta curve to vertical and vertical. TE_0 excitation corresponds to a horizontal polarizer orientation, while TM_0 excitation corresponds to a vertical orientation As expected, maximum transmission occurs when the two polarizers are aligned. However, when they are crossed, the transmission is not zero, which indicates that even when TE is

excited, a residual fraction of TM is measured. This is not due to fibers, since a free-space setup was used with a PER of the setup above 40 dB.

Figure 4.17 (b) presents the transmission measured at the through port of the PSR device, using the same color coding as in the reference measurements. We chose to display the reference separately in order to also show the different cross transmissions measured on the reference and on the PSR ports (T_{ref}^{TE-TM} and T_{ref}^{TM-TE}), without overcrowding the graph. When TE is excited at the input and the output polarizer is TE-oriented, the transmission measured at the through port is maximal (blue curve), with negligible insertion loss across the entire wavelength range (< 1 dB compared to the reference). When TM is excited, the transmission at 1260 nm decreases at the through port (magenta curve), which is consistent with TM_0 being converted into TE_1 and subsequently coupled to TE_0 at the cross port. For the cross measurements at the through port ($T_{through}^{TE-TM}$ and T_{cross}^{TM-TE}), oscillations similar to those observed in the reference are present. Figure 4.17 (c) shows the transmission measured at the cross port. When TE is excited, the transmission remains low (blue curve), whereas when TM is excited and measured as TE, the transmission is high (black curve), confirming the desired PSR effect. The PCL and IL can be estimated by comparison with the reference waveguide. Since the device performs polarization rotation, the light enters as TM, is converted into TE inside the chip, and exits as TE. As shown in the reference waveguide measurement (Fig. 4.17 (a)), the transmission differs by about 2 dB depending on whether TE or TM is excited, due to polarization-dependent edge couplers and propagation losses. Therefore, we estimate the PCL of the PSR device by normalizing the transmission of the TM mode converted to TE and measured at the cross port (black curve) using the average transmission values of the reference between TE and TM modes.

The definitions of PCL, IL, and PER from Eq. 4.1-Eq. 4.4 are therefore adapted to characterize the device performance as follows:

$$PER_{TE} = 10 \cdot \log_{10} \frac{T_{through}^{TE-TE}}{T_{cross}^{TE-TE}} \tag{4.6}$$

$$PER_{TM} = 10 \cdot \log_{10} \frac{T_{cross}^{TM-TE}}{T_{through}^{TM-TM}} \tag{4.7}$$

$$IL_{TE} = -10 \cdot \log_{10} \frac{T_{through}^{TE-TE}}{T_{ref}^{TE-TE}} \tag{4.8}$$

$$PCL = -10 \cdot \log_{10} \frac{T_{cross}^{TM-TE}}{0.5 \cdot (T_{ref}^{TE-TE} + T_{ref}^{TM-TM})} \tag{4.9}$$

The fabricated device demonstrates at λ = 1260 nm a PER of 28 dB for TE and 20 dB for TM , with IL below 1 dB for TE in the entire wavelength range and about 2 dB of PCL at 1260 nm. A comparison with state-of-the-art SiN PSRs is shown in the table. 4.1. Both SiN devices reported in the table are based on the mode evolution method. Our device

achieves comparable PCL and PER performance. Importantly, it also exhibits the shortest footprint among the compared designs while simplify the fabrication since it require only one lithography to be done instead of two as in the devices reported in [93] and [96].

5 - Silicon-based polarization rotator

5.1 . Design and simulation

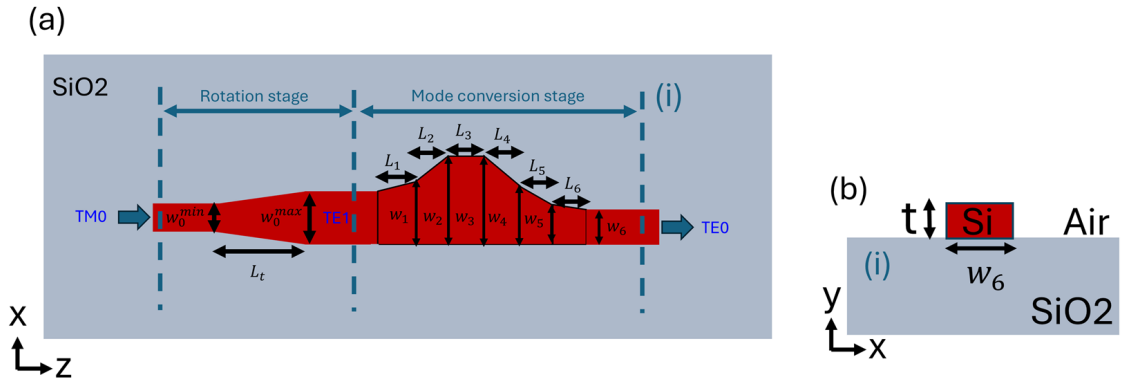

Figure 5.1: (a) Schematic of our PR device based on the mode evolution technique and a TE_1-to-TE_0 mode converter using cascaded asymmetric waveguide. (b) Cross-sectional view at position (i), indicated by the dashed line near the output.

A polarization rotator (PR) is a device that rotates the polarization state of the input light but, unlike a PSR, it does not provides a separation stage. We decided to treat it separately from PSRs because polarization rotation is sometimes required to be implemented as a stand-alone functionality in polarization-diversified circuits [97, 98]. What we consider as a PR here is a device designed to operate on a given input polarization at a time. For instance, a device could be designed specifically to rotate TM_0 into TE_0. For this reason, it is often used in combination with a PBS or a polarization filter to isolate a single polarization before the rotation stage.

Here, we choose to work on an SOI platform and at a wavelength of 1550 nm. This choice was made in order to demonstrate a proof-of-concept device exploiting the high index contrast between the silicon core and the air cladding that strongly favors the emergence of hybrid modes. In the literature, most of the work focuses on splitter-rotators, mainly because a PSR can be used as a PR. In our case, we propose to implement a dedicated PR using the mode evolution method. The devices were fabricated in the cleanroom of the C2N on a 220-nm-thick SOI platform with a 3-μm SiO_2 buried oxide. No upper SiO_2 cladding was deposited.

Figure 5.1 (a) shows a schematic of the device, which consists of two main sections. First, a tapered waveguide is used to rotate the TM_0 mode into the TE_1 mode, using the mode evolution technique described in Sec. 4.1.3. This is possible because, we use a waveguide without an upper cladding to generate the required asymmetry (see Fig. 5.1 (b)). Compared to the SiN platform, the higher index contrast offered by SOI is expected to

strongly enhance mode hybridization, resulting in a much shorter taper length than what we got with the PSR in Sec. 4.2.

The second section of the device needs to convert the higher-order TE_1 mode into the fundamental TE_0 mode, as shown in Fig. 5.1 (a). We propose to combine the polarization rotation stage with the TE_1–to–TE_0 mode converter proposed in Ref. [99] and based on an asymmetric taper structure (see Fig. 5.2). By introducing a lateral asymmetry, the waveguide creates different effective path lengths for the anti-phase components of the TE_1 mode, causing them to become in-phase and thus generating a TE_0 mode at the output. This step is fundamental for achieving the desired polarization rotation and was therefore carefully considered in the design.

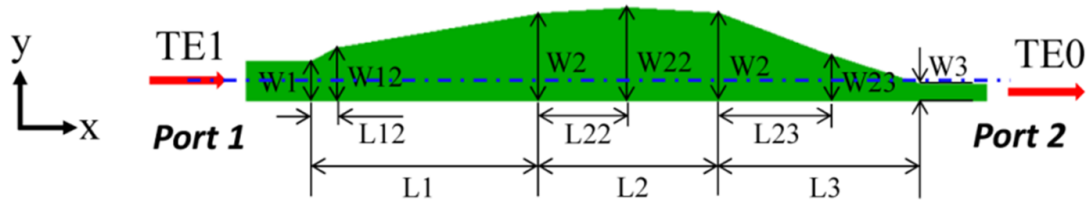

Figure 5.2: TE_1-to-TE_0 mode converter geometry based on a cascade of asymmetric tapers, as proposed in [99].

Once the operating principle of the device was defined, we carried out simulations to determine the appropriate design parameters. We first calculated the effective indices of the modes as a function of the waveguide width using a mode solver (see results in Fig. 5.3 for TE_0, TE_1, and TM_0 modes). Two widths, w_1^{min} and w_1^{max}, were selected around the hybridization region, such that the orientation θ (see Eq. 4.5) of the TM_0 mode is close to 0° and that of the TE_1 mode after the taper at w_1^{max} is close to 90°. The chosen widths are $w_1^{min} = 0.55\ \mu$m and $w_1^{max} = 0.75\ \mu$m. With these widths fixed, an EME simulation was performed on the structure represented in Fig. 5.4 (a) to evaluate the conversion from TM_0 to TE_1. Results are shown in Fig. 5.4 (b). For a taper length $L_t > 60\ \mu$m, the TM_0 mode is fully converted into the TE_1 mode. We therefore set the taper length slightly beyond this limit, at $L_t = 80\ \mu$m.

For the TE_1-to-TE_0 conversion stage, ref. [99] employed 2D FDTD simulations combined with a particle swarm optimization algorithm. Given the number of parameters in the structure, $\vec{w} = [w_1, w_2, w_3, w_4, w_5]$ and $\vec{L} = [L_1, L_2, L_3, L_4, L_5, L_6]$ (see fig.5.1 (a)), such approach appears necessary. However, due to the vertical asymmetry of our waveguide (due to the lack of upper cladding), 2D FDTD simulations are not suitable, as the effective index approximation used to extrapolate from 2D to 3D is generally inaccurate for vertically asymmetric waveguides. For this reason, we relied on EME simulations for the optimization, with each section of the TE_1-to-TE_0 mode converter stage taken as an EME cell comprising five sub-cells sections. The input and output widths of the mode converter, w_0^{max} and w_6, are fixed: $w_0^{max} = 0.75\ \mu$m is imposed by the rotation section, while

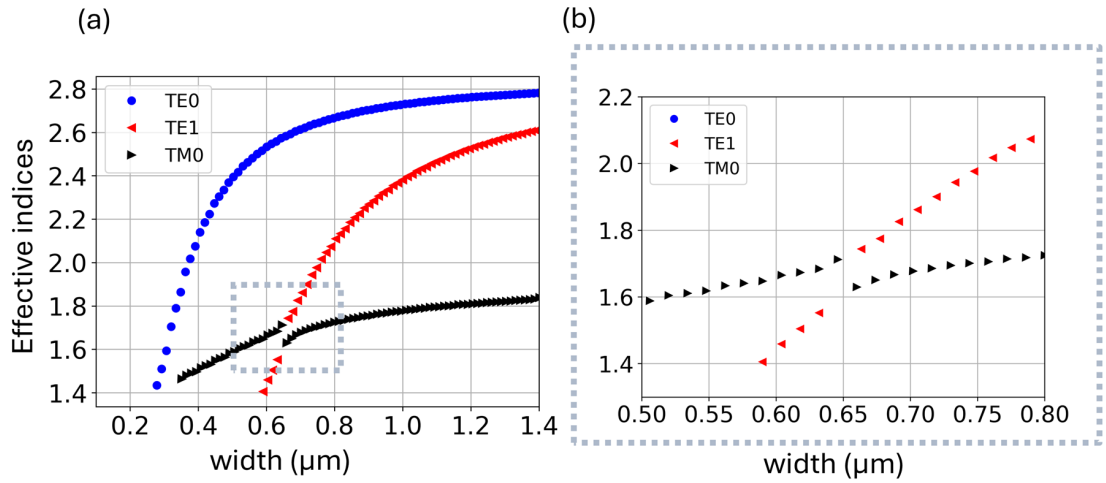

Figure 5.3: (a) Effective indices of an isolated SOI waveguide, calculated on the SOI platform with air cladding and SiO_2 buried oxide, as a function of waveguide width for the TE_0, TE_1, and TM_0 modes. (b) Zoom-in of the anti-crossing region between the TE_1 and TM_0 modes.

$w_6 = 0.5\ \mu$m is chosen to ensure single-mode operation (see Fig. 5.3 (a)). A genetic algorithm is employed for the optimization. The figure of merit is the TE_1-to-TE_0 conversion efficiency as a function of the 11 structural parameters, namely $\vec{w} = [w_1, w_2, w_3, w_4, w_5]$ and $\vec{L} = [L_1, L_2, L_3, L_4, L_5, L_6]$. For the optimization, a population size of 10 was chosen, and after about 100 generations the optimization began to converge, ultimately reaching a maximum conversion efficiency of approximately 78 % after 300 generations. This value was obtained with the following optimized parameters: $\vec{w} = [0.92, 0.88, 1.5, 2.05, 1.44]\ \mu$m and $\vec{L} = [1.58, 5.12, 1.99, 2.85, 5.63, 9.41]\ \mu$m.

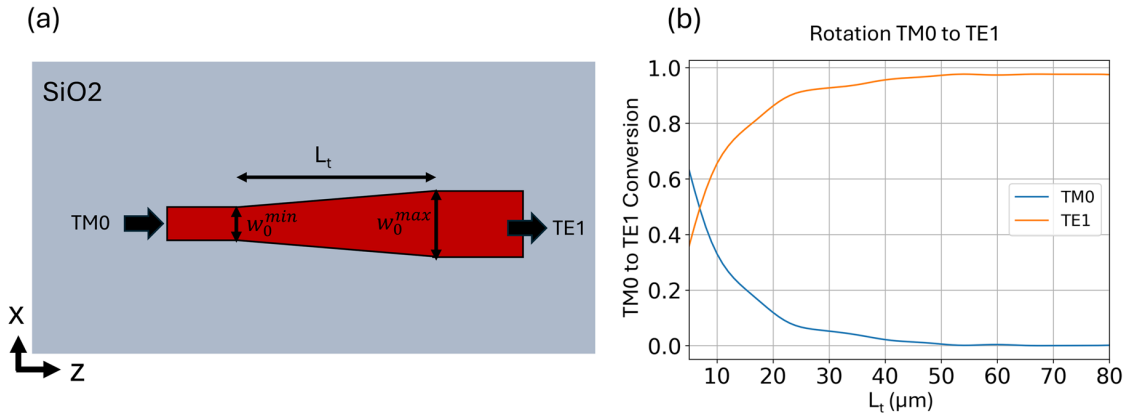

Figure 5.4: (a) Schematic of the taper structure in the rotation stage shown in Fig. 5.1 (a). (b) TM_0-to-TE_1 mode conversion calculated using EME as a function of the taper length (L_t) for the taper illustrated in panel (a).

Since the optimization was performed using EME simulations, we seek to validate the

results using 3D FDTD simulations. The simulation of the TE_1-to-TE_0 conversion are made for two versions of the mode converter: the first Fig. 5.5 (a) with abrupt width transitions in the cascaded asymmetric taper, and the second Fig. 5.5 (b) where splines were used to smooth these transitions. The results in Fig. 5.5 (c) shows that a conversion efficiency of 78 % at $\lambda = 1550$ nm is obtained using EME simulations on the device without spline, while the 3D FDTD simulation achieves for the same device a slightly higher efficiency of 79.5 %, but with a small blue-shift to $\lambda = 1540$ nm. Moreover, the bandwidth predicted by the 3D FDTD simulation is broader than that obtained from EME: over the 1500–1600 nm range, the transmission remains above 70 %, whereas in the EME simulation it stays above 70 % only between 1527 and 1583 nm. Finally, the conversion efficiency of the structure calculated from 3D FDTD simulations was further improved and re-centered at $\lambda = 1550$ nm by introducing spline, achieving up to 87 % efficiency and maintaining transmission above 80 % across the entire wavelength range.

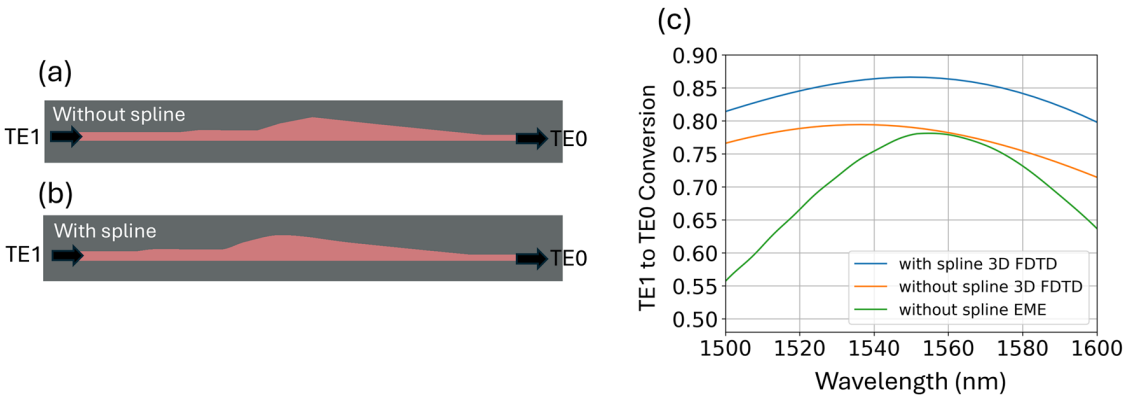

Figure 5.5: (a) TE_1-to-TE_0 mode converter. (b) Same as (a), but with the taper transitions smoothed using splines. (c) TE_1-to-TE_0 conversion as a function of wavelength, calculated using EME simulations for the geometry shown in panel (a), and 3D FDTD simulations for both geometries in panels (a) and (b).

We also investigated how the optimized TE_1-to-TE_0 mode converter behaves when a TE_0 mode is injected at the input, as shown in Fig. 5.6 (a), using 3D FDTD simulations. The results are presented in Fig. 5.6 (b), showing the transmission of the input TE_0 mode into the output TE_0 and TM_0 modes. At $\lambda = 1550$ nm, only 21% of the input TE_0 power remains at the output, with 7% converted into TE_0 and 14% into TM_0. This demonstrates that the cascade of asymmetric tapers acts as a filter for TE_0 excitation. This result is particularly interesting, since, as explained in Sec. 5.1, a PR device generally operates in combination with a PBS or a filter in order to work with a single polarization. Here, however, the filtering effect is already provided by the device itself. Moreover, when combining this polarization rotator with a PBS to realize a PSR, the intrinsic filtering effect of the rotator can help reducing crosstalk.

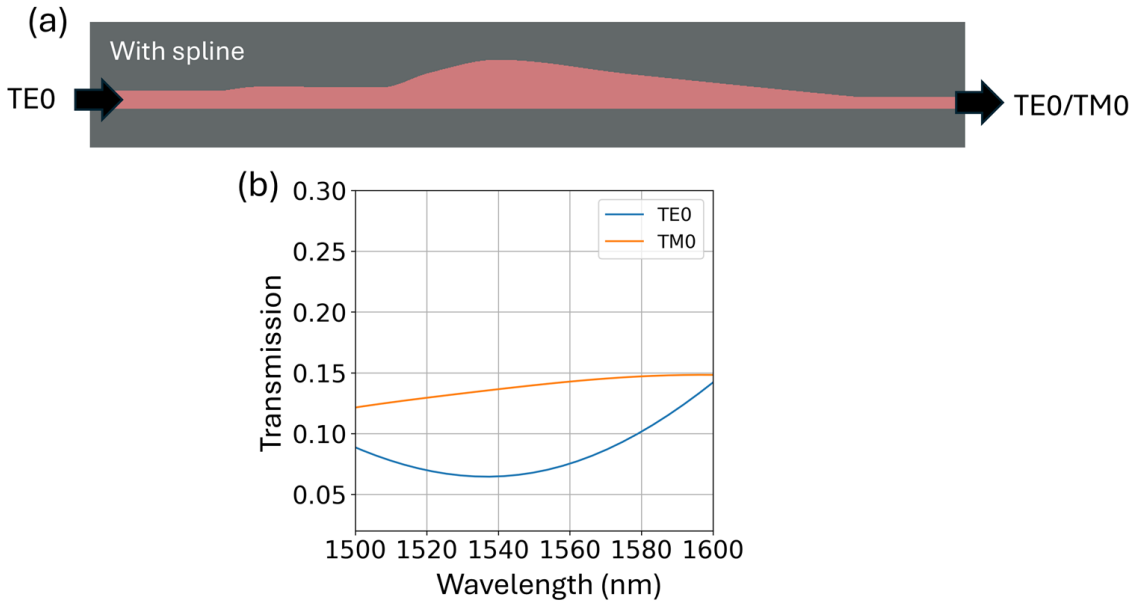

Figure 5.6: (a) TE_1-to-TE_0 mode converter geometry with taper transitions smoothed using splines, this time with the TE_0 mode injected at the input. (b) TE_0-to-TE_0 and TE_0-to-TM_0 conversion calculated using 3D FDTD simulations for the geometry shown in panel (a).

5.2 . Experimental result

The fabricated devices were measured using a fiber-based setup similar to the first one used for the characterization of the PBSs, Fig. 5.7. In this case, we added however a polarizer between the output PM fiber and the detector to analyze the output polarization, as shown in Fig. 2.28. In this configuration, the input polarization can be controlled via the fiber-bench polarizer and the polarization controller before injection into the chip. The polarization controller is used to optimize the polarization such that it is aligned with the polarizer. This polarization is then injected into the chip through a PM lensed fiber, while another PM lensed fiber collects the output light. The light is subsequently sent through the output polarizer, enabling us to determine which polarization is received, and is finally directed to the detector for measurement.

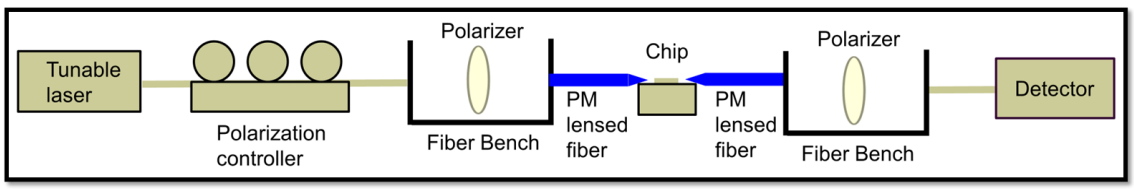

Figure 5.7: Lensed fiber experimental setup for PR device characterization.

Figure 5.8 shows the characterization results of a reference waveguide (without the polarization rotator): (a), when TE polarization (i.e., when the input polarizer is aligned

horizontally) is injected, and (b), when TM polarization (i.e., when the input polarizer is aligned vertically) is injected. The blue curves correspond to measurements obtained when the output polarizer is aligned for the TE polarization while the red curves to the output polarizer aligned for the TM polarization. From these results, it can be seen that in the case of TE injection, the maximum measurable extinction ratio is only about 10 dB on average. However, this is not critical, since the device is expected to behave as a filter when TE polarization is injected. In the case of TM polarization, a higher extinction ratio of at least 15 dB is measured across the full wavelength range, with values exceeding 20 dB at certain wavelengths.

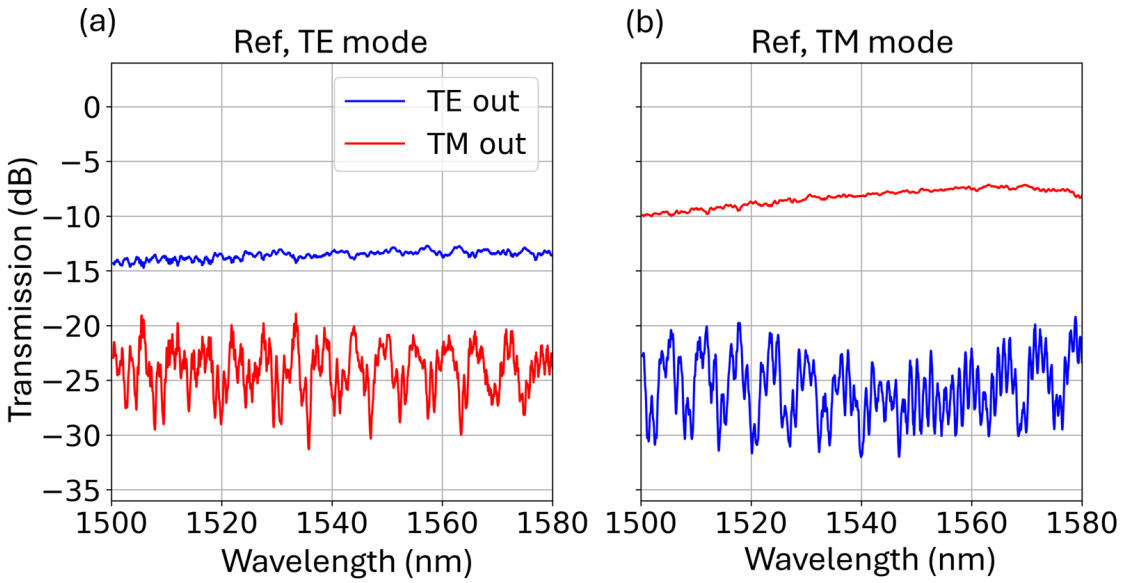

Figure 5.8: Reference waveguide measurements without PR device: (a) TE mode excitation, (b) TM mode excitation.

As can be observed, TE and TM transmissions are not identical because the edge couplers exhibit different performances depending on the polarization. This is reflected in the definition of the performance metrics used to characterize the polarization rotator:

$$PER = 10 \cdot \log_{10} \frac{T^{TM-TE}}{T^{TM-TM}}, \tag{5.1}$$

$$PCL = -10 \cdot \log_{10} \frac{T^{TM-TE}}{T_{ref}^{avg}}, \tag{5.2}$$

$$IL = -10 \cdot \log_{10} \frac{T^{TE-TE}}{T_{ref}^{TE-TE}}. \tag{5.3}$$

Here, T^{TM-TE} is the transmission measured when TM polarization is injected and then output polarizer is oriented horizontally. T^{TM-TM} is the transmission measured when TM

is injected and the output polarizer is vertically oriented. T_{ref}^{TM-TM} is the reference transmission measured when TM is injected and detected as TM. Finally, T_{ref}^{avg} is the average reference transmission for TE and TM injection measured with the output polarizer oriented accordingly:

$$T_{ref}^{avg} = 0.5 \cdot (T_{ref}^{TE-TE} + T_{ref}^{TM-TM}) \tag{5.4}$$

The PER measures the polarization purity, the PCL quantifies the efficiency of converting TM into TE, and the insertion loss indicates the amount of TE filtered by the device. The PR device performs a rotation: the input mode is TM, which is converted into TE, and the light exits as TE. For this reason, the average reference transmission is used to evaluate the PCL of the PR device, since the reference transmission (Fig. 5.8 (a) and (b)) differs by more than 5 dB depending on whether TE or TM is excited.

Figure 5.9 shows the normalized transmission for the polarization rotator, (a) for TE excitation, and (b) for TM excitation. The blue curves correspond to the output polarizer aligned horizontally and the red curves to the output polarized aligned vertically.. The normalization is carried out as follows: when TE is excited and measured in TE, the transmission is normalized by T_{ref}^{TE-TE}; when TM is injected and measured in TM, it is normalized by T_{ref}^{TM-TM}; for cross-polarization measurements, the normalization uses the average reference transmission T_{ref}^{avg}. Reference transmissions are those reported in Fig. 5.8.

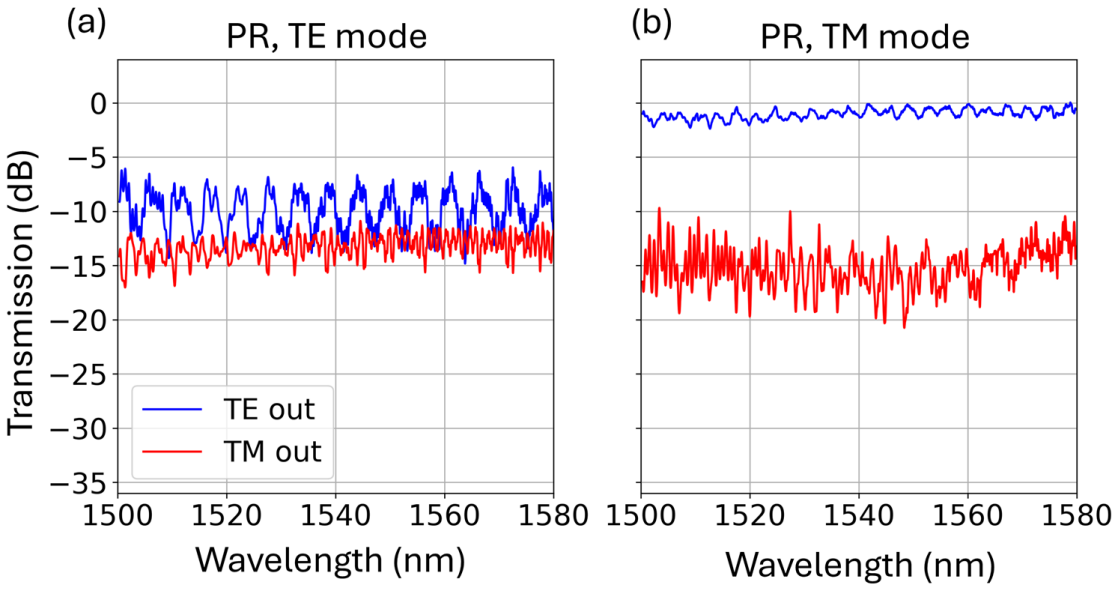

Figure 5.9: Normalized transmission measurements of the PR device are shown for (a) TE mode excitation and (b) TM mode excitation. The normalization is performed using the reference waveguide data presented in Fig. 5.8. For same-polarization measurements (TE to TE or TM to TM), the normalization is done with respect to the corresponding polarization reference. For cross-polarization measurements (TE to TM or TM to TE), the normalization is based on the average transmission of the TE-TE and TM–TM reference waveguide measurements.

The blue curve in Fig. 5.9 (b), with TM injection, corresponds directly to the negative

Table 5.1: Performance comparison of several PR and PSR devices on SOI and SiN platforms, where the PER definition used here corresponds to that of PR devices, as given in Eq. 5.1.

	Wavelength range (nm)	**Min PCL (dB)**	**Max PER (dB)**	**Size (µm)**	**BW$_{PCL<1dB}$ (nm)**	**platform**	**Asymmetry**	**Device**	**Ref**
SWG ADC	1500 - 1600	0.13	-	50	100	SOI	Air cladding	PSR	[94]
ADC	1480 - 1580	0.6	15	37	50	SOI	Air cladding	PSR	[88]
Tapered ADC	1520 - 1620	0.1	-	140	30	SOI	Air cladding	PSR	[89]
Bi-level taper and ADC	1530 - 1565	0.3	-	170	25	SOI	Dual etch	PSR	[91]
Asymmetric silicon waveguide	1520 - 1620	> 3	13	3	-	SOI	Dual etch	PR	[90]
Bi-level asymmetric taper	1420 - 1620	< 0.5	15	300	200	SOI	Dual etch	PR	[100]
Bi-level asymmetric taper	1525 - 1570	< 1	17	37.5	45	SOI	Dual etch	PR	[101]
Hybrid bi-layer taper	1480 - 1580	0.7	> 30	420	80	SiN on SOI	Dual etch	PR	[98]
Dual-etching tapered ADC	1510 - 1580	0.18	-	28	45	SOI	Dual etch and Air cladding	PSR	[95]
Bi-level taper, ADC and MMI	1530 - 1565	~ 1.5	24	2900	-	SiN	Dual etch	PSR	[93]
Bi-level taper and tapered ADC	1260 - 1360	~ 1.5	-	750	-	SiN	Dual etch	PSR	[96]
BI-level taper and asymmetric taper mode converter	740 - 860	< 0.5	25	800	100	SiN	Dual etch	PR	[17]
Taper and asymmetric taper mode converter	1500 - 1580	0.5	15	110	50	SOI	Air cladding	PR	Our work

PCL measurement, which remains below 1 dB across the full 1530–1580 nm range and reach a minimum value of 0.5 dB at λ = 1550 nm. The PER is directly obtained from the difference between the blue and red curves when TM is injected. A PER greater than 10 dB is observed over the entire range, reaching about 15 dB around $\lambda = 1550$ nm. It should be noted that the PER is currently limited by the experimental setup: the reference waveguides show a limited extinction ratio, most likely due to the use of polarization maintaining fibers. Further measurements will be carried out to confirm this. When TE polarization is injected, as expected, the measured transmissions at the output, both in TE and TM, are relatively low (i.e., about -10 dB). The device therefore effectively acts as a TE filter.

Table 5.1 summarizes the performance of different PSR and PR devices fabricated on SOI and SiN platforms. The PER reported here differs from that presented in Table 4.1, as in this case it corresponds to the PER defined for PR devices in Eq. 5.1. Our device compares well with the existing state of the art. When compared to the most similar device, which combines a taper and an ADC in SOI [91], the minimum PCL is nearly identical; however, our device achieves a wider bandwidth while being more compact. Furthermore, for our device the fabrication is simpler since it requires only a single lithography step. In [17], the same concept is implemented on a SiN platform. While the reported PCL and PER are better than ours, the device length is about 800 μm compared to only 110 μm in our case likely due to the use of SiN instead of Si. Additionally, their device requires a dual-etch process, further complicating fabrication.

6 - Conclusion

Controlling light polarization is a central elements in optical systems since many optical functions, such as filtering, multiplexing, or sensing depend critically on the ability to control and separate polarization states. In the context of integrated photonics, polarization management becomes even more important since waveguides inherently exhibit birefringence and polarization dependent behavior.

To meet the growing demand for compact, high-performance, and CMOS-compatible photonic devices, silicon-based platforms have emerged as a promising solution. Silicon-on-insulator (SOI) technology, in particular, offers strong optical confinement, high refractive index contrast, and compatibility with large-scale semiconductor fabrication. The use of silicon presents a major drawback, as its absorption largely grows for wavelengths below 1200 nm, a wavelength range where many applications operate. This limitation has driven the emergence of the silicon nitride (SiN) platform, which allows access to shorter wavelengths while also potentially enabling the fabrication of waveguides with lower propagation losses compared to silicon.

The SiN platform exhibits lower birefringence than the silicon one, yet it is still significant enough to require polarization control devices. However, despite the advantage of silicon nitride operating efficiently at sub-telecom wavelengths, only limited research has been devoted to polarization control in this spectral range. In general, polarizarion control devices exploit birefringence to manipulate the polarization state, which often makes SiN-based components larger and more susceptible to crosstalk than their silicon counterparts. Therefore, the objective of this thesis was to design and demonstrate high-performance and compact SiN-based devices operating at wavelengths that have been scarcely explored so far.

To address these challenges, we first focused on the realization of polarization beam splitters. Two devices were designed: one operating at 950 nm and the other at 1300 nm. The design and fabrication of an initial generation of PBSs based on directional couplers was performed. To improve fabrication tolerance, a second version incorporating tapered waveguides was also designed and fabricated. The fabricated devices successfully achieved polarization separation with very low insertion losses. The tapered versions demonstrated improved fabrication tolerance, albeit at the cost of slightly reduced peak performance.

From these results, two key issues emerged, motivating the development of a second generation of devices. The first was the difficulty in accurately measuring the insertion losses, as they were extremely low; the second was the presence of oscillations in the transmission spectra. To address these issues, cascaded PBS structures were fabricated at both 950 nm and 1300 nm to allow precise loss characterization. Additionally, the geometry of the central waveguide, suspected to act as a Fabry–Pérot cavity and thus cause spectral oscillations was modified. This second generation of devices enabled accurate determination of the insertion losses for the device operating around 1300 nm, which

were found to be as low as 0.1 dB for both TE_0 and TM_0 modes representing, to the best of our knowledge, the lowest losses ever reported for a SiN-based PBS. Unfortunately, similar characterization could not be completed for the 950 nm devices due to fabrication induced deviations in waveguide width, which caused a redshift of the operating wavelength beyond the tunable range of our laser source (910–980 nm).

As for the oscillations, they persisted even after redesign, suggesting that they originated from the experimental setup, particularly the use of polarization maintaining lensed fibers. Tests confirmed that the fibers limited measurement accuracy. Consequently, a new free-space setup using microscope objectives instead of fibers was developed to enable precise measurement of device performance, especially regarding the polarization extinction ratio.

Once this setup was implemented, new measurements of the previously fabricated devices were carried out. The non tapered PBS operating at 950 nm achieved a PER above 10 dB across the 910–955 nm range, with peak PER values of 29 dB and 24 dB for TE and TM modes, respectively, at 930 nm, and insertion losses below 1 dB. These results are comparable to other SiN PBSs reported in the literature, while being achieved with a compact device footprint of only 110 μm. The tapered version of the same design exhibited PER values above 10 dB across the full 910–980 nm laser range, with maximum PER values of 26 dB and 16 dB for TE and TM modes, respectively, and insertion losses below 1 dB. A stability analysis of the TM mode PER under various biases confirmed that the tapered device was significantly more robust than the non tapered one, although at the cost of increased device length (242 μm) and reduced maximum PER.

For the 1300 nm PBSs, the non tapered device showed a PER above 10 dB across the 1275–1335 nm range, with a maximum PER of 32 dB for both TE and TM modes at 1310 nm. At this wavelength, insertion losses were measured at only 0.1 dB for a 124 μm long device, demonstrating an exceptionally compact design with record-low losses and high PER, among the best reported for SiN-based PBSs. Similarly to the 950 nm case, the tapered version exhibited improved stability at the expense of reduced performance and larger footprint. It maintained PER values above 10 dB over the 1250–1350 nm range and insertion losses below 1 dB across the same range. However, its maximum PER values were reduced to 29 dB (TE) and 22 dB (TM), with an increased device length of 290 μm.

In the second part of this work, we developed a new PBS design based on subwavelength structures. The use of SWGs provided two major advantages. First, the device footprint was reduced by nearly half, as there was no longer any need to rely on higher-order modes for polarization separation. Second, in an SWG structure without an upper cladding, the TM_0 mode can become unguided, which further enhances polarization selectivity.

Based on these principles, the devices were designed and fabricated. The measured maximum PERs reached 24 dB and 26 dB for the TE and TM modes, respectively, at λ = 955 nm, with insertion losses below 1 dB. All of this was achieved with an extremely compact footprint of only 63 μm, representing the shortest SiN-based PBS reported to date, while maintaining PER performances comparable to the state of the art in SiN PBS technology.

For the SWG PBS operating around 1300 nm, a fabrication deviation caused a red shift of the operating wavelength to 1355 nm. Nevertheless, the device exhibited PER values of 20 dB and 30 dB for the TE_0 and TM_0 modes, respectively, with insertion losses below 1 dB. The total device length was only 88 μm, comparable to the shortest SiN PBS ever reported (the angled-MMI design [56]) with a length of 80 μm.

In the third part of this work, we focused on polarization splitter–rotator devices, for which we provided a detailed description of the operating principles behind the most common implementation methods. We then designed and fabricated a PSR based on the mode-evolution technique, operating around 1300 nm. The fabricated devices exhibited a minimum PCL of approximately 2 dB at a wavelength of 1260 nm. The IL remained below 1 dB over the entire wavelength range from 1240 to 1280 nm. The maximum polarization extinction ratios (PER) reached 28 dB and 20 dB for the TE_0 and TM_0 modes, respectively, both at λ = 1260 nm. The PSR demonstrated performance comparable to other SiN-based devices reported in the literature, while achieving a shorter device length of 685 μm. Furthermore, fabrication was simplified since the process required only a single lithography step, unlike most existing SiN PSR implementations.

In the fourth part, a final device was presented: a polarization rotator (PR) implemented on an SOI platform as part of a project conducted at C2N. The PR was designed using the mode evolution approach combined with a mode converter. The fabricated device achieved a PCL below 1 dB across the wavelength range from 1530 to 1580 nm, with a minimum value of 0.5 dB at 1550 nm. The PER is about 15 dB for a device length of 110 μm. These results compare favorably with the state of the art in SOI-based PSR and PR devices, while also offering a simplified fabrication process since, here again, no double lithography step was required.

We have thus demonstrated high-performance and compact SiN-based devices operating at wavelengths that have rarely been explored. To further advance this work, one promising direction would be to leverage subwavelength grating structures, not only to achieve even more compact devices, but also to broaden their operational bandwidth.

In the case of the PSR, a two-step etching process combined with air cladding could be investigated to realize an even more compact design. Moreover, given the range of devices developed throughout this work, it would be possible to envision an integrated polarization management devices, where the state of polarization could be actively controlled on-chip using MZIs with integrated heaters, in combination with the fabricated PBS and PSR devices.

7 - Appendix: Polarization-maintaining fibers

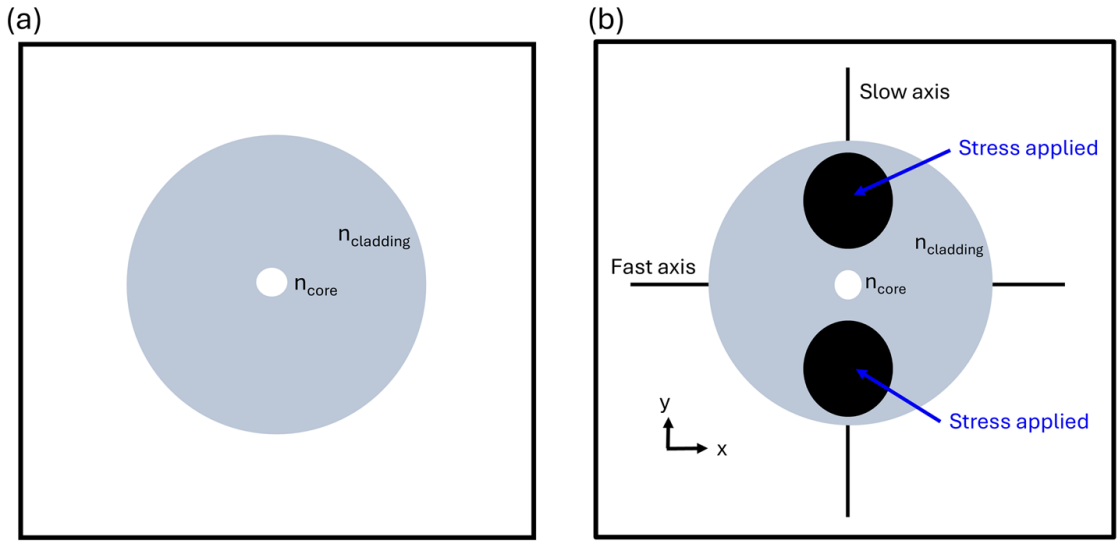

Figure 7.1: (a) Schematic of the cross section of a conventional rotationally symmetric fiber. (b) Schematic of the cross section of a specific type of polarization-maintaining fiber known as a PANDA fiber.

In conventional optical fibers, the geometry is fully rotationally symmetric (see Fig. 7.1 (a)), which ideally results in zero birefringence. Under such conditions, the polarization state of the light should be preserved. However, in practice this is not the case. Real fibers exhibit intrinsic imperfections such as slight ellipticity of the fiber core, which introduce birefringence. Additional birefringence may also be induced by external perturbations, for example bending of the fiber. These phenomena alter the polarization state over relatively short fiber lengths, sometimes even below one meter. Moreover, the resulting polarization changes are not only random and unpredictable, but also strongly dependent on the wavelength and on the temperature of the fiber along its entire length [102].

To address these limitations, alternative fiber designs known as polarization-maintaining fibers have been proposed. One such example is shown in Fig. 7.1 (b). This is called a Panda fiber which relies on mechanical stress, for instance by incorporating stress rods made from a different glass. The induced stress creates a birefringence and defines a fast and a slow axis in the fiber (see Fig. 7.1 (b)). A “polarization-maintaining” fiber does not preserve an arbitrary input polarization state something that might appear counter-intuitive given its name. For example, if circularly polarized light is injected, the circular polarization will not be maintained at the output. However, the fiber does preserve the polarization of linearly polarized light that is aligned with either the fast or the slow axis [102].

The purpose of introducing this birefringence is essentially to suppress the influence of small random birefringence variations, such as those caused by moderate bending. These random perturbations may slightly modify the local polarization state, but they generally do not produce significant effects over longer fiber lengths [102].

As a reminder, in Sec. 2.4.1, oscillations were observed during the characterization of the setup using a polarization-maintaining fiber, as shown in Fig. 2.41. To explain this behavior, we consider the case where the injected polarization is primarily aligned with the fast axis of the fiber (along x in Fig. 7.1 (b)), but slightly misaligned by a small angle $\theta \ll 1$. The normalized field (E_{af}) after the fiber can then be expressed as:

$$E_{af} = \begin{pmatrix} \cos(\theta) \\ \sin(\theta) e^{i\Delta\varphi} \end{pmatrix} \tag{7.1}$$

where $\Delta\varphi$ is the accumulated phase difference between the two axes, given by:

$$\Delta\varphi = \frac{2\pi}{\lambda}(n_{\text{slow}} - n_{\text{fast}})L_{\text{fiber}}, \tag{7.2}$$

with n_{fast} and n_{slow} the effective refractive indices of the fast and slow axes, respectively, and L_{fiber} the fiber length.

The field then passes through an analyzer whose transmission axis is mainly aligned with the slow axis of the fiber but slightly misaligned by a small angle $\alpha \ll 1$. The transmitted intensity is given by the squared magnitude of the projection of the field onto the analyzer's transmission axis:

$$I = \left| \begin{pmatrix} -\sin(\alpha) & \cos(\alpha) \end{pmatrix} \begin{pmatrix} \cos(\theta) \\ \sin(\theta) e^{i\Delta\varphi} \end{pmatrix} \right|^2 \tag{7.3}$$

which expands to:

$$I = \sin(\alpha)^2 \cos^2(\theta) + \cos^2(\alpha) \sin^2(\theta) - 2 \sin(\alpha) cos(\alpha) \sin(\theta) cos(\theta) \cos(\Delta\varphi) \tag{7.4}$$

Using the small-angle approximation, Eq. 7.4 simplifies to:

$$I = \alpha^2 + \theta^2 - 2\alpha\theta \cos(\Delta\varphi) \tag{7.5}$$

showing that even a very small misalignment of the input polarization from the fast axis, combined with a slight misalignment of the analyzer, causes the transmitted intensity to oscillate as a function of wavelength. The oscillation period is given by:

$$\Delta\lambda = \frac{\lambda^2}{(n_{\text{slow}} - n_{\text{fast}})L_{\text{fiber}}}. \tag{7.6}$$

These oscillations correspond to those observed in Fig. 2.41.

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